



TIA STANDARD

Project 25 Fixed Station Interface Messages and Procedures

**TIA-102.BAHA-A
(Revision of TIA-102.BAHA)**

June 2017

**TELECOMMUNICATIONS
INDUSTRY ASSOCIATION**

tiaonline.org

NOTICE

TIA Engineering Standards and Publications are designed to serve the public interest through eliminating misunderstandings between manufacturers and purchasers, facilitating interchangeability and improvement of products, and assisting the purchaser in selecting and obtaining with minimum delay the proper product for their particular need. The existence of such Standards and Publications shall not in any respect preclude any member or non-member of TIA from manufacturing or selling products not conforming to such Standards and Publications. Neither shall the existence of such Standards and Publications preclude their voluntary use by Non-TIA members, either domestically or internationally.

Standards and Publications are adopted by TIA in accordance with the American National Standards Institute (ANSI) patent policy. By such action, TIA does not assume any liability to any patent owner, nor does it assume any obligation whatever to parties adopting the Standard or Publication.

This Standard does not purport to address all safety problems associated with its use or all applicable regulatory requirements. It is the responsibility of the user of this Standard to establish appropriate safety and health practices and to determine the applicability of regulatory limitations before its use.

Any use of trademarks in this document are for information purposes and do not constitute an endorsement by TIA or this committee of the products or services of the company.

(From Project No. TIA-PN-102.BAHA-A-R1, formulated under the cognizance of the TIA TR-8 Mobile and Personal Private Radio TR-8.19 Subcommittee on Wireline System Interfaces).

Published by
©TELECOMMUNICATIONS INDUSTRY ASSOCIATION
Technology & Standards Department
1320 N. Courthouse Road, Suite 200
Arlington, VA 22201 U.S.A.

**PRICE: Please refer to current Catalog of
TIA TELECOMMUNICATIONS INDUSTRY ASSOCIATION STANDARDS
AND ENGINEERING PUBLICATIONS
or call IHS, USA and Canada
(1-877-413-5187) International (303-397-2896)
or search online at <http://www.tiaonline.org/standards/catalog/>**

All rights reserved
Printed in U.S.A.

NOTICE OF COPYRIGHT

This document is copyrighted by the TIA.

Reproduction of these documents either in hard copy or soft copy (including posting on the web) is prohibited without copyright permission. For copyright permission to reproduce portions of this document, please contact the TIA Standards Department or go to the TIA website (www.tiaonline.org) for details on how to request permission. Details are located at:

<http://www.tiaonline.org/standards/catalog/info.cfm#copyright>

or

Telecommunications Industry Association
Technology & Standards Department
1320 N. Courthouse Road, Suite 200
Arlington, VA 22201 USA
+1.703.907.7700

Organizations may obtain permission to reproduce a limited number of copies by entering into a license agreement. For information, contact

IHS
15 Inverness Way East
Englewood, CO 80112-5704
or call
USA and Canada (1.800.525.7052)
International (303.790.0600)

NOTICE OF DISCLAIMER AND LIMITATION OF LIABILITY

The document to which this Notice is affixed (the "Document") has been prepared by one or more Engineering Committees or Formulating Groups of the Telecommunications Industry Association ("TIA"). TIA is not the author of the Document contents, but publishes and claims copyright to the Document pursuant to licenses and permission granted by the authors of the contents.

TIA Engineering Committees and Formulating Groups are expected to conduct their affairs in accordance with the TIA Procedures for American National Standards and TIA Engineering Committee Operating Procedures, the current and predecessor versions of which are available at <http://www.tiaonline.org/standards/ec-procedures>) TIA's function is to administer the process, but not the content, of document preparation in accordance with the Manual and, when appropriate, the policies and procedures of the American National Standards Institute ("ANSI"). TIA does not evaluate, test, verify or investigate the information, accuracy, soundness, or credibility of the contents of the Document. In publishing the Document, TIA disclaims any undertaking to perform any duty owed to or for anyone.

If the Document is identified or marked as a project number (PN) document, or as a standards proposal (SP) document, persons or parties reading or in any way interested in the Document are cautioned that: (a) the Document is a proposal; (b) there is no assurance that the Document will be approved by any Committee of TIA or any other body in its present or any other form; (c) the Document may be amended, modified or changed in the standards development or any editing process.

The use or practice of contents of this Document may involve the use of intellectual property rights ("IPR"), including pending or issued patents, or copyrights, owned by one or more parties. TIA makes no search or investigation for IPR. When IPR consisting of patents and published pending patent applications are claimed and called to TIA's attention, a statement from the holder thereof is requested, all in accordance with the Manual. TIA takes no position with reference to, and disclaims any obligation to investigate or inquire into, the scope or validity of any claims of IPR. TIA will neither be a party to discussions of any licensing terms or conditions, which are instead left to the parties involved, nor will TIA opine or judge whether proposed licensing terms or conditions are reasonable or non-discriminatory. TIA does not warrant or represent that procedures or practices suggested or provided in the Manual have been complied with as respects the Document or its contents.

If the Document contains one or more Normative References to a document published by another organization ("other SSO") engaged in the formulation, development or publication of standards (whether designated as a standard, specification, recommendation or otherwise), whether such reference consists of mandatory, alternate or optional elements (as defined in the TIA Procedures for American National Standards) then (i) TIA disclaims any duty or obligation to search or investigate the records of any other SSO for IPR or letters of assurance relating to any such Normative Reference; (ii) TIA's policy of encouragement of voluntary disclosure (see TIA Procedures for American National Standards Annex C.1.2.3) of Essential Patent(s) and published pending patent applications shall apply; and (iii) Information as to claims of IPR in the records or publications of the other SSO shall not constitute identification to TIA of a claim of Essential Patent(s) or published pending patent applications.

TIA does not enforce or monitor compliance with the contents of the Document. TIA does not certify, inspect, test or otherwise investigate products, designs or services or any claims of compliance with the contents of the Document.

ALL WARRANTIES, EXPRESS OR IMPLIED, ARE DISCLAIMED, INCLUDING WITHOUT LIMITATION, ANY AND ALL WARRANTIES CONCERNING THE ACCURACY OF THE CONTENTS, ITS FITNESS OR APPROPRIATENESS FOR A PARTICULAR PURPOSE OR USE, ITS MERCHANTABILITY AND ITS NONINFRINGEMENT OF ANY THIRD PARTY'S INTELLECTUAL PROPERTY RIGHTS. TIA EXPRESSLY DISCLAIMS ANY AND ALL RESPONSIBILITIES FOR THE ACCURACY OF THE CONTENTS AND MAKES NO REPRESENTATIONS OR WARRANTIES REGARDING THE CONTENT'S COMPLIANCE WITH ANY APPLICABLE STATUTE, RULE OR REGULATION, OR THE SAFETY OR HEALTH EFFECTS OF THE CONTENTS OR ANY PRODUCT OR SERVICE REFERRED TO IN THE DOCUMENT OR PRODUCED OR RENDERED TO COMPLY WITH THE CONTENTS.

TIA SHALL NOT BE LIABLE FOR ANY AND ALL DAMAGES, DIRECT OR INDIRECT, ARISING FROM OR RELATING TO ANY USE OF THE CONTENTS CONTAINED HEREIN, INCLUDING WITHOUT LIMITATION ANY AND ALL INDIRECT, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES (INCLUDING DAMAGES FOR LOSS OF BUSINESS, LOSS OF PROFITS, LITIGATION, OR THE LIKE), WHETHER BASED UPON BREACH OF CONTRACT, BREACH OF WARRANTY, TORT (INCLUDING NEGLIGENCE), PRODUCT LIABILITY OR OTHERWISE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THE FOREGOING NEGATION OF DAMAGES IS A FUNDAMENTAL ELEMENT OF THE USE OF THE CONTENTS HEREOF, AND THESE CONTENTS WOULD NOT BE PUBLISHED BY TIA WITHOUT SUCH LIMITATIONS.

Copyright Telecommunications Industry Association
Provided by IHS under license with TIA
No reproduction or networking permitted without license from IHS Markit

FOREWORD

(This foreword is not part of this document.)

This document has been created in response to a request by the APCO/NASTD/FED Project 25 Steering Committee as provided for in a Memorandum of Understanding (MOU) dated April 1992 and amended December 1993. The MOU states APCO/NASTD/FED shall devise a Common System Standard for digital public safety communications (the Standard) commonly referred to as Project 25 or P25, and that TIA, as agreed to by the membership of the appropriate TIA Engineering Committee, will provide technical assistance in the development of documentation for the Standard. If the abbreviation "P25" or the wording "Project 25" appears on the cover sheet of this document when published that indicates the APCO/NASTD/FED Project 25 Steering Committee has adopted this document as part of the Standard. The appearance of the abbreviation "P25" or the wording "Project 25" on the cover sheet of this document or in the title of any document referenced herein should not be interpreted as limiting the applicability of the information contained in any document to "P25" or "Project 25" implementations exclusively.

This document has been developed by TR8.19 (Wireline Interfaces) with inputs from the APCO Project 25 Interface Committee (APIC), the APIC Fixed Station Interface Task Group, and TIA Industry members. This document is being published to provide technical information on the emerging digital techniques for Land Mobile Radio Service.

This document is the first revision to the published Fixed Station Interface Messages & Procedures document, TIA102.BAHA.

This document defines a Conventional Fixed Station Interface (CFSI). The CFSI is the interface between a conventional fixed station, which is defined in this document as a functional component of a Fixed Station Subsystem, and a Conventional Fixed Station Host (CFSH). A CFSH is defined in this document as a functional component of either an RF Subsystem (RFSS) or a Console Subsystem. For the purposes of this document the CFSI is defined as a specialization of a generic Fixed Station Interface (FSI) described in TSB-102-C. Future related efforts are planned to develop new TIA documents defining compliance tests (e.g., measurement methods, performance recommendations, and interoperability tests) for the Fixed Station Interface protocols defined in this and future related documents.

PATENT IDENTIFICATION

The reader's attention is called to the possibility that compliance with this document may require the use of one or more inventions covered by patent rights.

By publication of this document, no position is taken with respect to the validity of those claims or any patent rights in connection therewith. The patent holders so far identified have, we believe, filed statements of willingness to grant licenses under those rights on reasonable and nondiscriminatory terms and conditions to applicants desiring to obtain such licenses.

The following patent holders and patents have been identified in accordance with the TIA intellectual property rights policy:

No patents have been identified.

TIA shall not be responsible for identifying patents for which licenses may be required by this document or for conducting inquiries into the legal validity or scope of those patents that are brought to its attention. TIA will neither be a party to discussion of any licensing terms or conditions, which are left to the parties involved, nor will TIA opine or judge whether proposed licensing terms or conditions are reasonable or non-discriminatory.

TRADEMARK IDENTIFICATION

The following trademarks are referenced in this specification:

IMBE™ is a trademark of Digital Voice Systems, Inc.*

* Use of the trademark identified in this document is not an endorsement by TIA.

Contents

1 SCOPE	1
2 DOCUMENT ORGANIZATION	3
3 REFERENCES	4
3.1 NORMATIVE REFERENCES.....	4
3.2 INFORMATIVE REFERENCES	5
4 DEFINITIONS	6
5 ABBREVIATIONS AND ACRONYMS.....	9
6 CFSI ARCHITECTURE	12
6.1 CORE CFSI FUNCTIONS	12
6.1.1 Conventional Analog Fixed Stations.....	12
6.1.2 Conventional Digital Fixed Stations	12
6.1.3 Mixed Mode Fixed Stations	12
6.2 CFSI ARCHITECTURE AND OVERVIEW	13
6.2.1 Analog Fixed Station Interface.....	14
6.2.2 Digital Fixed Station Interface – Protocol Version 1	14
6.2.3 Digital Fixed Station Interface – Protocol Version 2	16
7 CFSI ANALOG INTERFACE	18
7.1 E & M.....	18
7.2 TONE REMOTE CONTROL.....	19
7.2.1 Simplex Wireline Interface for Configuration 6	20
7.2.2 Non-Transmit Sequence	21
7.2.3 Transmit Sequence.....	21
7.2.4 Guard Tones	21
7.2.5 Function Tone	22
7.2.6 Speech.....	25
7.2.7 Sequence Off Time	25
7.3 INTERCOM.....	26
8 CFSI DIGITAL INTERFACE	27
8.1 INTERNET PROTOCOL OVERVIEW	27
8.1.1 DFSI Lower Layers	28
8.1.2 Network Layer	28
8.1.3 Transport Layer.....	29
8.1.4 Transport and Network Layer Parameters.....	30
8.2 DFSI CONFIGURATION ITEMS.....	32
8.3 DFSI CONTROL SERVICE.....	33
8.3.1 Control Messages	33
8.3.2 Control Service Procedures	56
8.4 DFSI VOICE CONVEYANCE SERVICE	78
8.4.1 Voice Conveyance Connection Establishment Protocol Version 2	78
8.4.2 Voice Conveyance Connection Detachment Procedures Protocol Version 2	79
8.4.3 Structure of Voice Conveyance Payload	80
8.4.4 Voice Conveyance RTP Payload, Block PTs	85
8.4.5 Voice Conveyance Service Procedures	103
9 DATA CONVEYANCE PROCEDURES.....	113

TIA-102.BAHA-A

9.1	DATA CONVEYANCE CONNECTION ESTABLISHMENT PROTOCOL VERSION 2	113
9.2	DATA CONVEYANCE CONNECTION DETACHMENT PROCEDURES PROTOCOL VERSION 2	114
9.3	DATA CONVEYANCE MESSAGE	115
9.3.1	Data Conveyance Header	115
9.3.2	Data Conveyance Payload	117
9.4	DATA CONVEYANCE PROCEDURES	121
9.4.1	Transfer of Data Conveyance messages	121
10	INFORMATIVE ANNEX.....	127
10.1	RESERVED WORDS	127

Figures

Figure 1 – CFSI Architecture	13
Figure 2 – Conventional Analog Fixed Station Interface	14
Figure 3 – Conventional Digital Fixed Station Interface – Protocol Version 1	15
Figure 4 – Conventional Digital Fixed Station Interface – Protocol Version 2	17
Figure 5 – Possible E&M Interface Configurations.....	19
Figure 6 – Possible TRC Interface Configurations	20
Figure 7 – Non-Transmit TRC Sequence	21
Figure 8 – Transmit TRC Sequence	21
Figure 9 – Internet Architecture Layers and the OSI Layers	27
Figure 10 – DFSI Internet Protocol Suite including Data Conveyance	28
Figure 11 – Data for the Establish Connection Command, Message Version 1	35
Figure 12 - Data for the Establish Connection Command, Message Version 2	35
Figure 13 – Response Data for the Connection Command, Response Version 1.....	38
Figure 14 - Response Data for the Connection Command, Response Version 2	39
Figure 15 - Optional Response Data for CONTROL_NAK_CONNECTED.....	40
Figure 16 - Response data for CONTROL_NAK_OPT_UNSUPP	40
Figure 17 – Message for Control Service Heartbeat	40
Figure 18 – Message for Control Service Acknowledgement	42
Figure 19 - Message for Single Block Control, for all Protocol Versions.....	43
Figure 20 – Message for Single Block Control, Protocol Version 2	43
Figure 21 – Message for Manufacturer Extension	44
Figure 22 – Command for Channel Selection	45
Figure 23 - Response Data for Channel Selection, Report Version 2.....	45
Figure 24 – Command for Repeat Mode Selection	46
Figure 25 – Message for Squelch Control, Message Version 1 or 2.....	47
Figure 26 – Command for Report Selections	48
Figure 27 – Response Data for Report Selections.....	48
Figure 28 – Command for Disconnect Connection, Protocol Version 1	49
Figure 29 – Command for Disconnect Connection Protocol Version 2	49
Figure 30 - Command for Scan Enable	50
Figure 31 – Command for Receiver Control.....	50
Figure 32 - Command for Transmitter Control	52
Figure 33 - Multiple Block Control Message	53
Figure 34 – FSC_START_DATA message format.....	54
Figure 35 – FSC_STOP_DATA message format.....	55
Figure 36 – FSC_CAI_BUSY_BITS Message Format	55
Figure 37 – FS Control Service State Model	57
Figure 38 – Host Control Client State Model	58
Figure 39 – Overview of DFSI Voice Conveyance Packet	81
Figure 40 – DFSI Voice Conveyance Payload Header	82
Figure 41 – Control Octet	82
Figure 42 – DFSI Voice Conveyance Payload Header – Compact form	82
Figure 43 – DFSI Voice Conveyance Block Header – Verbose form.....	83
Figure 44 – Structure of Payload Header and Information Blocks	83
Figure 45 – Format of Block Type Field	83
Figure 46 – Common Fields for CAI Blocks	86
Figure 47 – Format for CAI Frames 1, 2, 10, and 11	88
Figure 48 – CAI block with Link Control Word Field.....	89
Figure 49 – CAI Block with a Low Speed Data Field.....	90
Figure 50 – CAI Blocks with Encryption Sync Field	91
Figure 51 – General Format of Link Control Word Block Payload	92

Figure 52 – Example Format of Link Control Word Block Payload (LCF= 0)	92
Figure 53 – Example Format of Link Control Word Block Payload (LCF= 3)	92
Figure 54 –Format of Encryption Sync Word Block Payload	93
Figure 55 –Format of Low Speed Data Block Payload	93
Figure 56 –Format of Decoded Header Block Payload	94
Figure 57 – CAI Voice Header Part 1	95
Figure 58 – CAI Voice Header Part 2	96
Figure 59 - Start of Stream Block.....	97
Figure 60 – Start of Stream Block.....	99
Figure 61 – Receiver Report Block	100
Figure 62 – Voter Control Block	101
Figure 63 - Transmitter Report Block	102
Figure 64 – Format for Manufacturer Customizations	103
Figure 65- Start of stream message sequence - inbound digital	107
Figure 66 - Start of stream message sequence - Analog	108
Figure 67 – Start of Stream Message Sequence – Outbound Voice for all FSI Protocol versions ...	109
Figure 68 – Start of Stream with the First ACK Lost for all FSI Protocol versions.....	109
Figure 69 – Packet with Analog Transparent Data	111
Figure 70 – End of Stream	112
Figure 71 – End of Stream Time Out	112
Figure 72 – FSI Data Conveyance Message	115
Figure 73 – Data Conveyance Message Header	116
Figure 74 – DC Payload Block Count for FSI Data Conveyance Message	118
Figure 75 – Data Conveyance Block Type Octet.....	118
Figure 76 - CAI Transmission Error Count.....	119
Figure 77 – Inbound Data Conveyance Message Assembly.....	121
Figure 78 – Outbound Data Conveyance Message Modulation	122

Tables

Table 1 – TRC Function Tones vs. Possible FS Functions.....	23
Table 2 – Possible FS Functions vs. TRC Function Tones.....	24
Table 3 – DSCP Values.....	30
Table 4 – Configuration Items for the DFSI.....	32
Table 5 – Control Service Message Set Mnemonics	33
Table 6 Options Field in Connection Comment.....	37
Table 7 - Values for FSC_CONNECT “Connection Type” parameter	38
Table 8 – Response Code Values.....	42
Table 9 - Squelch Mode Control Word	47
Table 10- Receiver Command Values	51
Table 11 - Transmitter Command Values	52
Table 12 Channel Selection Behavior Summary for FSI Protocol Version 2	72
Table 13 - Block Payload Type FSI Protocol Versions.....	85
Table 14 – CAI Frame Types	87
Table 15 - Valid Squelch Codes	97
Table 16 – Receiver Status Receiver Report Codes.....	100
Table 17 – Voter Control Command.....	101
Table 18 - Transmitter Status Report Codes.....	102
Table 19 - Data Conveyance Block Types	119
Table 20 - Block Payload Type Reservations	127

This page intentionally left blank

1 Scope

This document defines a Conventional Fixed Station Interface (CFSI). The CFSI is the interface between a conventional fixed station, which is defined in this document as a functional component of a Fixed Station Subsystem, and a Conventional Fixed Station Host (CFSH). A CFSH is defined in this document as a functional component of either an RF Subsystem (RFSS) or a Console Subsystem.¹ Clause 6 describes these elements and their relationship to each other. For the purposes of this scope statement, a fixed station may be either an “analog fixed station” or a “digital fixed station”. An “analog fixed station” supports conventional FM RF resources or their equivalent. A “digital fixed station” supports the Common Air Interface (CAI) for conventional implementations ([8]).

This document fully describes the CFSI including its architecture (clause 6), and protocols (clauses 7 and 8). Clause 7 describes a basic analog audio and control interface, while clause 8 describes a more capable Internet Protocol (IP) ([1]) based digital interface. This interface has been designed to work with IPv4. Use of this interface with IPv6 is for further study.

The functional scope of the CFSI as described herein comprises the following CFSI Analog and CFSI Digital Interface provisions.

The CFSI Analog Interface provides:

1. Transport of clear audio between a fixed station and its host² providing capabilities for full-duplex, half-duplex and simplex communications at the discretion of the fixed station.
2. Transport of E&M control signaling between a fixed station and its host to provide a simple “Push-to-talk” and “COR” capability.
3. Transport of Tone Remote Control (TRC) control signaling from a host to a fixed station to provide a variety of control functions including transmit channel control, squelch control, monitor control, clear/secure controls, analog/digital mode controls, etc.
4. An “Intercom” capability allowing for the transport of audio between the fixed station and its host without initiating an RF transmission.

The CFSI Digital Interface provides:

1. IP based capabilities equivalent to those provided via the Analog Interface.

¹ For the purposes of this document the CFSI is defined as a specialization of the generic Fixed Station Interface (FSI) described in TSB102-C (see [19]).

² Throughout this document the terms “Host” or “host” refer to the CFSH as defined in this document.

2. Transport of encrypted audio between a digital fixed station and its host.
3. Transport of Caller-ID information between a digital fixed station and its host. This includes Unit ID from field units to consoles, and "Console ID" from consoles to field units.
4. Transport of Talk-group information between a digital fixed station and its host.
5. Transport of NAC code information between a digital fixed station and its host.
6. Transport of Emergency Alert messages from the digital fixed station to its host.
7. Transport of Emergency indications along with the voice stream between the digital fixed station and its host.
8. Transport of Received voter identification from the digital fixed station to its host.
9. Advanced Control of the fixed station by its host including the disabling of all squelch (monitor function).
10. Transport of intercom audio to and from the fixed station.
11. For Protocol Version 2 interfaces, transport of Packet Data between a digital fixed station and its host.

2 Document Organization

This document defines the Conventional Fixed Station Interface (CFSI).

The document is organized into eight clauses that are structured as follows:

- Clause 1 provides a general description of the CFSI, including identification of key user requirements;
- Clause 2 identifies the document organization and structure;
- Clause 3 identifies the normative and informative references;
- Clause 4 defines key terms;
- Clause 5 provides a list of acronyms and abbreviations;
- Clause 6 defines the CFSI architecture;
- Clause 7 defines the CFSI analog interface;
- Clause 8 defines the CFSI digital interface.
- Clause 9 defines the CFSI packet data interface

3 References

The following documents contain provisions which, through reference herein, constitute provisions of this document. At the time of publication, the editions indicated were valid. All documents are subject to revision, and parties to agreements based on this document are encouraged to investigate the possibility of applying the most recent editions of the documents indicated below. ANSI and TIA maintain registers of currently valid national standards maintained by them. The appearance of “Project 25” in references below indicates the TIA document has been adopted by the APCO/NASTD/FED Project 25 Steering Committee as part of its Project 25 standard, i.e. “the Standard.”

3.1 Normative References

The following referenced documents are valid as of the date of this document. Please also refer to relevant updates or replacements to these documents offered by their publishing bodies.

- [1] RFC 791, *Internet Protocol*, IETF, September 1981.
- [2] TIA-603-D, *Land Mobile FM or PM – Communications Equipment – Measurement and Performance Standards*, TIA, August 2010.
- [3] GR-506-CORE Issue 2, *Signaling for Analog Interfaces*, Telcordia, December 2006.
- [4] ANSI/TIA-968-B, *Technical Requirements for Connection of Terminal Equipment To the Telephone Network*, Global Engineering Documents, Englewood CO, August 2009. (This is the privatization of FCC Title 47 CFR Part 68.)
- [5] TIA-102.BACA-B, *Project 25 Inter-RF Subsystem Interface Messages and Procedures for Voice Services, Mobility Management, and RFSS Capability Polling Services*, TIA, November 2011
- [6] RFC 768, *User Datagram Protocol*, IETF, August 1980.
- [7] RFC 3550, *RTP: A Transport Protocol for Real-Time Applications*, IETF, July 2003.
- [8] TIA-102.BAAA-A, *Project 25 FDMA Common Air Interface*, TIA, September 2003.
- [9] ANSI/TIA-102.AABC-D, *Project 25 Trunking Control Channel Messages*, TIA, April 2015.
- [10] TIA-102.AABF-D, *Project 25 Link Control Word Formats and Messages*, TIA, April 2015.
- [11] TIA-102.BABA-A, *Project 25 Vocoder Description*, TIA, February 2014.
- [12] ITU-T Recommendation G.711, *Pulse Code Modulation (PCM) Of Voice Frequencies*, International Telecommunication Union, November 1988.

[13] RFC 3246, *An Expedited Forwarding PHB (Per-Hop Behavior)*, IETF, March 2002.

[21] TIA-102.BAAC-C, *Project 25 Common Air Interface Reserved Values*, TIA, April 2011.

[22] TIA-102.BAAD-B *Project 25 Conventional Procedures*, TIA, August 2015.

3.2 Informative References

[14] TIA-102.AABG, *Project 25 Conventional Control Messages*, TIA, April 2009.

[15] RFC 793, *Transmission Control Protocol*, IETF, September 1981.

[16] RFC 2475, *An Architecture for Differentiated Services*, IETF, December 1998.

[17] TSB-102.BACC-B, *APCO Project 25 Inter-RF Subsystem Interface Overview*, TIA, November 2011.

[18] RFC 3551, *RTP Profile for Audio and Video Conferences with Minimal Control*, IETF, July 2003.

[19] TSB-102-C, *Project 25 TIA-102 Documentation Suite Overview*, TIA, March 2016.

[20] TIA-102.AABD-B, *Project 25 Trunking Procedures*, TIA, November 2014.

[23] TSSC-20150707_008_TIA Style Guide 2015_Final Revision *TIA Style Manual*, TIA, July, 2015

[24] TIA-102.BACE, *Project 25 Inter RF Subsystem Interface (ISSI) Messages and Procedures for Conventional Operation*, TIA, June 2008

[25] TIA-102.BAEA-C, *Project 25 Data Overview and Specification*, TIA, December 2015

4 Definitions

\$.....This prefix is used to indicate a hex value parameter.

%.....This prefix is used to indicate a binary value parameter.

Base Station.....Equipment within a fixed station subsystem that performs at least the Base Radio functions of power amplifier (PA), RF front-end, modulation and demodulation. It may perform other fixed station subsystem functions also. It is normally connected to a console or RFSS through a Fixed Station Interface.

Block PayloadThe actual data contained in the block. The block data may be voice, CAI data blocks, control words, low speed data, etc.

Block Payload Type, Block PTA number which describes the basic payload type as defined by the IETF (when E bit is set to 0) or, it may be a profile specific type (when E bit is set to 1). In this specification, E = 1 and the Block Payload Type is carried in either a Voice Conveyance or Data Conveyance message payload.

Block TypeA block identifier used to indicate what type of data is contained in the block, such as CAI voice, IMBE only, link control word, encryption sync word, CAI data, etc.

Busy Bits.....A set of two binary bits used by the Fixed Station Subsystem to indicate inbound channel status on the CAI.

CAI (Common Air Interface)A radio to radio signal path defined in terms of access method, modulation scheme, vocoding method, channel data rate, and channel data format.

CAI Data Block.....A CAI Data Block is a single formatted block of information protected by a trellis code that may be part of a related sequence of blocks (a data packet) transmitted over the Common Air Interface. Types of CAI Data Blocks include Confirmed Header, Confirmed Data block,

Confirmed Last block, Unconfirmed Header, Unconfirmed Data block, Unconfirmed Last block, or Confirmed Response.

CDCSS	Continuous Digital Controlled Squelch System which allows a receiver to block out all signals except those encoded with the selected sub-audible digital pattern.
CFSH.....	Conventional Fixed Station Host: Functional component of either an RF Subsystem (RFSS) or a Console Subsystem.
CFSS.....	See Fixed Station Subsystem.
Console.....	A device used by a dispatcher to communicate with radios in the field. These range from simple devices that resemble telephones to advanced touch screen computer based subsystems that control multiple transmitters and can patch multiple call groups together.
Conventional Fixed Station.....	See Fixed Station
CTCSS.....	Continuous Tone-Coded Squelch System which allows a receiver to block out all signals except those encoded with the selected sub-audible tone.
Fixed Data Network	A network of fixed computer equipment for wide-area data communications.
Fixed Station.....	Short-cut for 'Fixed Station Subsystem'
Fixed Station Subsystem	Set of equipment that is bounded by a Conventional Air Interface and the Fixed Station Interface. Equipment within the Fixed Station Subsystem boundary may include, but is not limited to; Conventional Base Station(s), Repeater(s), Voter(s), Distribution Function(s), and other RF fixed equipment. The RF equipment contained in a Fixed Station Subsystem may also transmit or receive conventional FM.
Fixed Station Host.....	The control equipment which is situated at the opposite end of the Fixed Station Interface from the Fixed Station. Examples of Fixed Station Hosts include Consoles and RFSSs.

IMBE	Improved Multi-Band Excitation. A low data rate Voice codec developed by Digital Voice Systems, Inc. (DVSI)
Mobile.....	A Subscriber Unit intended for vehicular operation.
Portable.....	A Subscriber Unit intended for hand-held operation.
Repeater	Equipment within a fixed station subsystem that performs at least the radio functions of power amplifier (PA), RF front-end, modulation and demodulation. It may include a limited number of functions such as NAC detection, CTCSS/DCS detection to enable repeater functions etc. It need not include a connection to a console or RFSS through a Fixed Station Interface.
RTP Header	The part of the RTP header which includes the following information: version (V), padding (P), extension (X), CSRC count (CC) {when inserted by a mixer}, marker (M), payload type (PT), sequence number, timestamp, SSRC, and CSRC.
Socket.....	A term used to represent the combination of an IP Address and UDP Port Number.
SSRC.....	Synchronization SouRCe: The SSRC identifier used by the fixed station on all RTP transmissions sent to the CFSH.
Status Symbols.....	See Busy Bits.
Tone Remote Control	A legacy method of controlling fixed station functions using in-band tones over analog bearers.
Wildcard	TRC functions that can be used for generic purposes via the terminating fixed station. Equipment that support Wildcard functions can use these to control additional fixed station functions or to control equipment external to the fixed station – for example start and stop of power generators.

5 Abbreviations and Acronyms

ACK	Acknowledgement
AFSI.....	Analog Fixed Station Interface
ANSI	American National Standards Institute
CAI.....	Common Air Interface
CAI BSA.....	Common Air Interface Bundle Sent Acknowledgement
CAI BSR.....	Common Air Interface Bundle Sent Request
CDCSS	Continuous Digital Controlled Squelch System
CFM	Conventional FM
CFSH	Conventional Fixed Station Host
CFSI.....	Conventional Fixed Station Interface
COR.....	Carrier On Relay
CS.....	Connected State
CTCSS.....	Continuous Tone Controlled Squelch System
DARPA	Defense Advanced Research Projects Agency
DC.....	Data Conveyance
DFSI.....	Digital Fixed Station Interface
DSCP	Differentiated Services Code Point
DUID	Data Unit IDentification
DVSI	Digital Voice Systems Inc.
E&M	Ear & Mouth SignalingEIA Electronics Industry Association
FDMA.....	Frequency Division Multiple Access
FEC.....	Forward Error Correction
FM.....	Frequency Modulation
FS	Fixed Station (shortcut for Fixed Station Subsystem)
FSI	Fixed Station Interface
FSI TA.....	Fixed Station Interface Transport Acknowledgement
FSI TR	Fixed Station Interface Transport Request

FSS	Fixed Station Subsystem
FSSI	Fixed Station Subsystem Interface
FT	Frame Type
HLGT	High-level Guard Tone
IANA	Internet Assigned Numbers Authority
ICMP	Internet Control Message Protocol
ID	IDentification
IETF	Internet Engineering Task Force
IMBE™	Improved Multi-band Excitation
IP	Internet Protocol
ISSI	Inter-RF Subsystem Interface
LDU1/2	Logical Link Data Unit 1/2
LLGT	Low-level Guard Tone
LSB	Least Significant Byte
mA	milliamp
MBC	Multi Block Control
MFID	ManuFacturer IDentification
MSB	Most Significant Byte
msec	millisecond
N/A	Not Applicable
NAC	Network Access Code
NAK	Negative Acknowledgement
NID	Network Identifier
OSI	Open Systems Interconnection
P25	Project 25
PA	Power Amplifier
PID	Protocol IDentifier
PT	Payload Type
PTT	Push-to-talk
RF	Radio Frequency
RFC	Request for Comments
RFSS	RF Subsystem
RTCP	Real-time Transport Control Protocol

RTP.....	Real-time Transport Protocol
SBC	Single Block Control
SF	Super Frame
sec	second
SSRC.....	Synchronization SouRCe
TCP.....	Transport Control Protocol
TIA	Telecommunications Industry Association
TRC	Tone Remote Control
TSB.....	Telecommunications Systems Bulletin
UDP	User Datagram Protocol
VC.....	Voice Conveyance

NOTE: Abbreviations for specific fields of the RTP, UDP and IP protocols can be found in the relevant references.

6 CFSI Architecture

This section describes the architecture of the Conventional Fixed Station Interface (CFSI) and is primarily informative in nature. Normative statements contained herein, however, are intended to be a part of this specification.

6.1 Core CFSI Functions

6.1.1 Conventional Analog Fixed Stations

Conventional analog fixed stations provide mobile-to-mobile and mobile-to-console communications compliant with TIA/EIA 603 ([2]). Conventional stations may “repeat” audio from their inbound channel to their outbound channel, or may only forward audio received from the fixed equipment.

Conventional analog fixed stations may provide CTCSS or CDCSS squelch capabilities, or may be set to pass all received audio. Consoles are typically able to control the squelch characteristics, transmit and receive frequencies, and repeat mode of conventional stations.

Conventional analog fixed stations may use either the Analog or Digital Fixed Station Interface.

6.1.2 Conventional Digital Fixed Stations

Conventional digital fixed stations provide similar capabilities to those provided by analog fixed stations, but they use the P25 CAI rather than a TIA/EIA 603 air interface. Instead of CTCSS and CDCSS squelch, the Conventional Digital Fixed Stations provide squelch mechanisms based on Network Access Code (NAC) and talk-group. Digital fixed stations may also operate clear or encrypted.

NOTE: The digital fixed station interface does not control secure mode and only exerts limited control over the squelch mode.

Conventional digital fixed stations may use either the Analog or Digital Fixed Station Interface.

6.1.3 Mixed Mode Fixed Stations

Certain fixed stations may provide both analog and digital modes. These stations also typically support console controls to switch between modes.

Mixed mode fixed stations may use either the Analog or Digital Fixed Station Interface.

6.2 CFSI Architecture and Overview

Figure 1 illustrates the architecture and context of the conventional FSI. Within the subsystem “cloud”, the interconnection of the elements is manufacturer specific and not described in this specification.

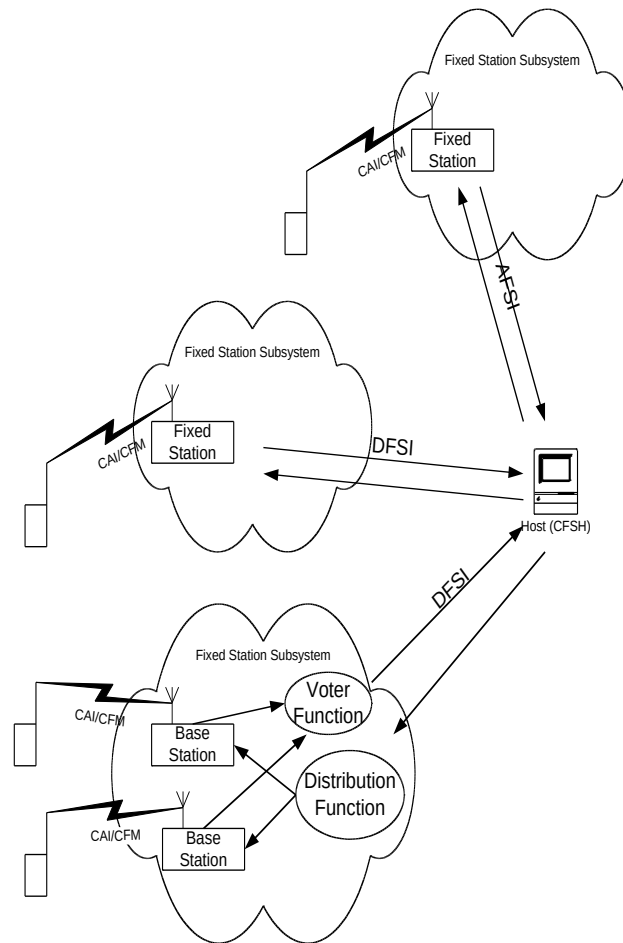


Figure 1 – CFSI Architecture

Outside of the cloud, however, each fixed station is connected via either an Analog Fixed Station Interface (AFSI) or a Digital Fixed Station Interface (DFSI) to one host, which may be either a Console Subsystem (of which a console is a functional element) or an RFSS. Any host may support a multiplicity of fixed station interfaces of any variety. The analog FSI supports only the simple model of a conventional FM (CFM) station. In contrast, the Digital FSI supports more complex models appropriate to the P25 CAI.

NOTE: Unless otherwise indicated, for the purposes of this document “host” or “Host” refers to a Conventional Fixed Station Host (CFSH) as defined in this document.

Each fixed station supports one inbound and one outbound audio stream at any particular time. The composition of the stream is determined by the source. This architecture should not be construed as prohibiting multi-drop consoles, particularly in the analog case. Additional operator positions are transparent to the interface as they reside behind the abstraction presented by the CFSH.

6.2.1 Analog Fixed Station Interface

The analog fixed station interface models an analog station as a single fixed station connected via a 2 or 4 wire audio interface with 1 or 2 additional optional control signals to a single host. Control may be provided using the optional E&M (COR/PTT) signalling on the control signals, or through in-band tones. Figure 2 illustrates the conventional Analog Fixed Station Interface (AFSI).

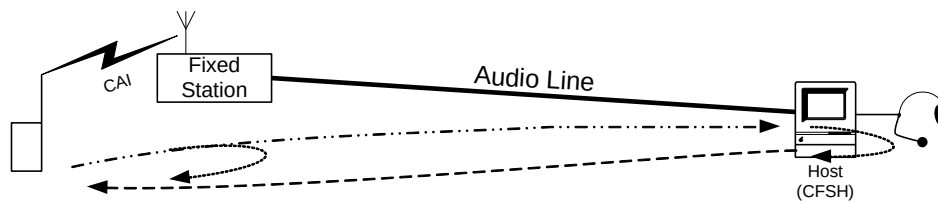


Figure 2 – Conventional Analog Fixed Station Interface

Audio from the Host is sent to the fixed station where it is transmitted to subscriber units. Likewise, audio received by the fixed station from subscribers is transmitted to the host. Two repeat modes may be supported:

1. The fixed station may be configured to retransmit on the outbound air interface the voice stream it is receiving via the inbound air interface. This configuration would be overridden by a transmission from the CFSH.
2. If the connection is full-duplex, the CFSH may send the audio received from the fixed station back to the fixed station for retransmission so that other subscriber units can hear.

6.2.2 Digital Fixed Station Interface – Protocol Version 1

The Digital Fixed Station Interface employs Internet protocols to provide connectivity between a conventional fixed station host (CFSH) and a fixed station. The Protocol Version 1 DFSI has two parts, a Control Service and a Voice Conveyance Service. The control service is modeled as a point-to-point connection between the host and a fixed station control service. The Voice Conveyance service is modeled as a point to multi-point connection between a host and one or more end-points sharing a single destination IP address and port, but with potentially different source IP

addresses and ports. Thus a CFSH sends RTP based voice packets to a single address, but may receive them from different addresses. This arrangement is illustrated in Figure 3.

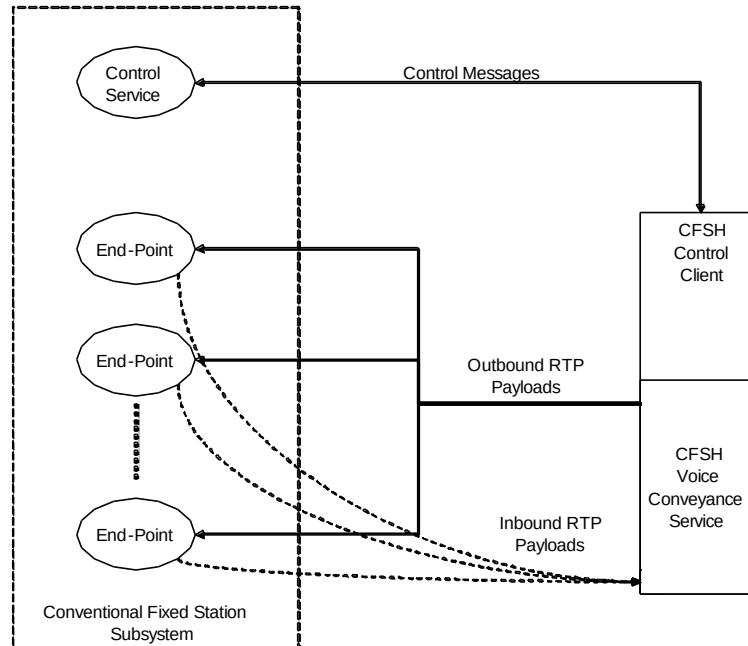


Figure 3 – Conventional Digital Fixed Station Interface – Protocol Version 1

6.2.2.1 Control Service – Protocol Version 1

The Protocol Version 1 control service is concerned with providing control functions analogous to those provided by the analog fixed station interface, and with establishing and maintaining connectivity with the fixed station. The fixed station service is provided at a well-known UDP port and the host port is variable. The well-known fixed station control service port has a default value specified in 8.2, but may be programmable on a station-by-station basis.

The control service is also responsible for:

1. Control of the fixed station by its host including squelch disable.
2. Transport of emergency alert and other conventional control messages from the fixed station to its host.

6.2.2.2 Voice Conveyance Service – Protocol Version 1

The voice conveyance service is concerned with the passing of RTP payloads between the host and the fixed station. Depending on the air interface supported by the fixed station, the RTP payload may carry either Common Air Interface voice

(encoded with IMBE and potentially encrypted) or waveform coded voice. Both the CFSH and the fixed station learn the UDP/RTP ports to be used for the Voice Conveyance Service upon connectivity establishment by the control service. As with the Analog Fixed Station Interface, two repeat modes may be supported.

1. The fixed station may be configured to retransmit on the outbound air interface the voice stream it is receiving via the inbound air interface. This configuration would be overridden by a transmission from the CFSH.
2. The CFSH may send the audio received from the fixed station back to the fixed station for retransmission so that other subscriber units can hear.

The voice conveyance service is responsible for:

1. Transport of full-duplex clear audio between the fixed station and its host.
2. Conveyance of PTT signaling between the fixed station and its host.
3. Providing intercom capabilities between the fixed station and its host.
4. Transport of encrypted audio between the fixed station and its host.
5. Transport of Caller-ID information between the fixed station and its host.
6. Transport of Talk-group information between the fixed station and its host.
7. Transport of NAC code information between the fixed station and its host.
8. Transport of Emergency Call indicators between the fixed station and its host.
9. Transport of voter identification from the fixed station to its host.

6.2.3 Digital Fixed Station Interface – Protocol Version 2

This revision adds support for Packet Data which requires Protocol Version 2 of the DFSI. To add Packet Data to the DFSI it was necessary to expand the interface in support of separate Control Service, Voice Conveyance, and Data Conveyance connections. The separation of the three connections is only applicable to FSI Protocol Version 2 and above. The Control Service is modeled as a point-to-point connection between the host and a fixed station control service. The Voice Conveyance service is modeled as a point to multi-point connection between a host and one or more end-points sharing a single destination IP address and port, but with potentially different source IP addresses and ports. Thus a CFSH sends RTP based voice packets to a single address, but may receive them from different addresses. The Data Conveyance service is modeled as a point to multi-point connection between a CFSH, and one or more end-points within a CFSS sharing a single destination IP address and port, similar to the Voice Conveyance service. Point-to-multipoint is supported in order to allow functionality such as a simulcast architecture where there is no distribution function within the FSS. Point-to-multipoint also enables a group data function where the same message would be transmitted by multiple transmitters.

The Protocol Version 2 Control connection protocol continues to utilize UDP/IP, the Voice Conveyance connection utilizes RTP/UDP/IP and the Data Conveyance connection utilizes UDP/IP.

The Protocol Version 2 DFSI architecture is shown in Figure 4. Protocol Version 2 DFSI equipment supports the Control Service and either the Voice Conveyance Service, or the Data Conveyance Service, or both conveyance services.

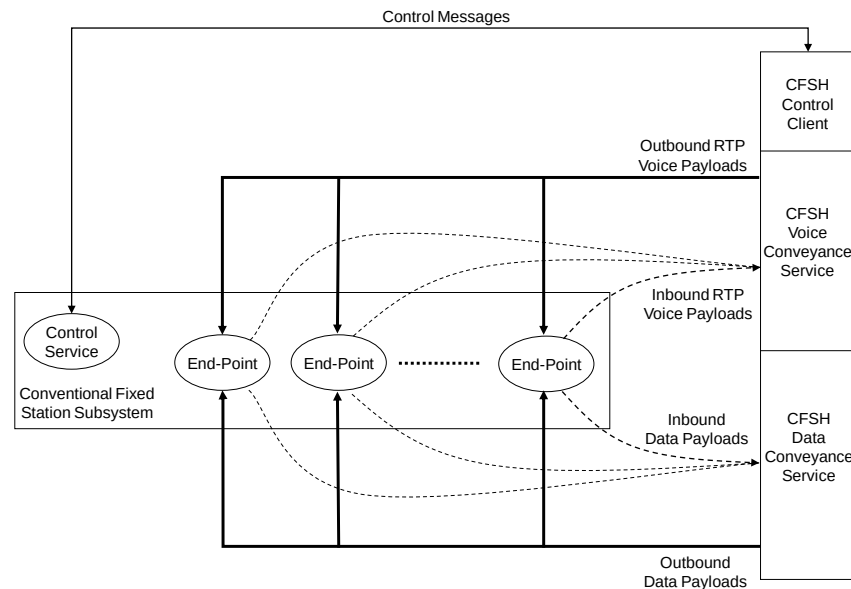


Figure 4 – Conventional Digital Fixed Station Interface – Protocol Version 2

6.2.3.1 Control Service – Protocol Version 2

In addition to the Protocol Version 1 functionality described in clause 6.2.2.1, the Protocol Version 2 Control Service also controls the setup and teardown of both the Voice and Data Conveyance connections.

6.2.3.2 Voice Conveyance Service – Protocol Version 2

The Protocol Version 2 Voice Conveyance Service provides the same functionality as the Protocol Version 1 Voice Conveyance service as described in clause 6.2.2.2.

6.2.3.3 Data Conveyance Service – Protocol Version 2

The Data Conveyance Service is only available with Protocol Version 2 and provides transport of packet data payloads between the Host and fixed station. See clause 9.3.2.1.2 regarding the Data Conveyance Service and repeat.

7 CFSI Analog Interface

The Fixed Station (FS) and Host may support the use of legacy analog circuits to control the fixed station.

7.1 E & M

The intent of this requirement is to be compatible with E & M interfaces used in legacy analog fixed stations.

The circuit shall be comprised of these signal pairs:

- A “Transmit” pair to convey analog audio to the fixed station transmitter.
- A “Receive” pair to convey analog audio from the fixed station receiver.
- An “E” pair to control the RF transmission of audio on the Transmit pair via the fixed station.
- An “M” pair to indicate the presence of signal being received by the fixed station. The host may use the “M” pair as an indication a signal is being received by the fixed station.

In the case where the FS is a Repeater, the transmit and receive pairs shall be a full duplex 4-wire circuit.

The FS shall accept an audio signal range of –30 dBm to +10 dBm on the transmit audio pair. Over this range the transmitter shall be capable of providing full perceived audio modulation in either analog or digital RF modes. When receiving full perceived audio modulation the FS shall be capable of generating to the receive audio pair an audio signal range that encompasses –30 dBm to the maximum limit permitted by leased circuit compliance (e.g. FCC Part 68 [4]).

NOTE: Some legacy analog equipment use signal levels on the audio pairs as high as +10 dBm to compensate for long line losses. However, +10 dBm levels may conflict with present compliance requirements.

The E & M pairs may support E & M telephony requirements as outlined in Section 9 of Telcordia GR-506 ([3]), However, the following configuration shall be available to the installer ensuring support of legacy equipment. The E-pair shall have a configuration that expects the Host end to provide a dry contact closure while RF transmission of the audio on the Transmit pair is desired. The M-pair shall have a configuration that provides a dry contact closure while a compatible RF carrier is being received. The E & M pairs shall work over a range of 5 to 150 mA current with 50 ohm maximum contact resistance while the contacts are closed, and 10 to 60 Vdc with 5 megohm minimum resistance while the contacts are open. The E-pair shall limit its current to this range.

The electrical characteristics of the signal pairs shall comply with the applicable requirements of Section 4 of ANSI/TIA-968-B ([4]). ANSI/TIA-968-B is the basis for US Government 47 CFR Part 68 (FCC Part 68).

The Transmit and Receive pairs may be capable of being combined on a single pair which may be simplex, rather than full duplex.

Figure 5 shows the possible E & M interface configurations. In configurations which use a combined Transmit and Receive pair in a simplex mode, Transmit (FS E-pair active) shall have priority over Receive and shall silence Receive pair audio and shall make the FS M-pair inactive for the duration of the FS E-pair activation.

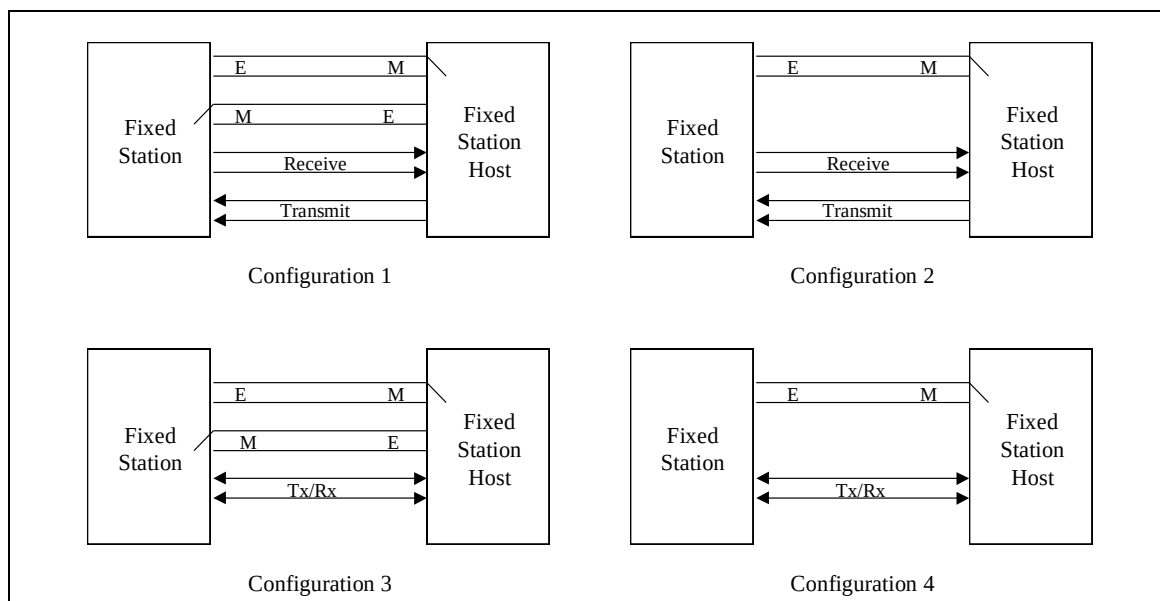


Figure 5 – Possible E&M Interface Configurations

7.2 Tone Remote Control

The intent of this requirement is to be compatible with Tone Remote Control (TRC) interfaces used in legacy, pre-P25 analog fixed stations.

If the FS is equipped with a TRC interface, the circuit shall be comprised of these signal pairs:

- A “Transmit” pair to convey analog audio to the fixed station transmitter, and to convey in-band control tones to the fixed station.
- A “Receive” pair to convey analog audio from the fixed station receiver.

In the case where the FS is a Repeater, the transmit and receive pairs shall be a full duplex 4-wire circuit.

The tone decoder associated with the transmit pair shall have the following characteristics:

- Frequencies within +/- 10 Hz of those specified in this section shall be accepted, and frequencies outside +/- 20 Hz shall be rejected.
- Durations within -10% (no upper limit) of those specified in this section shall be accepted, and durations less than - 50 % (no upper limit) shall be rejected.

The electrical characteristics of the signal pairs shall comply with the applicable requirements of Section 4 of ANSI/TIA-968-B ([4]). ANSI/TIA-968-B is the basis for US Government 47 CFR Part 68 (FCC Part 68).

7.2.1 Simplex Wireline Interface for Configuration 6

The Transmit and Receive pairs may be capable of being combined on a single pair which may be simplex, rather than full duplex.

Figure 6 shows the possible TRC interface configurations. In configurations which use a combined Transmit and Receive pair in a simplex mode, the fixed station shall be able to decode an incoming TRC sequence even while Receive audio is present. If the TRC sequence is a transmit sequence the fixed station shall mute any Receive audio on the audio pair for the duration of the transmission.

NOTE: When using a 2-wire, simplex interface it is not possible for a console to take-over repeat audio.

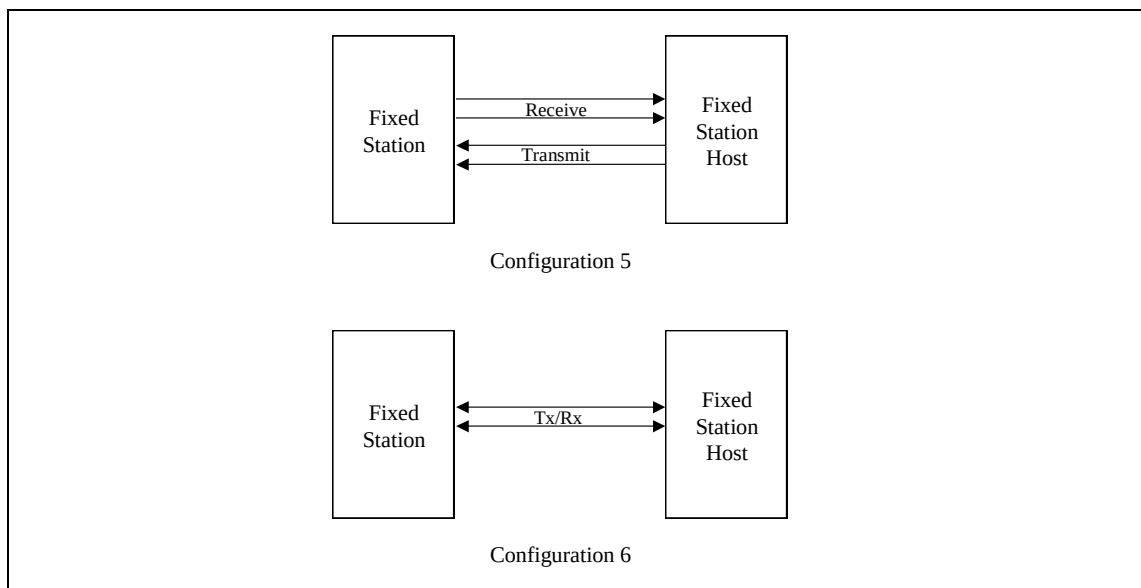


Figure 6 – Possible TRC Interface Configurations

In addition to audio, the Transmit pair shall contain in-band TRC tones described in the sections that follow.

7.2.2 Non-Transmit Sequence

Figure 7 shows a TRC sequence that is used to control the fixed station without causing speech transmission over the RF channel. The sequence begins with a burst of High Level Guard Tone (HLGT) followed by a burst of Function Tone.

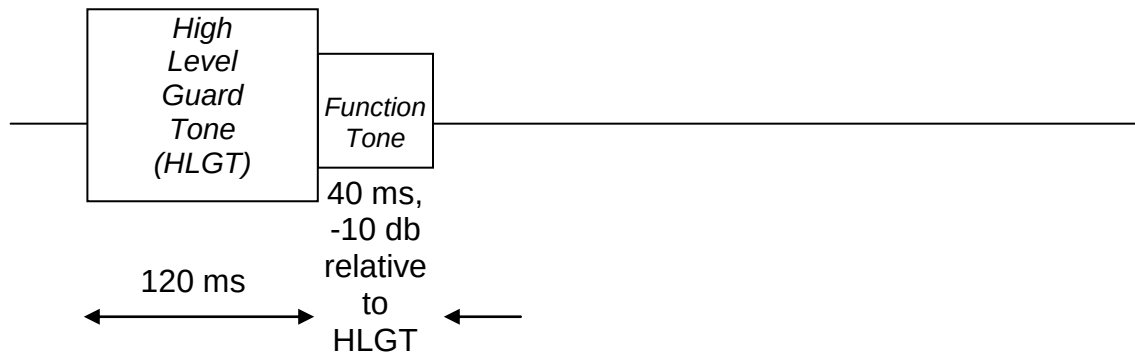


Figure 7 – Non-Transmit TRC Sequence

7.2.3 Transmit Sequence

Figure 8 shows a TRC sequence that is used to control a fixed station RF transmission. The sequence begins with a burst of High Level Guard Tone (HLGT) followed by a burst of Function Tone, followed by Low Level Guard Tone (LLGT) summed with transmit speech and is present for the duration of the speech transmission.

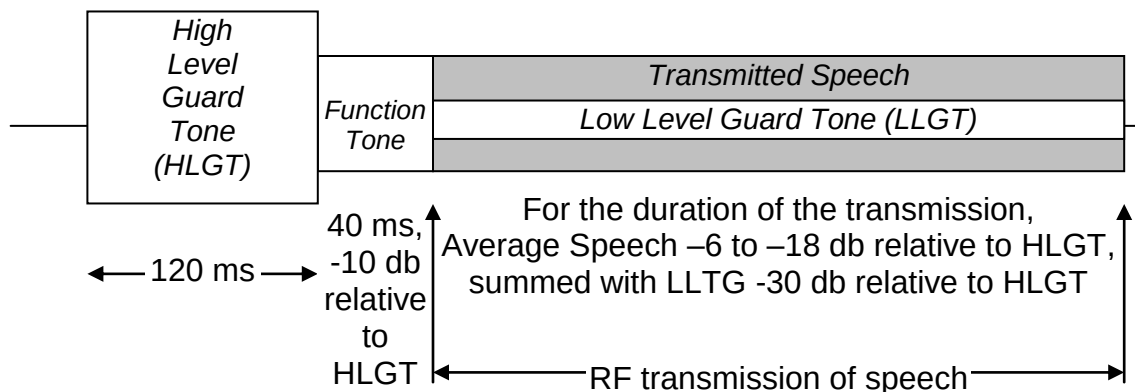


Figure 8 – Transmit TRC Sequence

7.2.4 Guard Tones

The duration of the High Level Guard Tone (HLGT) shall be nominally 120 milliseconds. The Low Level Guard Tone (LLGT) shall be nominally 30 db below HLGT and shall be summed with the speech intended for transmission. The LLGT should be filtered out prior to audio modulation of the RF transmission. The duration of the LLGT shall be for the duration of the speech transmission. The tone frequency used for the HLGT and LLGT shall be the same, and shall be one of the following field selectable values: 2100 Hz, 2175 Hz or 2325 Hz, with 2175 Hz being the

default. The Transmit pair of the FS shall be capable of accepting a HLGT range of – 30 dBm to +10 dBm, and shall reject a range less than -36 dBm.

NOTE: If the FS has an AFSI transmit input level adjustment, the level accept/reject range shall be considered compliant if lower limits can be achieved at higher input level adjustment sensitivity, and higher limits can be achieved at lower input level adjustment sensitivity.

NOTE: The Host equipment generating the Guard Tone should maintain a tolerance of +/- 5 Hz. Some legacy analog host equipment use signal levels on the audio pairs as high as +10 dBm to compensate for long line losses. However, +10 dBm levels may conflict with present compliance requirements (e.g. FCC Part 68 – [4]).

7.2.5 Function Tone

The duration of the Function Tone shall be nominally 40 milliseconds and shall be nominally 10 db below HLGT. The possible Function Tones shall be among the 15 tones listed Table 1. Table 2 outlines the functions that shall be supported by the FS and the recommended Functions Tone to be used for each of those functions. Table 1 shows six recommended mappings of Function Tones, one or more of which are generally compatible with legacy Host equipment. Manufacturers may support additional Function Tones and may support additional features controlled by any of the Function Tones. However, when the fixed station is configured to support features listed in Table 1, it shall be possible to use the Function Tones designated in the Tables 1 and 2 to control those features. The Transmit pair of the FS shall accept and reject Function Tone levels that are 10 dB less than the HLTG acceptance and rejection range.

Table 1 – TRC Function Tones vs. Possible FS Functions

Tone (Hz)	Simple Keying Command	Keying ^a	Function ^a	Function ^b	Wild Card	Other
None			Intercom			
2050			Monitor without receive code ^{b,e}	Monitor without receive code ^{b,e}		
1950	Xmit on selected channel ^a , mode ^{c,d} and/or code ^{b,d}	Xmit on chan 1 ^d	Select chan 1			
1850	Xmit on channel ^a 1 without code ^{b,d}	Xmit on chan 2 ^d	Select chan 2			
1750		Xmit on chan 7 ^d	Select chan 7	Select code E	Second receiver Off	Receiver maximum squelch
1650		Xmit on chan 8 ^d	Select chan 8	Select code F	Second receiver On	Receiver minimum squelch
1550				Select code G	Repeat Off	Scan Disable ^f
1450				Select code H	Repeat On	Scan Enable ^f
1350		Xmit on chan 3 ^d	Select chan 3	Select code A	Wildcard 1 On	Select Digital RF Mode ^c
1250		Xmit on chan 4 ^d	Select chan 4	Select code B	Wildcard 1 Off	Select Analog RF Mode ^c
1150		Xmit on chan 5 ^d	Select chan 5	Select code C	Wildcard 2 On	Select Coded Speech
1050		Xmit on chan 6 ^d	Select chan 6	Select code D	Wildcard 2 Off	Select Clear Speech
950						Special
850						Special
750						Special
650						Special

Table Notes:

a – “Channels” may be either conventional frequencies, or talk groups.

b – “Codes” may be either analog CTCSS/DCS (subaudible squelch), or digital NAC.

c – Although the interface may be analog, the RF mode may be either analog or digital. This selection is controlling the Transmitter mode. It is up to the implementers to decide how receive mode is controlled (e.g. autosensing).

d – Individual channel parameters may include default code, secure mode (coded/clear) and RF mode (analog/digital) settings. Unless the channel parameters include default codes and modes, the transmission shall use the previously selected code, secure mode and RF mode.

e – Monitor (listen to all channel traffic without regard to code) is canceled by any Transmit or Channel Selection TRC sequence.

f - Scan Enable/Disable may be moved to Wildcard I (On/Off) (i.e. 1350/1250) or Wildcard II (On/Off) (i.e. 1150/1050) if Scan and Repeat Enable functions are required for a given configuration.

Table 2 – Possible FS Functions vs. TRC Function Tones

Function	TRC Sequence Type	Function Tone (Hz)
Intercom	None	None
Monitor without receive code ^{b,e}	Non-Transmit	2050
Transmit on selected channel ^a , mode ^{c,d} and/or code ^{b,d}	Transmit	1950
Transmit on channel 1 ^d	Transmit	1950
Transmit on channel ^a 1 without code ^{b,d}	Transmit	1850
Transmit on channel 2 ^d	Transmit	1850
Transmit on channel 3 ^d	Transmit	1350
Transmit on channel 4 ^d	Transmit	1250
Transmit on channel 5 ^d	Transmit	1150
Transmit on channel 6 ^d	Transmit	1050
Transmit on channel 7 ^d	Transmit	1750
Transmit on channel 8 ^d	Transmit	1650
Scan Enable ^f	Non-Transmit	1450
Scan Disable ^f	Non-Transmit	1550
Select channel 1	Non-Transmit	1950
Select channel 2	Non-Transmit	1850
Select channel 3	Non-Transmit	1350
Select channel 4	Non-Transmit	1250
Select channel 5	Non-Transmit	1150
Select channel 6	Non-Transmit	1050
Select channel 7	Non-Transmit	1750
Select channel 8	Non-Transmit	1650
Select code A	Non-Transmit	1350
Select code B	Non-Transmit	1250
Select code C	Non-Transmit	1150

Function	TRC Sequence Type	Function Tone (Hz)
Select code D	Non-Transmit	1050
Select code E	Non-Transmit	1750
Select code F	Non-Transmit	1650
Select code G	Non-Transmit	1550
Select code H	Non-Transmit	1450
Select Digital RF Mode ^c	Non-Transmit	1350
Select Analog RF Mode ^c	Non-Transmit	1250
Select Coded Speech	Non-Transmit	1150
Select Clear Speech	Non-Transmit	1050
Repeat On	Non-Transmit	1450
Repeat Off	Non-Transmit	1550
Second receiver On	Non-Transmit	1650
Second receiver Off	Non-Transmit	1750
Receiver maximum squelch	Non-Transmit	1750
Receiver minimum squelch	Non-Transmit	1650
Wildcard 1 On	Non-Transmit	1350
Wildcard 1 Off	Non-Transmit	1250
Wildcard 2 On	Non-Transmit	1150
Wildcard 2 Off	Non-Transmit	1050
Special	Transmit or Non-Transmit	950
Special	Transmit or Non-Transmit	850
Special	Transmit or Non-Transmit	750
Special	Transmit or Non-Transmit	650

7.2.6 Speech

Speech intended for RF transmission by the fixed station shall be summed with LLGT, and shall have an average level that is –6 to –18 db below HLG. The Transmit pair of the FS shall accept and reject LLGT levels that are 30 dB less than the HLG acceptance and rejection range.

NOTE: The Host equipment generating the tones should also filter the Guard Tone frequency from the speech prior to summing the speech with the LLGT otherwise phase coincidence may cause apparent gaps in the presence of the LLGT resulting in unintentionally dropping the RF transmission. It may be desirable to be able to adjust the speech level relative to the tone levels within the fixed station to ensure greatest compatibility with legacy Host equipment.

7.2.7 Sequence Off Time

Following either type of TRC sequence there shall be a minimum of 300 ms before another TRC sequence is sent.

7.3 Intercom

Speech may be sent between the Host and the fixed station without any accompanying TRC sequence or E & M-pair keying. This may be used for intercom purposes only, rather than to transmit RF.

8 CFSI Digital Interface

Note that throughout this description, the conventions of Appendix B of RFC 791 ([1]) are used, that is, for multi-byte fields, information is transmitted most significant byte first and bit 0 represents the most significant bit of each byte

8.1 Internet Protocol Overview

The Internet architecture layers (see Figure 9 for a mapping to the OSI model) are used to construct the Digital fixed station Interface (DFSI) protocol stack (illustrated in Figure 10). The non-proprietary protocols that make up the Internet protocol suite have become widely implemented because of the success of the Internet, and are supported by the widest variety of network equipment vendors. This allows the networks to connect DFSI equipment to be built with common off the shelf equipment, either by a user organization or a specialist network provider. Because the Internet architecture layers were developed much earlier than the Open Systems Interconnection (OSI) model they can only be roughly mapped to it (see Figure 9).

Internet Architecture Layers	OSI Layers
Application	Application
	Presentation
	Session
Transport (i.e., Host-to-Host)	Transport
Network (i.e., Internet)	Network
Lower Layers (i.e., Data Link and Physical)	Data Link
	Physical

Figure 9 – Internet Architecture Layers and the OSI Layers

Internet Architecture Layers**DFSI Protocol Suite**

Application	Control Protocol	Voice Conveyance Protocol		Data Conveyance Protocol
Transport		RTP	RTCP	
Network	UDP			
Link	IP (Version 4)			
Physical	Ethernet or other protocols specified in this standard			

Figure 10 – DFSI Internet Protocol Suite including Data Conveyance

The following sections describe each of the layers of the Digital FSI (DFSI) Internet Protocol Suite, through the Transport Layer.

8.1.1 DFSI Lower Layers

These layers correspond to the physical and data link layers of the OSI layers. The Internet protocols, designed to be hardware independent, do not cover these lowest two OSI layers. As a result, the protocols of the Internet suite are widely used to connect heterogeneous subsystems.

The following specific lower layer requirements apply to the Digital fixed station Interface:

- Equipment conforming to DFSI should offer the option of Ethernet 100 Base-T with a RJ-45 connector as the physical and data link layers.
- Equipment may offer any physical and link layer protocols that support the internet protocol, in addition to Ethernet 100 Base-T.

The network carrying the DFSI may change data link and physical protocols.

8.1.2 Network Layer

This layer corresponds to the OSI network layer, supporting routing of the packets through the network from the source host to the destination host or hosts. At this

layer of the Internet, the main protocol is the Internet Protocol (IP). In addition, there are several control protocols.

IP is a connectionless packet protocol. Packets are groups of information transmitted as a unit to and from the upper layer protocols on sending and receiving hosts, and delivered from the sending to the receiving host over interconnected subsystems. Each IP packet includes an IP header that includes source and destination IP addresses.

There are two styles of IP address, unicast and multicast. Unicast addresses allow packets to be routed to a single, particular device. Multicast addresses allow packets to be routed to an arbitrary collection of devices. IP multicast offers a convenient mechanism to distribute data or voice to multiple hosts which suppliers of fixed station subsystems may use to implement the distribution function shown in Figure 1. Equipment conforming to the DFSI shall support unicast and may support multicast IP addresses. Address selection is at the discretion of infrastructure designers. Note that regardless of address selection, this interface does not support multiple hosts controlling a single station.

8.1.3 Transport Layer

This layer corresponds roughly to the OSI transport layer. Protocols at this layer communicate with peer processes in other hosts or networked devices. Two key host-to-host protocols at this layer are the User Datagram Protocol (UDP) ([6]) and Transmission Control Protocol (TCP) ([15]).

TCP is not well-suited to real-time applications, and it cannot be used with IP multicast, whereas UDP is a simple connectionless datagram protocol. Therefore, TCP is not used in the DFSI.

8.1.3.1 RTP

The main feature that UDP provides is multiplexing of application protocols using the mechanism of port numbers which, along with an integrity checksum, are the only contents of the UDP header.

UDP does not provide "reliable" transport, there are no mechanisms to recover from lost packets or prevent packets from being delivered out of order. To regain some of the transport layer facilities that are not provided in UDP without incurring the penalties of TCP, the Real-time Transport Protocol (RTP) ([7]) is often used. RTP provides detection of lost packets and out-of-order packets.

8.1.3.2 Requirement

Equipment conforming to the DFSI shall transport the Control service over UDP only. Equipment conforming to the DFSI shall transport the Voice conveyance

Service over RTP on UDP. Equipment conforming to the DFSI shall transport the Data Conveyance service over UDP only.

8.1.3.3 RTCP Considerations

The Real-time Transport Control Protocol (RTCP) is a companion to RTP. Equipment may send RTCP reports as described in RFC 3550 ([7]), but shall not depend on other equipment sending such reports.

8.1.4 Transport and Network Layer Parameters

The following sub-clauses specify the values of transport and network layer parameters for the DFSI.

8.1.4.1 IP Address

In accordance with 8.2, the IP addresses of the fixed station are configured in the host and the IP address of the host is learned by the fixed station. Specific addresses for both unicast and multicast operation are at the discretion of infrastructure designers.

8.1.4.2 DSCP Value

So that Differentiated Service enabled network equipment as described in RFC 2475 ([16]) can give this traffic appropriate handling, equipment using this interface should set the Differentiated Services Code Point (DSCP) field of the IP headers as in Table 3.

Table 3 – DSCP Values

Voice Traffic	101110	Expedited forwarding (EF).
Control Traffic	100010	Assured forwarding class 4 Low drop

More details of expedited forwarding can be found in RFC 3246 ([13]).

8.1.4.3 UDP Port Number

In accordance with 8.2, the FS Control UDP Port is configured in the host, and each end learns the other ports through the connection establishment procedure. Specified port assignments are at the discretion of subsystem designers.

8.1.4.4 RTP Padding Flag P

Implementers should set this to the appropriate value according to RFC 3550 ([7]).

8.1.4.5 RTP Header Extension Flag X

Payload independent header extension is not used by the DFSI so X = 0.

8.1.4.6 RTP Payload Type

This interface shall use the dynamic payload type; type 100 (decimal).

8.1.4.7 RTP Marker Bit

The marker bit of the RTP header shall be set to 0.

8.1.4.8 RTP Time Stamp

In accordance with typical sampling rates used by voice codecs, 8 kHz shall be used as the clock rate for the timestamp.

8.1.4.9 CSRC

Radio networks rather than combining all signals in a conference bridge typically use techniques such as voting so that only one input voice stream contributes to the transmitted voice stream. There is therefore no need for a list of contributing synchronization sources. Therefore there shall be no Contributing Source (CSRC) list in DFSI RTP packets, and therefore the RTP header parameter CC = 0.

8.1.4.10 SSRC

Synchronization Sources (SSRCs) are selected by the host and assigned to the fixed stations through the connection protocol as described in 8.3. The SSRC shall be selected by the host to be unique in the context of the specific connection at any particular time.

8.2 DFSI Configuration Items

The following shall be preconfigured as indicated in Table 4. Not shown are the cases where the host or fixed station is configured with its own information and the other entity learns of that information via the FSC_Connect or the FSC_ACK message.

Table 4 – Configuration Items for the DFSI

Configuration Item	Entity ³	Purpose	Default Value
Fixed Station Control IP Address	B	IP Address at which the fixed station subsystem supplies the Control Service.	N/A
Fixed Station Voice IP Address	B for PV 1, S for PV 2	IP Address at which the fixed station subsystem receives. outbound voice conveyance service traffic. It may be the same as or differ from the Fixed Station Control or Data Conveyance IP Address. Provided to Host in Connection Establishment Response for Protocol Version 2.	None
Fixed Station Control Port	B	UDP Port at which the fixed station supplies the Control Service.	50006
Fixed Station Voice Conveyance Port	S	UDP Port at which RTP traffic is conveyed to the fixed station. Provisioned in Connection Establishment Response. In accordance with RFC 3550 [7], the port number shall be even.	None
Fixed Station RTCP Voice Conveyance port	S	UDP port at which RTCP traffic may be conveyed. Note that RTCP traffic is not required in the DFSI. In accordance with RFC 3550 [7], the port number shall be the next higher odd number from the “Fixed Station Voice Conveyance Port”.	None
Host IP Address	H	The IP Address of the Host.	None
Control Retry Timer	B	Retry period for control messages.	500 msec
Control Attempt Limit	B	Maximum number of attempts for control messages.	3
Connectivity Tx Timer	H	Periodicity of control heartbeats. This value is provisioned to the	30 sec

³ H = Host, S = Fixed Station, B = Both

Configuration Item	Entity ³	Purpose	Default Value
		fixed station in the “Establish Connection” message.	
FS Heartbeat Period (Version 1 only)	H	Periodicity of Fixed Station Heartbeats. This value is provisioned to the fixed station in the “Establish Connection” message.	30 sec
Host Heartbeat Period (Version 1 only)	H	Periodicity of Host Heartbeats. This value is provisioned to the fixed station in the “Establish Connection” message.	30 sec
Connectivity Loss Limit	B	Maximum number of heartbeat losses to allow.	2
Voter Reporting Period	S	Periodicity of voter reports.	10 sec
Transmitter Report Interval	S	Periodicity of transmitter reports. See 8.4.5.3	30 sec

8.3 DFSI Control Service

This document is designed to facilitate support of multiple versions of the FSI. All messages and procedures defined in this document apply to all FSI Protocol versions unless specifically stated otherwise.

A Host that supports the FSI may support a subset or all of the versions specified in this document. At a minimum a Host SHALL support FSI Protocol Version 1.

A fixed station that supports the FSI may support a subset or all of the versions specified in this document. At a minimum a fixed station SHALL support FSI Protocol Version 1.

8.3.1 Control Messages

Table 5 summarizes the Control Service message set.

Table 5 – Control Service Message Set Mnemonics

Mnemonic	Message ID	Protocol Version	Meaning/Purpose
FSC_CONNECT	0	1 and 2	Establish a connection with a fixed station
FSC_HEARTBEAT	1	1 and 2	Heartbeat/Connectivity Maintenance
FSC_ACK	2	1 and 2	Control Service Acknowledgement

FSC_SBC	3	1 and 2	Single Block Control
FSC_MAN_EXT	4	1 and 2	Manufacturer Extension
FSC_SEL_CHAN	5	1 and 2	Channel Selection Command
FSC_SEL_RPT	6	1 and 2	Repeat Mode Selection
FSC_SEL_SQUELCH	7	1 and 2	Squelch Mode Selection
FSC_REPORT_SEL	8	1 and 2	Report Selected Modes
FSC_DISCONNECT	9	1 and 2	Disconnect Control Service
FSC_SCAN	10	2	Scan Enable / Disable
FSC_RCV_CNTRL	11	2	Receiver Control Command
FSC_TX_CNTRL	12	2	Transmitter Control Command
FSC_MBC	13	2	Multi Block Control
FSC_START_DATA	14	2	Data Conveyance messages can be sent to the CFSS
FSC_STOP_DATA	15	2	Data Conveyance messages being sent to the CFSS need to be stopped
FSC_CAI_BUSY_BITS	16	2	CAI Status Symbols

Note: Protocol Version 2 means Protocol Version 2 and above.

All messages have a Message ID and Message version field and all messages except Heartbeat and Acknowledgement have a Message Correlation Tag. The Message ID and Message version fields are fixed values determined by this specification. The Correlation Tag shall be selected by the source of the message and echoed back in its acknowledgement. Retries of any message shall be indistinguishable from the original message.

8.3.1.1 Establish Connection

The Establish Connection command may be sent by the host to establish a connection with the fixed station. The FS learns the host Control IP address and Control Port from this FSC_CONNECT message that initiates the connection. It can only be sent by the host. Figure 11 illustrates the FSC_CONNECT message format and content for FSI Protocol Version 1. Figure 12 illustrates the FSC_CONNECT message format and content for FSI Protocol Version 2 and above.

Octet 0	Message ID	FSC_CONNECT
1	Message Version	1
2	Message Correlation Tag	
3	VC Base Port (MSB)	Bits 8-15
4	VC Base Port (LSB)	Bits 0-7
5	VC SSRC (MSB)	Bits 24-31
6	VC SSRC (Next MS bits)	Bits 16-23
7	VC SSRC (Next MS bits)	Bits 8-15
8	VC SSRC (LSB)	Bits 0-7
9	FS Heartbeat Period	
10	Host Heartbeat Period	
Bit	0 1 2 3 4 5 6 7	

Figure 11 – Data for the Establish Connection Command, Message Version 1

Octet 0	Message ID	FSC_CONNECT
1	Message Version	2
2	Message Correlation Tag	
3	CFSH Base Port (MSB)	Bits 8-15
4	CFSH Base Port (LSB)	Bits 0-7
5	CFSH SSRC (MSB)	Bits 24-31
6	CFSH SSRC (Next MS bits)	Bits 16-23
7	CFSH SSRC (Next MS bits)	Bits 8-15
8	CFSH SSRC (LSB)	Bits 0-7
9	Connectivity Tx Timer	
10	Protocol Version	
11	Reserved	BB SD RM Options
12	Connection Type	Cntrl, Voice or Data
13	CFSH IP Address (MSB)	Bits 24-31
14	CFSH IP Address (Next MS bits)	Bits 16-23
15	CFSH IP Address (Next MS bits)	Bits 8-15
16	CFSH IP Address (LSB)	Bits 0-7
17	CFSH_Outbound_Max_CAI_Data_Blocks	
18	CFSH_Inbound_Max_CAI_Data_Blocks	
Bit	0 1 2 3 4 5 6 7	

Figure 12 - Data for the Establish Connection Command, Message Version 2

Message Version – Message Version 1 is used when the host wants to establish a connection using Protocol Version 1. Message Version 2 is used when attempting to establish a connection using Protocol Version 2 and above.

VC Base Port (Protocol Version 1) - The host Voice Conveyance RTP Port. By rule, the RTCP port (unused but reserved) is ((VC Base Port) + 1).

CFSH Base Port (Protocol Version 2) - The CFSH Base Port field specifies the CFSH UDP Port Number of the identified connection. If the connection type is Control, then this is the UDP port number of the CFSH socket where the CFSH receives Control messages from the CFSS. If the connection type is Voice Conveyance, then this is the UDP port number of the CFSH socket where the CFSH receives Voice Conveyance messages from the CFSS. If the connection type is Data Conveyance, then this is the UDP port number of the CFSH socket where the CFSH receives Data Conveyance messages from the CFSS. Note that in the case of the Control connection type, the Base Port shall be the same as the source UDP Port of the message to enable the CFSS to continue to use the source socket of the FSC_CONNECT message as the CFSH Control socket. By rule, the RTCP port (unused but reserved) is ((CFSH Base Port) + 1).

VC SSRC (Protocol Version 1) –The SSRC identifier to be used by the fixed station on all RTP transmissions sent to the host. VC SSRC shall be selected by the host to be unique in the context of the host at any particular time.

CFSH SSRC (Protocol Version 2) – The SSRC field specifies the RTP Synchronization Source value. The SSRC field is a CFSH generated value that is used by the CFSS in all Voice Conveyance transmissions sent to the CFSH. This parameter is set when establishing the Voice Conveyance service. When the Connection Type is Control or Data Conveyance, then this field shall be set to all zeros and ignored by the CFSS.

FS Heartbeat Period – As previously defined for Protocol Version 1 in TIA-102.BAHA, this was the period at which the fixed station should send FSC_HEARTBEAT messages to the host control port. The FS Heartbeat Period is in units of seconds and is restricted to the range 5 to 255. Protocol Version 1 as defined in this document does not use this parameter in order to improve heartbeat operations. FS Heartbeat Period should be the same as the host's configured "Connectivity TX Timer" value.

Host Heartbeat Period – For Protocol Version 1 the period at which the host will send FSC_HEARTBEAT messages to the fixed station control port. The Host Heartbeat Period is in units of seconds and is restricted to the range 5 to 255. For Protocol Version 1 as defined in this document, Host Heartbeat Period shall be the same as the host's configured "Connectivity TX Timer" value. As defined in this document for Protocol Version 1, this is also the period of time in which the fixed station expects a heartbeat from the host.

Connectivity Tx Timer (Protocol Version 2) – The period at which the host or FS will send FSC_HEARTBEAT messages to the fixed station control port. The Connectivity Tx Timer is used solely by the Control Conveyance service and in that case is restricted to the range 5 to 255 seconds. When the Connection Type is Voice or Data Conveyance, this field shall be set to all zeros and ignored by the CFSS.

Protocol Version (Protocol Version 2) – Overall FSI Protocol version. This is the version requested by the connecting FSH. The Protocol Version field is only used when setting up the Control Connection and in that case is set to 2. The FSI Protocol Version field shall be set to all zeros and ignored by the CFSS when the Connection Type is Voice Conveyance or Data Conveyance. The value set for the Control Connection applies to the other subsequently established connections (Voice Conveyance and/or Data Conveyance).

Options (Protocol Version 2) – Allows a host to specify desired optional behavior. The RM and SD fields are only used when setting up the Voice Conveyance connection and shall be set to all zeros and ignored by the CFSS when the Connection Type is Control or Data Conveyance. The BB field is used by the CFSH to provide an indication to the CFSS to use the CAI Busy Bits provided by the CFSH (provided via the FSC_CAI_BUSY_BITS message) as the status symbols transmitted over the air for voice, control, and data. The BB field is only valid when setting up the Control Connection and shall be set to zero and ignored by the CFSS when the Connection Type is Voice or Data Conveyance.

Table 6 Options Field in Connection Comment

Mnemonic	Meaning/Purpose	Setting
RM	RTP Mode (see section 8.4.3.1.1)	0 – Verbose 1 – Compact
SD	Station Decode of Header, Link Control, Encryption Sync and Low Speed Data (see sections 8.4.4.2 through 8.4.4.5)	0 – Station does not send decoded blocks. 1 – Station sends decoded blocks.
BB	CAI Busy Bits	0 –CFSH does not provide CAI busy bits 1 – CFSS instructed to use CFSH provided CAI busy bits.

Connection Type (Protocol Version 2) - The Connection Type field is assigned one of the values in Table 7 depending on the connection type that is being setup in the FSC_CONNECT message.

Table 7 - Values for FSC_CONNECT “Connection Type” parameter

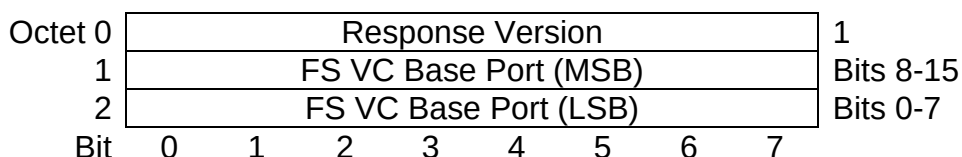
Connection Type parameter value	Description
\$10	Control Connection
\$20	Voice Conveyance Connection
\$40	Data Conveyance Connection

CFSH IP Address (Protocol Version 2) - The CFSH IP Address parameter conveys the CFSH IP Address of the specified connection. If the connection type is Control, then this is the IP Address of the CFSH socket where the CFSH receives Control messages from the CFSS. If the connection type is Voice Conveyance, then this is the IP Address of the CFSH socket where the CFSH receives Voice Conveyance messages from the CFSS. If the connection type is Data Conveyance, then this is the IP Address of the CFSH socket where the CFSH receives Data Conveyance messages from the CFSS. Note that in the case of the Control connection type, the CFSH IP Address shall be the same as the source IP address of the message to enable the CFSS to continue to use the source socket of the FSC_CONNECT message as the CFSH Control socket.

CFSH Outbound Max CAI_Data_Blocks (Protocol Version 2) – This parameter is used solely by the Data Conveyance Service. The parameter specifies the maximum number of CAI Data Blocks to be sent in a single outbound Data Conveyance message. The range of this parameter is 1 to 128. This parameter shall be set to all zeros and ignored by the CFSS when the Connection Type is Voice Conveyance or Control.

CFSH Inbound Max_CAI_Data_Blocks (Protocol Version 2) - This parameter is used solely by the Data Conveyance Service. The parameter specifies the maximum number of CAI Data Blocks to be sent in a single inbound Data Conveyance message. The range of this parameter is 1 to 128. This parameter shall be set to all zeros and ignored by the CFSS when the Connection Type is Voice Conveyance or Control.

The response data for this command shall be as shown in Figure 13 and Figure 14. Figure 13 defines the Response Data format and content for FSI Protocol Version 1 and Figure 14 defines the Response Data format and content for FSI Protocol Version 2 and above.

**Figure 13 – Response Data for the Connection Command, Response Version 1**

Octet 0	Response Version	2						
1	CFSS Base Port (MSB)	Bits 8-15						
2	CFSS Base Port (LSB)							
3	CFSS IP Address (MSB)	Bits 24-31						
4	CFSS IP Address (next MS bits)							
5	CFSS IP Address (next MS bits)	Bits 8-15						
6	CFSS IP Address (LSB)							
7	CFSS Outbound Max_CAI_Data_Blocks	Bits 0-7						
8	CFSS Inbound Max_CAI_Data_Blocks							
9	Protocol Version							
Bit	0	1	2	3	4	5	6	7

Figure 14 - Response Data for the Connection Command, Response Version 2

FS VC Base Port (Protocol Version 1 only) - The fixed station Voice Conveyance RTP Port. By rule, the RTCP port (unused but reserved) is ((FS VC Base Port) + 1).

The CFSS Base Port (Protocol Version 2) - This field specifies the UDP Port Number of the CFSS Control, Voice Conveyance or Data Conveyance socket.

The CFSS IP Address field (Protocol Version 2) – This field specifies the IP Address of the CFSS Voice Conveyance or Data Conveyance socket. The CFSS IP Address field in the Response Data is not used by the Control Connection service, and has a value of all zeros when an FSC_CONNECT message is used to establish the FSI Control connection.

The CFSSH uses the addressing information in the Response Data as the outbound destination socket for the specified Voice Conveyance or Data Conveyance connection.

CFSS Outbound Max_CAI_Data_Blocks (Protocol Version 2) - This parameter specifies the maximum number of CAI Data Blocks to be sent in a single outbound Data Conveyance message.

CFSS Inbound Max_CAI_Data_Blocks (Protocol Version 2) – This parameter specifies the maximum number of CAI Data Blocks to be sent in a single inbound Data Conveyance message.

The CFSS Outbound Max_CAI_Data_Blocks and the CFSS Inbound Max_CAI_Data_Block parameters are used solely by the Data Conveyance Service, and have a value of all zeros in the FSC_ACK message when sent in response to a Control or Voice Conveyance FSC_CONNECT message.

Protocol Version (Response Version 2) – When the station accepts the connection request it indicates the accepted protocol version. When responding with CONTROL_NAK_UNSUPP it indicates either the highest protocol version supported by the station or the lowest protocol version supported by the station. Which case is based on the type of conflict with the proposed protocol version.

Figure 15 defines Response Data for CONTROL_NAK_CONNECTED for FSI Protocol Version 2 and above.

Octet 0	Connected Host IP (MSB)	Bits 24-31
1	Connected Host IP ()	Bits 16-23
2	Connected Host IP ()	Bits 8-15
3	Connected Host IP (LSB)	Bits 0- 7
4	Connected Host Port (MSB)	Bits 8-15
5	Connected Host Port (MSB)	Bits 0- 7
Bit	0 1 2 3 4 5 6 7	

Figure 15 - Optional Response Data for CONTROL_NAK_CONNECTED

For CONTROL_NAK_CONNECTED response including connected host IP and Port the Ack version shall be set to 1.

Figure 16 shows the response data for the CONTROL_NAK_OPT_UNSUPP response. The response shall specify which of the requested options it cannot support by placing a 1 in the field(s).

Octet 0	Reserved	BB	SD	RM	Options
Bit	0 1 2 3 4	5	6	7	

Figure 16 - Response data for CONTROL_NAK_OPT_UNSUPP

8.3.1.2 Control Service Heartbeat

Figure 17 defines the Control Service Heartbeat message format and content for all FSI Protocol versions. This message is sent periodically by both the fixed station and the CFSH to establish and maintain connectivity of the Control server and client. This message is not sent on the Voice Conveyance or Data conveyance connections. No acknowledgement is sent in response to the Control Service Heartbeat.

Octet 0	Heartbeat Message ID	FSC_HEARTBEAT
1	Message Version	1
Bit	0 1 2 3 4 5 6 7	

Figure 17 – Message for Control Service Heartbeat

8.3.1.3 Control Service Acknowledgement

The Control Service Acknowledgement is used by both the client and the server to positively or negatively acknowledge the receipt of a control service message. It contains the identifying information of the message being acknowledged along with an Response Code and optional response data. Figure 18 defines the Control Service Acknowledgement message format and content for all FSI Protocol versions.

Octet 0	Message ID	FSC_ACK
1	Message Version	1
2	Acknowledged Message ID	From msg being ack'd.
3	Acknowledged Message Version	From msg being ack'd.
4	Acknowledged Message Correlation Tag	From msg being ack'd.
5	Response code	
6	Response Data Length	
7+	Response Data	
Bit	0 1 2 3 4 5 6 7	

Figure 18 – Message for Control Service Acknowledgement

Table 8 summarizes the Response Code values.

Table 8 – Response Code Values

Response Code	Mnemonic	Meaning
\$00	CONTROL_ACK	General Acknowledgement
\$01	CONTROL_NAK	Unspecified Negative Acknowledgement
\$02	CONTROL_NAK_CONNECTED	The server is connected to some other host
\$03	CONTROL_NAK_M_UNSUPP	Unsupported Manufacturer specific message.
\$04	CONTROL_NAK_V_UNSUPP	Unsupported Message Version
\$05	CONTROL_NAK_F_UNSUPP	The requested optional Function is not supported
\$06	CONTROL_NAK_PARMS	Bad or Unsupported Command Parameters
\$07	CONTROL_NAK_BUSY	The fixed station subsystem is currently busy with a function that is incompatible with the requested action.
\$08	CONTROL_NAK_P_UNSUPP	Unsupported Protocol Version
\$09	CONTROL_NAK_OPT_UNSUPP	Unsupported options
\$0A	CONTROL_NAK_CONNFAIL	Unable to establish requested connecton
\$0B	CONTROL_NAK_NO_CONTROL	Unable to establish a Voice or Data connection because there is no established Control connection

Response Codes \$00 through \$07 are defined for all FSI Protocol versions. Response Codes \$08 through \$0A are defined for FSI Protocol Version 2 and above.

The content of the Response Data is specified with the message definition given in 8.3.1.1 and 8.3.1.4-10.

8.3.1.4 Single Block Control Message

The single block control message is used to convey conventional control messages as specified in TIA-102.AABG ([14]) and ANSI/TIA-102.AABC-C ([9]) between a fixed station and its host. Emergency Alert is one possible single block control message. Either the fixed station or its host may send it. Figure 19 defines the SBC Control message for all FSI Protocol versions. Figure 20 defines an alternative SBC Control message format and content for FSI Protocol Version 2 and above.

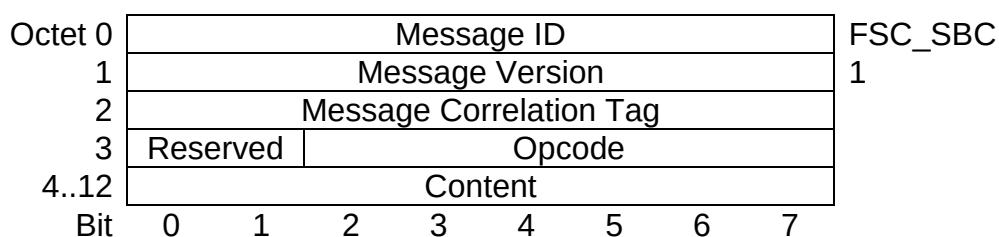


Figure 19 - Message for Single Block Control, for all Protocol Versions

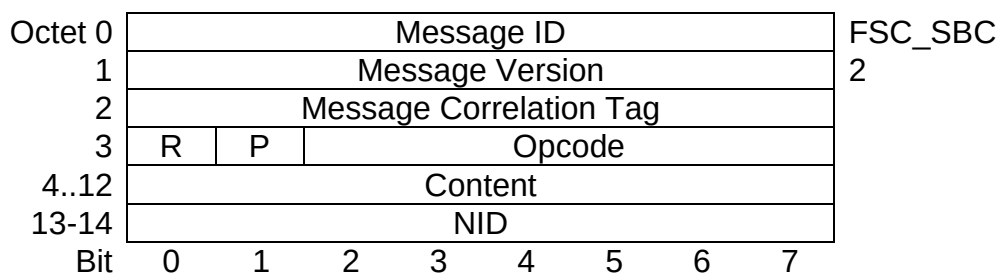


Figure 20 – Message for Single Block Control, Protocol Version 2

R = Reserved

P = Protected Mode from ANSI/TIA-102.AABC-C (See [9]). The use of this bit is a future work item, and until the work is completed, this bit SHOULD be ignored by the receiving device.

Opcode = Control Message Opcode from ANSI/TIA-102.AABC-C ([9]).

Content = Bytes 1-9 of the corresponding single block control message from ANSI/TIA-102.AABC-C ([9]).

NID = 16-bit NID received or to be transmitted with the SBC. Version 2 only. A version 2 station or host receiving a version 1 message shall assume a NID in accordance with local policy. The NID is composed of the NAC and DUID per TIA-102.BAAA-A [8].

This treatment of the NAC shall be consistent with the treatment of voice per paragraph 8.4.5.4.1.1.

The DUID for an SBC, per TIA-102.BAAC-C ([21]), is %0111 (trunking signaling data unit).

If the NAC of stream is either \$F7F or \$F7E, then the locally configured NAC shall be used.

The Acknowledgement for this command shall have no response data (Response Data Length = 0).

8.3.1.5 Manufacturer Extension Message

The Manufacturer Extension Message may be used by manufacturers to provide value added and experimental extensions to the protocol. Either the fixed station or its host may send it. Figure 21 defines the Manufacturer Extension message format and content for all FSI Protocol versions.

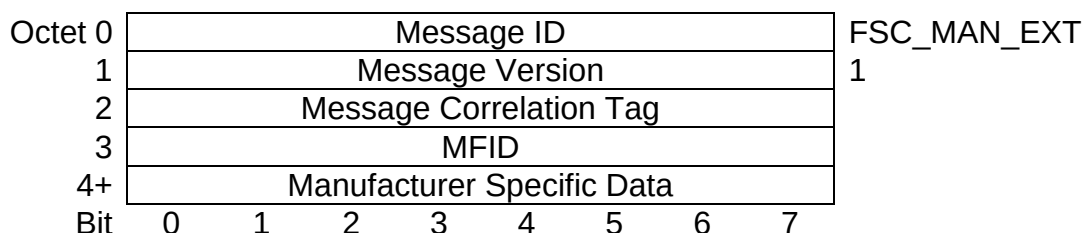


Figure 21 – Message for Manufacturer Extension

Message Version – Version of this extension message header. The version, if any, of the Manufacturer Specific Data shall be carried therein.

The response data for a supported manufacturer extension will be specified by the manufacturer making the extension.

The Acknowledgements for unsupported manufacturer extensions shall have no response data (Response Data Length = 0).

8.3.1.6 Channel Selection Command

The Channel Selection command may be used by the host to select a particular RF channel for receive and/or transmit operations. Only the host may send the Channel

Selection Command. Figure 22 defines the Channel Selection message format and content for all FSI Protocol versions.

Octet 0	Message ID	FSC_SEL_CHAN
1	Message Version	1
2	Message Correlation Tag	
3	Receive Channel Number	
4	Transmit Channel Number	
Bit	01234567	

Figure 22 – Command for Channel Selection

Receive Channel Number = Channel number for receive operations. The specific meaning of the Channel Number parameter may be manufacturer specific.

Transmit Channel Number – Channel Number for transmit operations. The specific meaning of the Channel Number parameter may be manufacturer specific.

The Acknowledgement for this command shall have no response data for FSI Protocol Version 1 (Response Data Length = 0). For FSI Protocol Version 2 and above the response data shall be as shown in Figure 23 (Response Data Length = 3). The response shall be sent as soon as the channel(s) are locked. For more detail on the use of this command see section 8.3.2.5.3 Channel Selection Command Procedure.

Octet 0	Report Version	2
1	Receive Channel Selection	
2	Transmit Channel Selection	
Bit	01234567	

Figure 23 - Response Data for Channel Selection, Report Version 2

Receive Channel Selection = Numeric representation of the selected channel.

Transmit Channel Selection = Numeric representation of the selected channel.

8.3.1.7 Repeat Mode Selection Command

The Repeat Mode selection shall allow a host to select between the repeat and non-repeat modes of a fixed station. Only the host may send the Repeat Mode Selection command. Figure 24 defines the Repeat Mode Selection Command message format and content for all FSI Protocol versions.

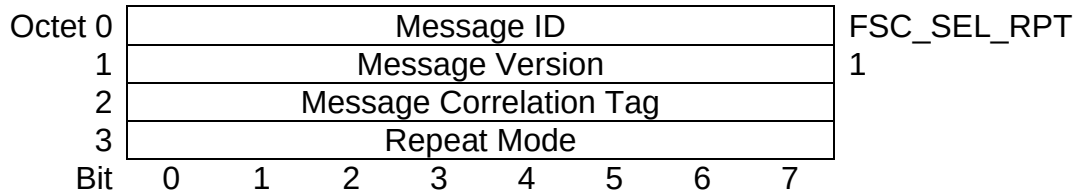


Figure 24 – Command for Repeat Mode Selection

Repeat Mode = 0, Repeat of inbound channel disabled;
 = 1, Repeat of inbound channel enabled;
 = otherwise, illegal.

The Acknowledgement for this command shall have no response data (Response Data Length = 0).

8.3.1.8 Squelch Control Message

The Squelch control message is used by the host to select the Squelch mode of the fixed station. Squelch mode determines the conditions under which the station will forward the audio stream to the host. A value of zero turns off all squelch overrides and returns the station to processing squelch as set by the configuration of the station. Figure 25 illustrates the message for Squelch control for all Protocol Versions. Bit 6 of Octet 3 is Reserved for Protocol Version 1 and is used for Carrier Bypass Mode in Protocol Version 2 and above.

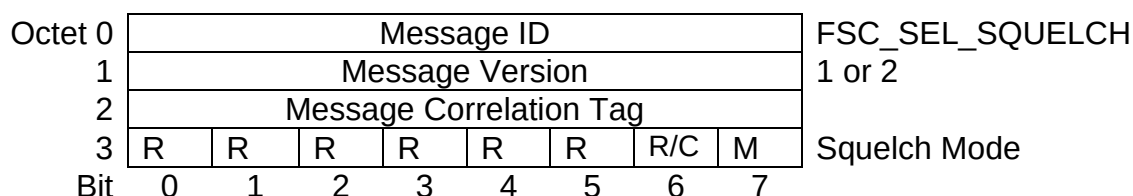


Figure 25 – Message for Squelch Control, Message Version 1 or 2

Table 9 - Squelch Mode Control Word

Mnemonic	Squelch Selection	Mode	Value
M	Monitor Mode (all Protocol Versions)	Off	0
		On	1
C	Carrier Bypass Mode (Protocol Version 2 only)	Off	0
		On	1
R	Reserved	-	-

Monitor Mode – This mode overrides configured squelch processing and is applicable to all Protocol Versions. When ‘On’, the station shall forward audio without regard to the presence or absence of the following signaling.

- Analog – sub-audible CTCSS / CDCSS tones
- Digital – NAC code

Carrier Bypass Mode – This mode overrides the normal carrier detection and may be provided when utilizing FSI Protocol Version 2 and above. When ‘On’, the station shall forward audio without regard to the presence or absence of carrier.

The station shall send an SOS (Start of Stream) when Carrier Bypass mode is initiated and, if the station is equipped with a decoder, send a SOS with the appropriate squelch code when it is detected. Should the code change, a new SOS shall be sent. Likewise, an SOS with the appropriate NID would be sent on detection of a digital bitsync. No EOS (End of Stream) shall be sent until Carrier Bypass mode is exited.

Note that Monitor Mode does not change the station’s normal functionality with respect to indicating the beginning and ending of a received transmission. That is, while in Monitor Mode the station sends an SOS whenever it detects the start of a transmission and an EOS whenever it detects the end of a transmission.

The above modes may be used together or independently for Protocol Version 2 and above.

See section 8.3.2.5.5 for procedures dealing with version compatibility.

The Acknowledgement for this command shall have no response data (Response Data Length = 0) for all FSI Protocol Versions.

Note that Squelch type and Squelch parameters may be set by configuration of the fixed station in accordance with local policy and manufacturer implementation.

8.3.1.9 Report Selections

The Report Selections command may be sent by the host to cause the fixed station to report settings defined in the DFSI. It may only be sent by the host. Figure 26 defines the Report Selection Command message format and content for all FSI Protocol versions.

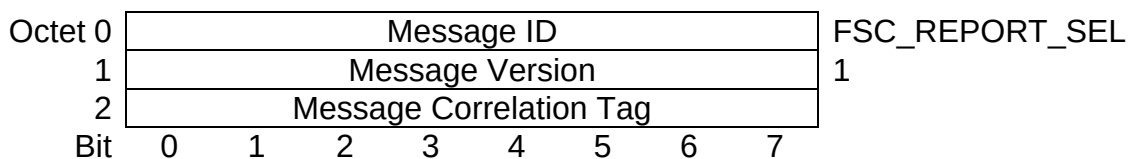


Figure 26 – Command for Report Selections

The positive acknowledgements of the Report Selections command for all FSI Protocol Versions shall contain the response data shown in Figure 27.

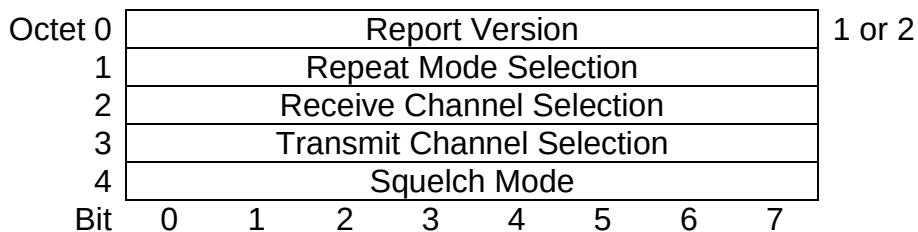


Figure 27 – Response Data for Report Selections

Report Version shall be 2 if the report contains Squelch Mode data with the Carrier Bypass Mode bit.

Repeat Mode Selection = 0, Do not repeat inbound audio;
= 1, Repeat inbound audio.

Receive Channel Selection = Numeric representation of the selected channel.

Transmit Channel Selection = Numeric representation of the selected channel.

Squelch Mode – See 8.3.1.8.

8.3.1.10 Disconnect

The Disconnect command shall allow a host to tear down the connection with the fixed station in an orderly way. Only the host may send the Disconnect command. Figure 28 illustrates the command for Disconnect connection. Figure 28 defines the Disconnect Command message format and content for FSI Protocol version 1.

Octet 0	Message ID	FSC_DISCONNECT
1	Message Version	1
2	Message Correlation Tag	
Bit	0 1 2 3 4 5 6 7	

Figure 28 – Command for Disconnect Connection, Protocol Version 1

The Protocol Version 2 version of this message adds the Connection Type parameter and identifies whether the Control, Voice Conveyance or Data Conveyance connection is to be disconnected. Values for this parameter are shown in Table 7.

Octet 0	Message ID	FSC_DISCONNECT
1	Message Version	2
2	Message Correlation Tag	
3	Connection Type	Control, Voice, Data
Bit	0 1 2 3 4 5 6 7	

Figure 29 – Command for Disconnect Connection Protocol Version 2

The CFSS acknowledgement to this message is the FSC_ACK message. A successful acknowledgment to the FSC_DISCONNECT message is a Response Code of \$00 (General Acknowledgment). The Acknowledged Message ID field is set to \$09 (FSC_DISCONNECT) and the Acknowledged Message Correlation Tag field is populated with the Message Correlation Tag from the associated FSC_DISCONNECT message. The Acknowledgement for this command shall have no response data (Response Data Length = 0).

8.3.1.11 Scan Enable/Disable Command

The Scan Enable/Disable command may be used by the host to enable scanning for receive operations. Only the host may send the Scan Enable/Disable Command. Figure 30 illustrates the command for enabling/disabling scanning. Figure 30 defines the Scan Enable/Disable Command message format and content for FSI Protocol Version 2 and above.

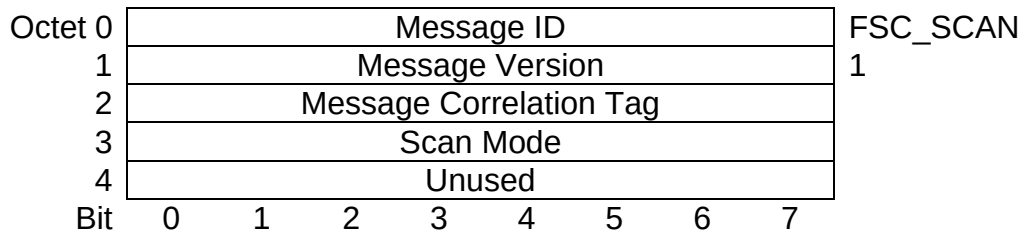


Figure 30 - Command for Scan Enable

The Scan Mode byte shall be set as follows:

- 0 = Disable scan on the inbound channel
- 1 = enable scan on the inbound channel
- All other values are invalid
- The unused byte SHALL be set to 0.

Note that scan parameters may be set by configuration of the fixed station in accordance with local policy and manufacturer implementation.

8.3.1.12 Receiver Control Command

The Receiver Control Command may be used by the host to select a particular radio receiver for receive operations. This may be used in voter applications. Only the host may send the Receiver Control Command. Figure 31 defines the Receiver Control Command message format and content for FSI Protocol Version 2 and above.

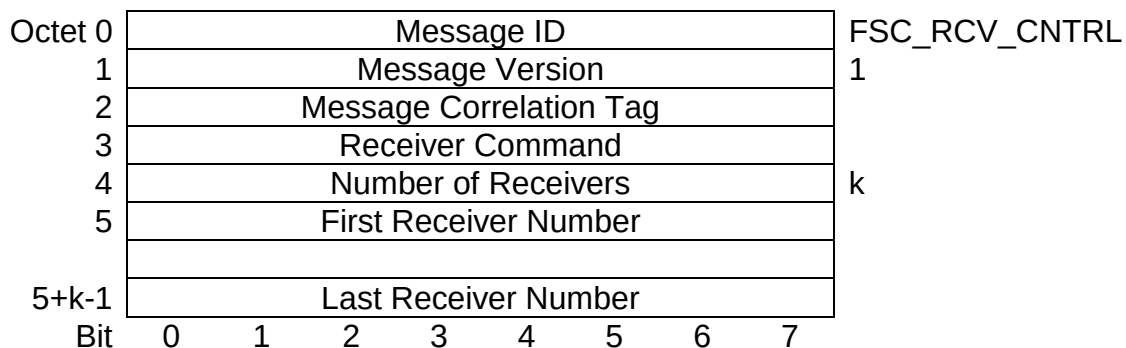


Figure 31 – Command for Receiver Control

Number of Receivers = The total number of receivers that are controlled by this command. For the NORMAL, ENABLE ALL and DISABLE ALL commands specific

receivers do not need to be identified and this field is set to zero. For the SELECT command this field should be set to one.

Receiver Number = Identifier of the receiver being controlled. The specific meaning of the Receiver Number parameter is manufacturer specific.

Receiver Command – Action to take on the specified receiver. The defined values are as follows.

Table 10- Receiver Command Values

Code	Mnemonic	Meaning
0	REPORT	Send the status of this receiver now.
1	NORMAL	Forward speech from the enabled receiver that has the best signal. This command cancels a SELECT. Receiver number is not relevant to this command and Number of Receivers should be set to zero.
2	SELECT	Forward only the specified receiver's speech regardless of how good the signal is. This command does not affect the enable / disable status of the receivers, but temporarily overrides the settings. A NORMAL command undoes this temporary override allowing the subsystem to revert to its normal rules and select from any enabled receiver. Only a single Receiver is specified when issuing a SELECT command. If a SELECT command is sent to an unselected receiver, the previous receiver is deselected. Only one receiver may be selected at a time.
3	DISABLE RCVR SET	Don't forward the listed receivers speech regardless of how strong the signal is. Disabled receivers are removed from the vote when the subsystem is operating in the NORMAL mode.
4	ENABLE RCVR SET	Allow the use of the listed receivers when in NORMAL mode. Enabled receivers participates in the vote. This can be used to individually return disabled receivers to service without affecting other disabled receivers.
5	DISABLE ALL	Mark all receivers as disabled. Receiver Number is not relevant to this command and Number of Receivers should be set to zero.
6	ENABLE ALL	Mark all receivers as enabled. Receiver Number is not relevant to this command and Number of Receivers should be set to zero.

The Acknowledgement for this command shall have no response data (Response Data Length = 0).

8.3.1.13 Transmitter Control Command

The Transmitter Control Command may be used by the host to select a particular radio transmitter for transmit operations. This may be used in voter applications. Only the host may send the Transmitter Control Command. Figure 32 defines the Transmitter Control Command message format and content for FSI Protocol Version 2 and above.

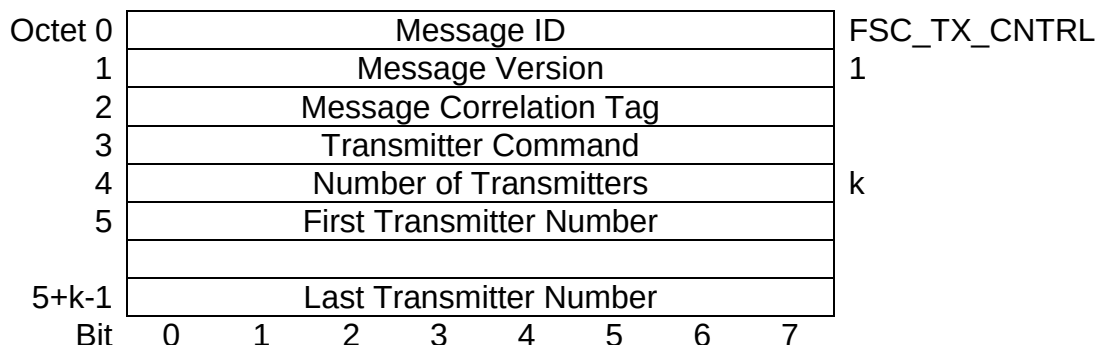


Figure 32 - Command for Transmitter Control

Number of Transmitters = The total number of transmitters are controlled by this command. For ENABLE ALL and DISABLE ALL commands specific transmitters do not need to be identified and this field is set to zero.

Transmitter Number = Identifier of the transmitter being controlled. The specific meaning of the Transmitter Number parameter may be manufacturer specific.

Transmitter Command – Action to take on the specified transmitter. The defined values are as follows.

Table 11 - Transmitter Command Values

Code	Mnemonic	Meaning
0	REPORT	Send the status of this transmitter now.
1	NORMAL	Allow the subsystem to automatically select among enabled transmitters. (Local steering control or utilize all enabled transmitters for multicast or simulcast.)
2	SELECT	This temporarily selects the listed transmitters without regard to the enable / disable status. A NORMAL command undoes this temporary override allowing the subsystem to revert to its normal rules and select from any enabled transmitter.
3	DISABLE	Do not use the listed transmitters regardless what the normal criteria would dictate. This takes the transmitters out of the available transmitters in NORMAL mode.
4	ENABLE	Allow the use of the listed transmitters when in NORMAL mode. This can be used to individually return the listed disabled transmitters to service without affecting other disabled transmitters.

5	DISABLE ALL	Mark all transmitters as disabled. Transmitter Number is not relevant to this command and Number of Transmitters should be set to zero.
6	ENABLE ALL	Mark all transmitters as enabled. Transmitter Number is not relevant to this command and Number of Transmitters should be set to zero.

The Acknowledgement for this command shall have no response data (Response Data Length = 0).

8.3.1.14 Multiple Block Control Message

The Multiple Block Control (MBC) Packet Format is used to convey more information than can be contained in the SBC format as specified in TIA-102.AABG ([14]) and ANSI/TIA-102.AABC-C ([9]). Either the fixed station or its host may send this packet. This message may contain up to 3 MBC blocks in addition to the header. Figure 33 illustrates the format of the Multiple Block Control Message.

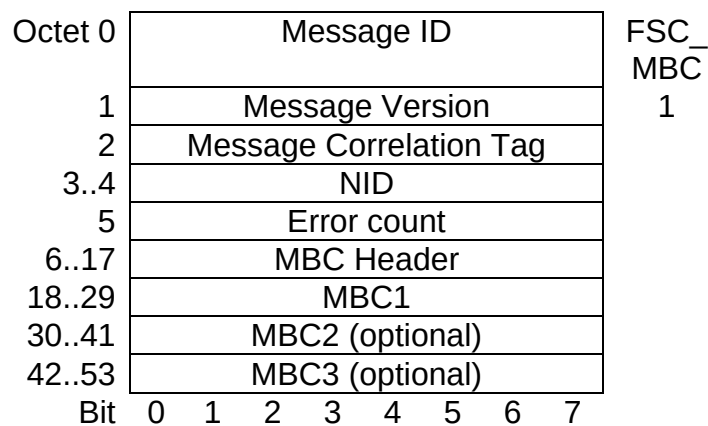


Figure 33 - Multiple Block Control Message

Message ID - FSC_MBC

NID - Network Identifier - 12 bit NAC plus 4 bit DUID

Error Count - Number of errors in Data Block (0xff = uncorrectable). Always 0 when originated from the host.

MBC Header - 12 byte header of the corresponding MBC message from [9].

The blocks to follow entry within the header identifies whether this message contains 1, 2 or 3 data blocks. [9] limits the length of an MBC message to 3 blocks to minimize the disruption to the control channel stream. The last 2 bytes of the header are a CRC.

The MBC Header contains a 6 bit Service Access Point (SAP) Identifier as defined in [9]. Two formats are defined: \$61 for Non-Protected Trunking Control and \$63 for Protected Trunking Control. Protected MBC is a future

work item, and until the work is completed, the CFSH and CFSS SHOULD utilize a SAP Identifier of \$61 only.

MBC1 - 12 byte MBC data block from [9]. The last 4 bytes of the last block contain a message CRC.

MBC2 - 12 byte MBC data block from [9] present if blocks to follow equals 2 or 3.

MBC3 - 12 byte MBC data block from [9] present if blocks to follow equals 3.

The Acknowledgement for this command shall have no response data (Response Data Length = 0).

8.3.1.15 FSC START DATA Message

This message is sent by the CFSS to the CFSH to signal that Data Conveyance messages may be sent across the FSI. It is sent to the Control socket of the CFSH and is accepted by the CFSH only if the Data Conveyance service is in the Connected State.

Figure 34 shows the format for the FSC_START_DATA message in FSI Protocol Version 2 and above.

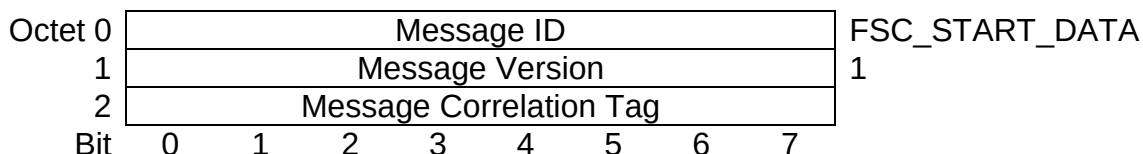


Figure 34 – FSC_START_DATA message format

The CFSH acknowledgement to this message is the FSC_ACK message. A successful acknowledgment to the FSC_START_DATA message is a Response Code of \$00 (General Acknowledgment). The Acknowledged Message ID field is set to \$0E (FSC_START_DATA) and the Acknowledged Message Correlation Tag field is populated with the Message Correlation Tag from the FSC_START_DATA message.

If the CFSH is expecting a FSC_START_DATA message and it is not received in a timely manner, the CFSH may take action based upon its local policy (i.e. a system alarm or other indication).

8.3.1.16 FSC STOP DATA Message

This message is sent by the CFSS to stop Data Conveyance messages from being sent across the FSI by the CFSH. It is sent to the Control socket of the CFSH and is

accepted by the CFSH only if the Data Conveyance service is in the Connected State.

Figure 35 shows the format for the FSC_STOP_DATA message in FSI Protocol Version 2 and above.

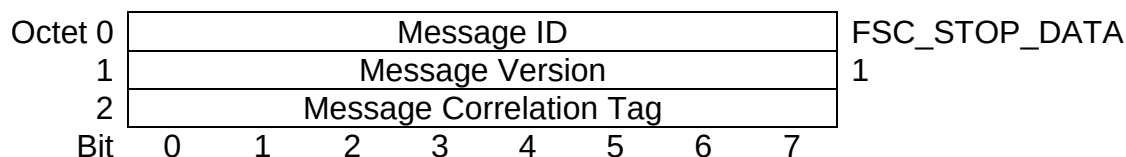


Figure 35 – FSC_STOP_DATA message format

The CFSH acknowledgement to this message is the FSC_ACK message. A successful acknowledgment to the FSC_STOP_DATA message is a Response Code of \$00 (General Acknowledgment). The Acknowledged Message ID field is set to \$0F (FSC_STOP_DATA) and the Acknowledged Message Correlation Tag field is populated with the Message Correlation Tag from the FSC_STOP_DATA message.

8.3.1.17 CAI Busy Bits Message

The FSC_CAI_BUSY_BITS message allows the CFSH to provide the CFSS a 2-bit Status Symbol. Figure 12 shows the format of the FSC_CAI_BUSY_BITS message. Bits 0 and 1 represent the Status Symbols defined in [8] Common Air Interface.

Figure 36 shows the format for the FSC CAI Busy Bits message in FSI Protocol Version 2 and above.

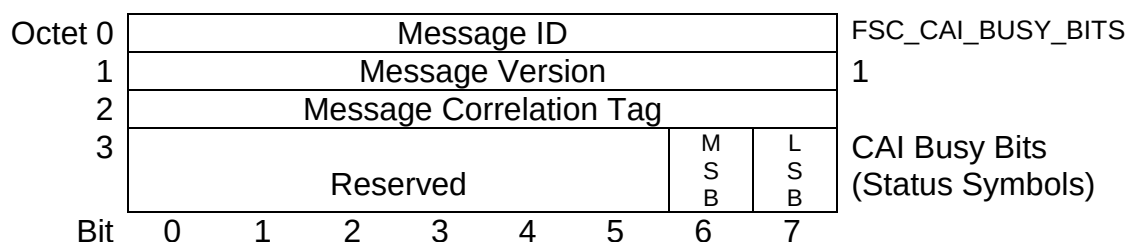


Figure 36 – FSC_CAI_BUSY_BITS Message Format

The CFSH may send the FSC_CAI_BUSY_BITS message to the CFSS. The information in the message is used with all types of messages sent on the CAI (voice, control, and data). Per reference [22] Conventional Procedures, when a repeater's receiver has detected activity, it SHALL assert a Busy indication on the next status symbol on the next slot boundary. When the CFSH is providing CAI busy bits to the CFSS, the determination of appropriate delay is governed by local policy.

If the Busy Bits Flag (BB) in the Control connection FSC_Connect message was asserted, the CFSS shall modulate the Busy Bits received in the FSC_CAI_BUSY_BITS message over the air. If the Busy Bits Flag is not asserted, then the CFSS

shall locally manage the Status Symbols in accordance with reference [22] Conventional Procedures. The ability for the CFSS to tell the CFSS which Status Symbols to use is provided for those cases where the CFSS does not have knowledge of activity on the receive frequency.

If instructed to utilize the provided CAI Busy Bits, the CFSS shall locally manage the busy bits in accordance with reference [22] Conventional Procedures until the first FSC_CAI_BUSY_BITS message is received. Once the CFSS begins receiving FSC_CAI_BUSY_BITS messages, the CFSS shall use the status bits from the last FSC_CAI_BUSY_BITS message received. If the CFSS no longer has an active Control connection it SHALL revert to local control. If the CFSS no longer has an active Voice Conveyance or Data Conveyance connection, it MAY revert to local control.

If the Busy Bits Flag (BB) in the Control connection FSC_Connect message was not asserted and the CFSS receives a FSC_CAI_BUSY_BITS message, the CFSS shall ignore it.

8.3.2 Control Service Procedures

8.3.2.1 Control Service Operational Description

Note that all state models represented in this section are for the purpose of describing the DFSI protocol and should not be construed as requiring or suggesting any particular implementation. For Protocol Version 2 the control service has to be in the CS *Connected* state before establishing the Voice Conveyance or Data Conveyance connections.

The state model for the fixed station control service is shown in Figure 37. In the CS *Not Connected* state, the fixed station operates in whatever configuration was last provided by a host. Emergency alerts received by the FS while in the CS *Not Connected* state are dealt with in accordance with local policy and manufacturer implementations. The FS transitions to the CS *Connected* state upon receipt of an FSC_CONNECT message. When using Protocol Version 2 the FSC_CONNECT message is for a Control Connection. In the CS *Connected* state, the fixed station is able to respond to control messages and forward Emergency Alerts to the host. When using Protocol Version 2 the fixed station can only establish a Voice Conveyance or Data Conveyance service with a host when it is in the CS *Connected* state. Failure to receive an acknowledgement (Retry Failure), the lack of a heartbeat for a sufficient period of time (Connectivity Timeout), the receipt of a FSC_DISCONNECT message, or other implementation specific connectivity loss triggers (such as the receipt of a ICMP Port Not Found notification) cause a transition back to the CS *Not Connected* state. For Protocol Version 2 the Voice Conveyance and Data Conveyance services are not available after the fixed station transitions to the CS *Not Connected* state.

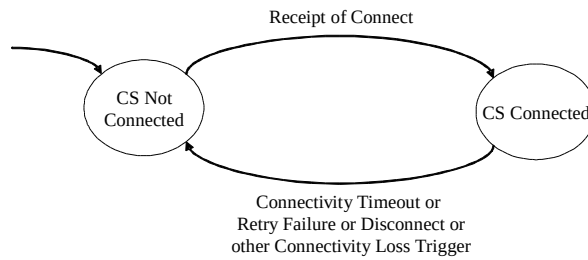


Figure 37 – FS Control Service State Model

The host side of the control service protocol is similar to the fixed station side, and is illustrated in Figure 38. Upon receipt of an implementation specific connection trigger, the CFSH sends FSC_CONNECT and enters the *CS Connecting* state. For Protocol Version 2 the connection trigger is for a Control Connection and the FSC_CONNECT message indicates Control Connection. Upon receipt of an acknowledgement to FSC_CONNECT, the host proceeds to the *CS Connected* state wherein FSC_HEARTBEATS and other messages may be sent. For Protocol Version 2, once in the *CS Connected* state, the host can respond to implementation specific triggers to establish a Voice Conveyance connection and/or Data Conveyance connection. As with the fixed station, failure of the retransmission protocol or connectivity protocol can cause a transition to the *CS Not Connected* state. Furthermore, the host can initiate a control disconnection by sending FSC_DISCONNECT and transitioning to the *CS Disconnecting* state. For Protocol Version 2 the FSC_DISCONNECT indicates Control Connection. In this case, receipt of the acknowledgement to the FSC_DISCONNECT (or a connectivity or retry failure) causes the transition to *CS Not Connected*. For Protocol Version 2, an established Voice Conveyance connection and Data Conveyance connections are disconnected when the host transitions to the *CS Not Connected* state and can be reestablished only after the Control Service connection is reestablished.

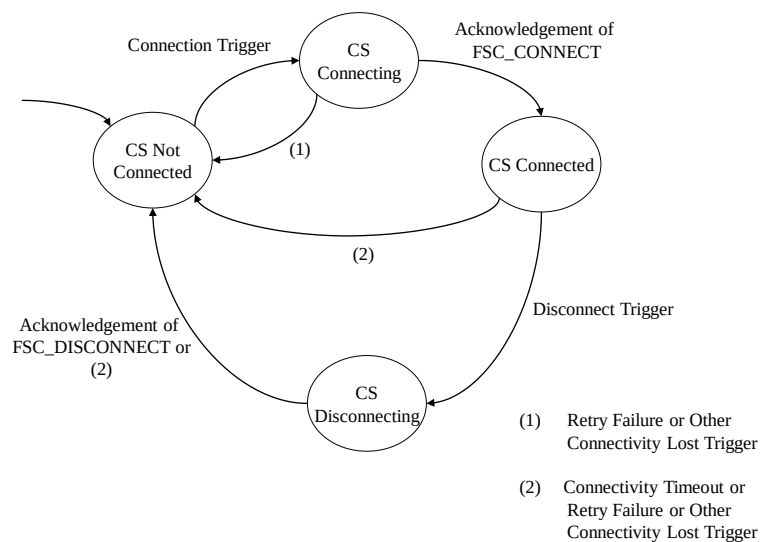


Figure 38 – Host Control Client State Model

8.3.2.2 Control Procedures – General Requirements

The following control procedures are subject to the following general requirements:

1. While in the *CS Not Connected* state, the fixed station shall silently discard all messages except for FSC_CONNECT and FSC_DISCONNECT.
2. While in the *CS Not Connected* state, the host shall silently discard all messages. As described previously, air interface messages received while in the *CS Not Connected* state are processed in accordance with local policy and specific manufacturer implementations.
3. While in the *CS Connecting* state, the host shall silently discard all messages except for FSC_ACK.
4. While in the *CS Not Connected* state, neither the fixed station nor the host shall send any messages to the other, with the exception of FSC_ACK in response to a FSC_DISCONNECT that might occur due to a dropped FSC_ACK.
5. While in the *CS Connecting* state, the host shall only send FSC_CONNECT messages.
6. FSC_ACK shall always be sent to the IP address and UDP port contained in the message being acknowledged.
7. The host shall send all messages except for FSC_ACK to the pre-configured IP address and UDP control port of the fixed station.
8. The fixed station shall send heartbeat messages to the IP address and UDP port derived from the FSC_CONNECT message that initiated the connection.
9. Subject to these restrictions and those in the following procedures, the host may send any control message defined in 8.3.1.
10. Subject to these restrictions and the following procedures, the fixed station may only send FSC_HEARTBEAT, FSC_ACK, FSC_MAN_EXT, FSC_MBC, and FSC_SBC messages to the host. If FSI Protocol Version 2, the fixed station shall have an established Data Conveyance connection before it sends FSC_STOP_DATA and FSC_START_DATA messages.
11. While in the *CS Not Connected* state, the fixed station and the host shall ensure the Voice Conveyance and Data Conveyance connections are disconnected.

The host shall ensure it is in the *CS Connected* state prior to establishing a Voice Conveyance or Data Conveyance connection with the fixed station with which it has the Control connection.

12. The fixed station shall ensure it is in the *CS Connected* state prior to establishing a Voice Conveyance or Data Conveyance connection with the host with which it has the Control connection.

8.3.2.3 Retry Protocol

With the exception of the Heartbeat and Acknowledgement messages, all control service messages shall be sent using the following retransmission protocol. Within the following description, “sender” refers to the host or fixed station sending the message, and “receiver” refers to the host or fixed station receiving the message.

1. Upon receiving a trigger for the transmission of a message, the sender shall:
 - a. Choose a correlation tag, format the message, and send it to the control service port of the receiver.
 - b. Start a retransmission timer with the value `ControlRetryTimer`.
 - c. Reset a retransmission counter to zero.
2. Upon expiration of the retransmission timer, the sender shall:
 - a. Increment the retransmission counter.
 - b. If the value of the retransmission counter is equal to `ControlAttemptLimit`, the retry protocol has failed. The sender shall
 - i. If the connectivity procedures of 8.3.2.4.3 and 8.3.2.4.4 are active, it shall discontinue connectivity procedures as described therein and revert to the *CS Not Connected* state.
 - ii. The sender shall implement recovery procedures in accordance with local policy.
 - c. If the value of the retransmission counter is less than `ControlAttemptLimit`, the sender shall reset the retransmission timer to the value `ControlRetryTimer` and resend the message.
3. Upon receipt of an acknowledgement, the sender shall:
 - a. If the Acknowledged Message Id, Acknowledged Message Version, and Acknowledged Message Correlation Tag match those of an outstanding message, the sender shall:
 - i. Cancel the retransmission timer of the outstanding message.
 - ii. Perform further processing on the reply in accordance with local policy.
 - b. Otherwise, the sender shall silently discard the acknowledgement.

8.3.2.4 Connectivity Procedures

8.3.2.4.1 Connection Establishment Procedures

1. Upon detecting a connectivity trigger while in the *CS Not Connected* state, the host shall:
 - a. Transition to the *CS Connecting* state.
 - b. Send the FSC_CONNECT message indicating the highest FSI Protocol Version it supports to the fixed station, applying the retransmission protocol of 8.3.2.3. Note that a host that attempts to connection with FSI Protocol Version 1 uses Message Version 1 of the FSC_CONNECT message.
2. Upon receipt of the FSC_ACK response to FSC_CONNECT,
 - a. If the status is CONTROL_ACK the host shall
 - i. Determine and save the Protocol Version received from the station and use it to limit the communications with that station to that level of this specification. For Protocol Version 2 and higher, the agreed upon protocol version shall apply to subsequent Voice and Data Conveyance connections between the same host and fixed station.
 - ii. Enter the *CS Connected* state, and initiate connectivity maintenance procedures as described in 8.3.2.4.3. And if it is operating in Protocol Version 2, the host may try to connect the Voice Conveyance and/or Data Conveyance services.
 - b. If the status is CONTROL_NAK_P_UNSUPP the host shall take one of the following actions.
 - i. Retry the connect using the version from the response message received from the fixed station if the host can support this version.

-or-

 - ii. Take actions as specified by local policy if it cannot support the requested version.
 - c. If the status is CONTROL_NAK_V_UNSUPP, the host shall retry the connect using the protocol version associated with the response message version.
 - d. If the status is CONTROL_NAK_OPT_UNSUPP and the response data indicates the Busy Bit (BB) option cannot be supported the host shall take one of the following actions:
 - i. Retry the connect without requesting the enablement of the Busy Bits option

-or-

- i. Take actions as specified by local policy for operation without the optional functionality
 - ii.
- 3. Upon failure of the retransmission protocol on the FSC_CONNECT, the host shall, if it is capable of doing so, attempt to connect with version 1 of the protocol to accommodate legacy units that do not fully support the version negotiation and apply the retransmission protocol of 8.3.2.3. Version 1 support SHALL be provided by all version 2 hosts. If this retry also fails, the host shall transition to the *CS Not Connected* state.
- 4. Upon receipt of some other implementation specific connectivity lost indication, the host shall Transition to the *CS Not Connected* state.
- 5. Upon receipt of FSC_CONNECT in the *CS Not Connected* state, the fixed station shall determine the requested FSI Protocol Version (through either the Protocol Version in the FSC_CONNECT message or by the fact that the FSC_CONNECT Message Version is 1).
 - a. If the fixed station can support the message version and protocol version and for Protocol Version 2 the requested option (e.g.BB =1 (Station uses Host provided CAI Busy Bits) in the FSC_CONNECT) it shall:
 - i. Reply with FSC_ACK with the CONTROL_ACK Response Code and the Protocol Version parameter of the response set to the version requested by the host.
 - ii. Transition to the *CS Connected* state, initiate connectivity maintenance procedures as described in 8.3.2.4.4, and if Protocol Version 1, allow Voice Conveyance services to operate using the SSRC contained within the FSC_CONNECT message as the SRC identifier for the RTP protocol. Note that for Protocol Version 2, once the Control service is connected, the Voice Conveyance and Data Conveyance services are set up separately by the Host using the FSC_CONNECT message.
 - b. If the fixed station can support the message version but cannot support the protocol version, it shall:
 - i. Respond with an FSC_ACK with the CONTROL_NAK_P_UNSUPP Response Code and the Protocol Version parameter of the response set to the highest supported if the request version is higher than can be supported, or the lowest supported version if the version is less than the lowest supported version.
 - ii. Remain in the *CS Not Connected* state.
 - c. If the fixed station cannot support the Message Version of the FSC_CONNECT request, it shall:
 - i. Reply with FSC_ACK with the CONTROL_NAK_V_UNSUPP Response Code and if it supports inclusion of response data, the Protocol Version parameter set to the highest version it supports.

- ii. Remain in the *CS Not Connected* State.
 - d. If the fixed station cannot support the requested option (BB =1 (Station uses Host provided CAI Busy Bits) in the FSC_CONNECT), it shall:
 - i. Reply with FSC_ACK with the CONTROL_NAK_OPT_UNSUPP Response Code with response data indicating it cannot support the BB option.
 - ii. Remain in the *CS Not Connected* State
6. For Protocol Version 1, upon receipt of FSC_CONNECT in the *CS Connected* state, the fixed station shall reply with FSC_ACK message with the CONTROL_ACK Response Code if and only if the FSC_CONNECT message originated at the same IP address and UDP port as the FSC_CONNECT that caused the transition into the *CS Connected* state. Otherwise, the fixed station shall respond with the FSC_ACK with the CONTROL_NAK_CONNECTED Response Code. Note that this paragraph also applies to Protocol Version 2 when the FSC_CONNECT message indicates it is for the Control Connection and the CFSS is already in the *CS Connected* state.

8.3.2.4.2 Connection Detachment Procedures

1. Upon detecting a disconnect trigger in the *CS Connected* state, the host shall:
 - a. Transition to the *CS Disconnecting* state.
 - b. For Protocol Version 2, instruct the fixed station to disconnect the Voice Conveyance connection if it is active, as defined in 8.4.2. The host may wait for the FSC_ACK or immediately proceed to the next step.
 - c. For Protocol Version 2, instruct the fixed station to disconnect the Data Conveyance connection if it is active, as defined in 9.2. The host may wait for the FSC_ACK or immediately proceed to the next step.
 - d. Send the FSC_DISCONNECT message for the Control Connection to the fixed station, applying the retransmission protocol of 8.3.2.3.
2. Upon receiving the FSC_DISCONNECT message in the *CS Not Connected* state, the fixed station shall respond with FSC_ACK with a Response Code of CONTROL_ACK.
3. Upon receiving a FSC_DISCONNECT message for the Control Connection in the *CS Connected* state, the fixed station shall:
 - a. Reply with FSC_ACK with a status code of CONTROL_NAK_CONNECTED if the IP address and port of the sender of the FSC_DISCONNECT do not match those of the sender of the FSC_CONNECT that caused the transition into *CS Connected*.
 - b. Otherwise,
 - i. Reply with FSC_ACK with a status code of CONTROL_ACK and,
 - ii. Enter the *CS Not Connected* state.
 - iii. Disconnect the Voice Conveyance connection if it is active per local policy.
 - iv. Disconnect the Data Conveyance connection if it is active per local policy.
 - v. Terminate connectivity procedures as described in 8.3.2.4.4.
4. Upon receiving a FSC_ACK with a status code of CONTROL_ACK in response to sending a FSC_DISCONNECT to a fixed station, a host shall:
 - a. Enter the *CS Not Connected* state.
 - b. Terminate connectivity procedures.

8.3.2.4.3 Host Connectivity Maintenance Procedures

The “ConnectivityTxTimer” value in this section refers to the configuration timer value “Connectivity Tx Timer” found in Table 4, and is related to parameters in the two versions of the FSC_CONNECT message as follows:

- For Protocol Version 1, the value of the “Host Heartbeat Period” parameter in the message version 1 FSC_CONNECT message as shown in Figure 11 shall be used as the “ConnectivityTxTimer” in these procedures.
- For Protocol Version 2, the value of the “Connectivity Tx Timer” parameter in the message version 2 FSC_CONNECT message shown in Figure 12 shall be used as the value as the “ConnectivityTxTimer” in these procedures.

1. To establish connectivity maintenance procedures, the host shall:
 - a. Start a connectivity timer with the value ConnectivityTxTimer.
 - b. Reset a connectivity counter to zero.
 - c. Set an indicator that says a heartbeat message from the fixed station is not pending.

This procedure can only be done in the *CS Connected* state.

2. To discontinue (terminate) connectivity maintenance procedures, the host shall:
 - a. Cancel the connectivity timer.
 - b. Reset the connectivity counter to zero.
 - c. Discontinue all pending retransmission procedures.
 - d. If operating in Protocol Version 1, discontinue the transmission of any and all packets on the Voice Conveyance service.
3. A host that receives a heartbeat message in the *CS Connected* state shall perform the following if and only if the heartbeat originates from the expected IP address and UDP port:
 - a. Reset the connectivity counter to zero.
 - b. Set an indicator that says a heartbeat message from the fixed station is not pending.
4. Upon receiving a heartbeat message in any other state, the host shall silently discard the message.
5. Upon expiration of the connectivity timer, the host shall:
 - a. If a heartbeat message from the fixed station is pending (not received since the last connectivity timer expiration), the host shall increment the connectivity counter.
 - b. If the connectivity counter is greater than ConnectivityLossLimit, connectivity has been lost. The host shall enter the *CS Not Connected* state and take recovery actions in accordance with local policy.
 - c. Otherwise, the host shall:

- i. Send a heartbeat to the fixed station.
 - ii. Set an indicator that says a heartbeat message from the fixed station is pending.
 - iii. Restart the connectivity timer with the value ConnectivityTxTimer.
6. If a heartbeat is received from other than the expected IP address and port then it shall silently discard the message.

8.3.2.4.4 Fixed Station Connectivity Maintenance Procedures

This procedure can only be done in the *CS Connected* state.

The “Connectivity Tx Timer” value in this section refers to the connectivity timer value that the fixed station receives in the FSC_CONNECT message. The timer value used depends on the message version (and therefore Protocol Version) of the FSC_CONNECT message:

- For Protocol Version 1, the “ConnectivityTxTimer” in these procedures shall be equated to the “Host Heartbeat Period” parameter received in the message version 1 FSC_CONNECT message shown in Figure 11. Note that a Fixed Station connected using Protocol Version 1 does not use the “FS Heartbeat Period” that is also received in the message version 1 FSC_CONNECT message.
 - For Protocol Version 2, the “ConnectivityTxTimer” in these procedures shall be equated to the “Connectivity Tx Timer” parameter received in the message version 2 FSC_CONNECT message shown in Figure 12.
1. To initiate connectivity maintenance procedures, the fixed station shall:
 - a. Reset a connectivity counter to zero.
 - b. Start a connectivity timer with the value ConnectivityTxTimer.
 2. To discontinue (terminate) connectivity maintenance procedures, the fixed station shall:
 - a. Cancel the connectivity timer.
 - b. Reset the connectivity counter to zero.
 - c. Discontinue all pending retransmission procedures.
 - d. If operating in Protocol Version 1, discontinue the transmission and any and all packets on the Voice Conveyance Service.
 - e. Go to the Disconnected state.
 3. Upon receipt of a heartbeat message in the *CS Connected* state, the fixed station shall:
 - a. If and only if the IP address and port of the message match those from the most recent successful FSC_CONNECT message, the fixed station shall:
 - i. Reset the connectivity counter to zero.
 - ii. Restart the connectivity timer with the value ConnectivityTxTimer.
 - iii. Send a heartbeat message to its host.
 - b. Otherwise, the fixed station shall silently discard the FSC_HEARTBEAT message.
 4. Upon receipt of a Heartbeat message, in any other state, the fixed station shall silently discard the message.

5. Upon expiration of the ConnectivityTxTimer, the fixed station shall:
 - a. Increment the connectivity counter.
 - b. If the connectivity counter is greater than ConnectivityLossLimit, connectivity has been lost. The fixed station shall enter the CS *Not Connected* state and take recovery actions in accordance with local policy.
 - c. Otherwise, the fixed station shall
 - i. start the connectivity timer with the value ConnectivityTxTimer.

8.3.2.5 Control Procedures – Specific Requirements

The following procedures related to actions performed following the validation of a message as described in 8.3.2.3. Each of these procedures results in the generation of an FSC_ACK message.

Upon receiving a message other than a heartbeat or acknowledgement:

1. If the receiver is a fixed station that is currently connected to a host, and the IP address or UDP port of the sender do not match those of the host, the receiver shall reply with FSC_ACK with the Response Code set to CONTROL_NAK_CONNECTED and no response data. The FS may optionally append the connected host IP address and port number in the NAK response data.
2. Otherwise, if the received message is FSC_MAN_EXT and the receiver does not support the message, the receiver shall reply with FSC_ACK with the Response Code set to CONTROL_NAK_M_UNSUPP and no response data.
3. Otherwise, if the received message is a known command with an unsupported message version, the receiver shall reply with FSC_ACK with the Response Code set to CONTROL_NAK_V_UNSUPP.
4. Otherwise, the receiver shall perform message specific processing as specified in the following subsections.

8.3.2.5.1 Single Block Control Message Procedure

The following procedure applies when a host receives a valid FSC_SBC command from a fixed station:

1. If the host does not support the Single Block Control message contained in the command:
 - a. The host shall reply with FSC_ACK with the Response Code set to CONTROL_NAK_F_UNSUPP and the response data length set to zero.
 - b. Upon receiving the FSC_ACK, the fixed station shall take such steps as may be required by its air interface and local policy.
2. Otherwise, the host:
 - a. Shall reply with FSC_ACK with the Response Code set to CONTROL_ACK and shall take such actions as may be required by the received message and local policy.

- b. In the event that the Single Block Control (SBC) message requires a response, the host shall formulate a Single Block Control message response (recommended to be implemented in accordance with the requirements of TIA-102.AABG ([14])), and shall transmit an appropriate FSC_SBC to the fixed station.

The following procedure applies when a fixed station receives a valid FSC_SBC command from its host:

1. If the fixed station does not support the Single Block Control message contained in the command:
 - a. The fixed station shall reply with FSC_ACK with the Response Code set to CONTROL_NAK_F_UNSUPP and the response data length set to zero.
 - b. Upon receiving the FSC_ACK, the host shall take such steps as may be required by local policy.
2. If the fixed station subsystem is transmitting a voice stream on the outbound CAI at the time that it receives the FSC_SBC from the CFSH:
 - a. The fixed station may reply with an FSC_ACK with the Response Code set to CONTROL_NAK_BUSY subject to local policy.
 - b. If the station responds per paragraph a, the fixed station should discard the FSC_SBC.
3. Otherwise, the fixed station shall respond with FSC_ACK with the Response Code set to CONTROL_ACK and shall further transmit the corresponding conventional message via the outbound air interface.

8.3.2.5.2 Manufacturer Extension Procedure

Processing of the FSC_MAN_EXT shall be manufacturer specific, but shall always result in the generation of FSC_ACK with an Response Code of CONTROL_ACK if the command was successfully executed, and an Response Code with some other value if the command was not successfully executed.

If a FSC_MAN_EXT is received that the receiver does not support, it shall send an acknowledgement with a NAK of CONTROL_NAK_M_UNSUPP and no response data.

8.3.2.5.3 Channel Selection Command Procedure

For FSI Protocol Version 1, upon receipt of a valid FSC_SEL_CHAN command, the fixed station subsystem shall:

1. Reply with FSC_ACK with an Response Code of CONTROL_NAK_PARMS if the Channel Number parameter is illegal.
2. Reply with FSC_ACK with an Response Code set to CONTROL_NAK_BUSY and discard the FSC_SEL_CHAN command, if the fixed station subsystem is currently transmitting voice.

- Otherwise, it shall set the receive channel selection as indicated by the command and reply with FSC_ACK with and Response Code of CONTROL_ACK. In both cases, the Response Data Length of FSC_ACK shall be zero.

The remaining procedures in this section apply to FSI Protocol Version 2 and above only.

Upon receipt of a valid FSC_SEL_CHAN command, the fixed station subsystem may reply with FSC_ACK with an Response Code set to CONTROL_NAK_BUSY and discard the FSC_SEL_CHAN command, if commanded to change the transmit channel and the fixed station subsystem is currently transmitting voice.

Otherwise the fixed station subsystem shall:

- Return the Receive and Transmit Channel Numbers currently selected after the command is processed and the channels are locked on frequency using the message defined in Figure 23 of section 8.3.1.6.
- Reply with FSC_ACK with an Response Code of CONTROL_NAK_PARMS if either of the Channel Number parameters cannot be honored, such as if the channel selection in the command is not a channel programmed into the fixed station subsystem.
- Reply with FSC_ACK with an Response Code of CONTROL_NAK_PARMS if the Channel Number parameters are different, and the station does not support independent selection of the receive and transmit channels.
- Leave the channel unchanged if the transmit or receive channel parameter matches the currently selected channel. A transmit or receive in progress shall not be interrupted if the corresponding channel number matches the current channel.

A summary of the behavior based on the parameters sent by the CFSH for station subsystems with and without independent control of the transmit and receive channels is shown in Table 12. The table assumes a CFS with current receive and transmit channels of R and T respectively.

Table 12 Channel Selection Behavior Summary for FSI Protocol Version 2

Receive Channel	Transmit Channel	Stations with Independent Control	Stations with Paired control
R	T	Send FSC_ACK with Response Code of CONTROL_ACK and Ignore message	Send FSC_ACK with Response Code of CONTROL_ACK and Ignore message

Receive Channel	Transmit Channel	Stations with Independent Control	Stations with Paired control
R	N	Set transmit channel to N and leave Rx unchanged. Send FSC_ACK with Response Code of CONTROL_ACK.	Send FSC_ACK with Response Code of CONTROL_NAK_PARMS.
N	T	Set receive channel to N and leave transmit unchanged. Send FSC_ACK with Response Code of CONTROL_ACK.	Send FSC_ACK with Response Code of CONTROL_NAK_PARMS
N	N	Set Tx and Rx channels to N. Send FSC_ACK with Response Code of CONTROL_ACK.	Set channel pair to N. Send FSC_ACK with Response Code of CONTROL_ACK.
N	M	Set Rx to N and Set Tx to M. Send FSC_ACK with Response Code of CONTROL_ACK.	Send FSC_ACK with Response Code of CONTROL_NAK_PARMS.

8.3.2.5.4 Repeat Mode Selection Procedure

Upon receipt of a valid FSC_SEL_RPT command, the fixed station shall:

1. Begin operating in the selected mode and reply with FSC_ACK with an Response Code of CONTROL_ACK.
2. If the Repeat Mode parameter is illegal, the fixed station shall reply with FSC_ACK with an Response Code of CONTROL_NAK_PARMS.
3. The Response Data Length of FSC_ACK shall be zero.

8.3.2.5.5 Squelch Control Procedure

Upon receipt of a valid Squelch Control command, the fixed station shall:

1. Reply with FSC_ACK with an Response Code of CONTROL_NAK_PARMS if it does not support the selected mode in its current operating mode. A FSI Protocol Version 1 station considers the “M” (Monitor) bit in the request. A FSI Protocol Version 2 and above station considers both the “M” bit and the “C” (CSQ) bit in the request.
2. Begin operating in the selected mode and reply with FSC_ACK with an Response Code of CONTROL_ACK. The Response Data Length of FSC_ACK shall be zero.

8.3.2.5.6 Report Selections Procedure

Upon receipt of a valid FSC_REPORT_SEL command, the fixed station shall:

1. Reply with FSC_ACK with an Response Code of CONTROL_ACK. The Response Data of the FSC_ACK shall be as indicated in 8.3.1.9.

8.3.2.5.7 Scan Enable/Disable Command Procedure

This procedure is only applicable to FSI Protocol Version 2 and above. Upon receipt of a valid FSC_SCAN command, the fixed station subsystem shall:

1. Return the Scan Mode currently selected after the command is processed.
2. Reply with FSC_ACK with an Response Code of CONTROL_NAK_PARMS if scan mode cannot be honored, such as if the there is only one logical channel programmed into the fixed station subsystem.

The fixed station subsystem may reply with FSC_ACK with an Response Code set to CONTROL_NAK_BUSY and discard the FSC_SCAN command, if commanded to enable scan and the fixed station subsystem is currently transmitting voice.

Otherwise, it shall set the scan mode as indicated by the command and reply with FSC_ACK and Response Code of CONTROL_ACK. In both cases, the Response Data shall include the scan mode after the command is processed.

8.3.2.5.8 Receiver Control Command Procedure

This procedure is applicable to FSI Protocol Version 2 and above. Receiver Control using Protocol Version 1 is implemented via the voice connection. If Receiver Control is implemented in Protocol Version 2 it shall conform to the following procedure:

Upon receipt of a valid FSC_RCV_CNTRL command, the fixed station subsystem shall:

1. Reply with FSC_ACK with an Response Code of CONTROL_NAK_PARMS if the Receiver Number parameter is not valid.
2. Otherwise, it shall:
 - Cause a receiver status update to be sent on the bearer plane.
 - Set the receiver number to the state indicated by the command.
 - Reply with FSC_ACK with and Response Code of CONTROL_ACK.

In all cases, the Response Data Length of FSC_ACK shall be zero.

8.3.2.5.9 Transmitter Control Command Procedure

This procedure is applicable to FSI Protocol Version 2 and above. If Transmitter Control is implemented in FSI Protocol Version 2 it shall conform to the following procedure:

Upon receipt of a valid FSC_TX_CNTRL command, the fixed station subsystem shall:

1. Reply with FSC_ACK with an Response Code of CONTROL_NAK_PARMS if the Transmitter Number parameter is not valid.
2. Otherwise, it shall:
 - Cause a transmitter status update to be sent on the bearer plane.
 - Set the transmitter number to the state indicated by the command.
 - Reply with FSC_ACK with an Response Code of CONTROL_ACK.

In all cases, the Response Data Length of FSC_ACK shall be zero.

8.3.2.5.10 **Unsupported Function Procedure**

If a request for a function is received that the receiver does not support, it shall:

1. Send an acknowledgement with an FSC_ACK with an ACK/NAK of CONTROL_NAK_F_UNSUPP and no response data.

8.3.2.5.11 **Multi Block Control Message Procedure**

The following procedure applies when a host receives a valid FSC_MBC command from a fixed station:

1. If the host does not support the Multi Block Control message contained in the command:
 - a. The host shall reply with FSC_ACK with the Response Code set to CONTROL_NAK_F_UNSUPP and the response data length set to zero.
 - b. Upon receiving the FSC_ACK, the fixed station shall take such steps as may be required by its air interface and local policy.
2. Otherwise, the host:
 - a. Shall reply with FSC_ACK with the Response Code set to CONTROL_ACK and shall take such actions as may be required by the received message and local policy.
 - b. In the event that the Multi Block Control (MBC) message requires a response, the host shall formulate a response (recommended to be implemented in accordance with the requirements of TIA-102.AABG ([14]), and shall transmit an appropriate FSC_MBC to the fixed station.

The following procedure applies when a fixed station receives a valid FSC_MBC command from its host:

1. If the fixed station does not support the Multi Block Control message contained in the command:

- a. The fixed station shall reply with FSC_ACK with the Response Code set to CONTROL_NAK_F_UNSUPP and the response data length set to zero.
 - b. Upon receiving the FSC_ACK, the host shall take such steps as may be required by local policy.
2. If the fixed station subsystem is transmitting a voice stream on the outbound CAI at the time that it receives the FSC_MBC from the CFSH:
 - a. The fixed station may reply with an FSC_ACK with the Response Code set to CONTROL_NAK_BUSY subject to local policy.
 - b. If the station responds per paragraph a, the fixed station should discard the FSC_MBC.
3. Otherwise, the fixed station shall respond with FSC_ACK with the Response Code set to CONTROL_ACK and shall further transmit the corresponding conventional message via the outbound air interface.

8.3.2.5.12 **START_DATA and STOP_DATA Message Procedures**

The following procedures apply when the host and fixed station also have Voice Conveyance and Data Conveyance connections.

The FSC_START_DATA and FSC_STOP_DATA are provided to enable arbitration between voice and data.

When both Voice and Data Conveyance connections exist on the same CFSS, a method to arbitrate between inbound and outbound traffic is required. For the FSI, arbitration is controlled by the CFSS.

Any time there is both voice and data, a situation is created where the CFSS may need to restrict the flow of FSI Data Conveyance messages in order to optimize the transfer of voice.

The CFSS may use control messages FSC_START_DATA and FSC_STOP_DATA to instruct the CFSH to hold off or allow the sending of Data Conveyance messages to the CFSS.

A CFSS method of managing voice and data arbitration on the FSI using these control messages may be done in the following manner:

1. The CFSS sends FSC_STOP_DATA to the CFSH when there is outbound voice on the FSI or repeated voice on the CAI.
2. When the CFSH receives the FSC_STOP_DATA, it completes the transmission of the current data fragment on the FSI (if any) and then stop.

3. The CFSS sends the FSC_START_DATA when outbound voice on the CAI is complete. This message indicates data messages can be sent.

8.4 DFSI Voice Conveyance Service

8.4.1 Voice Conveyance Connection Establishment Protocol Version 2

For Protocol Version 2, the CFSH shall establish the Voice Conveyance connection only after it establishes the Protocol Version 2 Control Connection.

1. Upon detecting a Voice Conveyance connectivity trigger while in the CS *Connected* state, the CFSH shall:
 - a. Send the FSC_CONNECT message indicating Voice Conveyance connection to the CFSS that it is connected to via the Control Connection, applying the retransmission protocol of 8.3.2.3.
2. Upon receipt of the FSC_ACK response to the FSC_CONNECT,
 - a. If the status is CONTROL_ACK the CFSH shall
 - i. Activate the Voice Conveyance connection based on local policy.
 - b. If the status is CONTROL_NAK_P_UNSUPP the CFSH shall:
 - i. Take actions as specified by local policy since protocol version is not relevant to the Voice Conveyance connection.
 - c. If the status is CONTROL_NAK_V_UNSUPP, the CFSH shall retry the connect using the version associated with the response message version.
 - d. If the status is CONTROL_NAK_CONNFALL the CFSH shall take actions as specified by local policy, this response indicates that a Voice Conveyance connection is not supported by the CFSS.
 - e. If the status is CONTROL_NAK_OPT_UNSUPP and the response data indicates the RTP Mode (RM) option and/or Station Decode (SD) option(s) cannot be supported the host shall take one of the following actions:
 - i. Retry the Voice Conveyance connect request without the options(s) that were indicated as not supported by the station

-or-

 - ii. Take actions as specified by local policy for operation without the optional functionality
3. Upon failure of the retransmission protocol on the FSC_CONNECT, the CFSH shall take actions as specified by local policy.
4. Upon receipt of FSC_CONNECT and it is not in the CS *Connected* state and the connection type is Voice Conveyance, the CFSS shall respond with a FSC_ACK with the Response Code of CONTROL_NAK_NO_CONTROL to indicate it cannot start the requested connection because it does not have a Control connection.

5. Upon receipt of FSC_CONNECT and it is in the *CS Connected* state and the connection type is Voice Conveyance and the CFSS does not support a Voice Conveyance connection, the CFSS shall respond with a FSC_ACK with the Response Code of CONTROL_NAK_CONNFAIL to indicate it cannot support the requested connection.
6. Upon receipt of FSC_CONNECT and it is in the *CS Connected* state and the connection type is Voice Conveyance and the source IP address and Port of the FSC_CONNECT is not the same as the FSC_CONNECT that set up the Control Connection, the CFSS shall respond with a FSC_ACK with the Response Code of CONTROL_NAK_CONNECTED to indicate it is connected to a different CFSH.
7. Upon receipt of FSC_CONNECT and it is in the *CS Connected* state and the connection type is Voice Conveyance and the CFSS supports a Voice Conveyance connection and the source IP address and Port of the FSC_CONNECT is the same as the FSC_CONNECT that set up the Control Connection:
 - a. If the CFSS can support the Message Version and the requested voice related optional functionality (e.g. RM =0 (Verbose) and/or SD=1 (Station sends decoded data) in FSC_CONNECT) it shall:
 - i. Reply with FSC_ACK with the CONTROL_ACK Response Code
 - ii. Activate the Voice Conveyance connection based on local policy.
 - b. If the CFSS cannot support the Message Version of the FSC_CONNECT request, it shall:
 - i. Reply with FSC_ACK with the CONTROL_NAK_V_UNSUPP Response Code.
 - c. If the CFSS cannot support one or both of the requested voice related optional functionalities (RM =0 (Verbose) and/or SD=1 (Station sends decoded data) in FSC_CONNECT), it shall:
 - i. Reply with FSC_ACK with the CONTROL_NAK_OPT_UNSUPP Response Code and response data indicating the option(s) it cannot support.
 - ii. Consider the Voice Conveyance Connection not connected.

8.4.2 Voice Conveyance Connection Detachment Procedures Protocol Version 2

1. Upon detecting a Voice Conveyance disconnect trigger while it has an Voice Conveyance connection with a CFSS, the CFSH shall:
 - a. Send the FSC_DISCONNECT message for the Voice Conveyance connection to the CFSS, applying the retransmission protocol of 8.3.2.3.

2. Upon receiving the FSC_DISCONNECT message for the Voice Conveyance connection while it does not have an active Voice Conveyance connection, the CFSS shall respond with FSC_ACK with a Response Code of CONTROL_ACK.
3. Upon receiving a FSC_DISCONNECT message for an active Voice Conveyance connection, the CFSS shall:
 - a. Reply with FSC_ACK with a status code of CONTROL_NAK_CONNECTED if the IP address and port of the sender of the FSC_DISCONNECT do not match those of the sender of the FSC_CONNECT that caused the activation of the Voice Conveyance connection.
 - b. Otherwise,
 - i. Reply with FSC_ACK with a status code of CONTROL_ACK and,
 - ii. Deactivate the Voice Conveyance connection per local policy.
4. Upon receiving a FSC_ACK with a status code of CONTROL_ACK in response to sending a FSC_DISCONNECT for a Voice Conveyance connection to a CFSS, a CFSS shall:
 - a. Deactivate the Voice Conveyance connection per local policy.
 - b. Discontinue the transmission of any and all packets on the Voice Conveyance connection.

8.4.3 Structure of Voice Conveyance Payload

The Voice Conveyance Payload shall consist of a payload header followed by blocks of data (Voice Conveyance Payload) as illustrated in Figure 39. The blocks of data carry P25 voice, signaling that is carried with voice on the P25 CAI and analog transparent audio. The Voice Conveyance Payload header shall indicate which of these blocks are present in the packet.

NOTE - It is assumed that if the Voice Conveyance Payload data is P25 CAI, the associated air interface format is P25 CAI. Similarly, it is assumed that if the Voice Conveyance Payload data is analog transparent audio, the associated air interface format is analog FM. It is a good idea to have an agreement in place between the host and fixed station manufacturers if either of these assumptions is not followed. Without an agreement the host and fixed station might not interoperate if one of them does not follow the previously described assumptions.

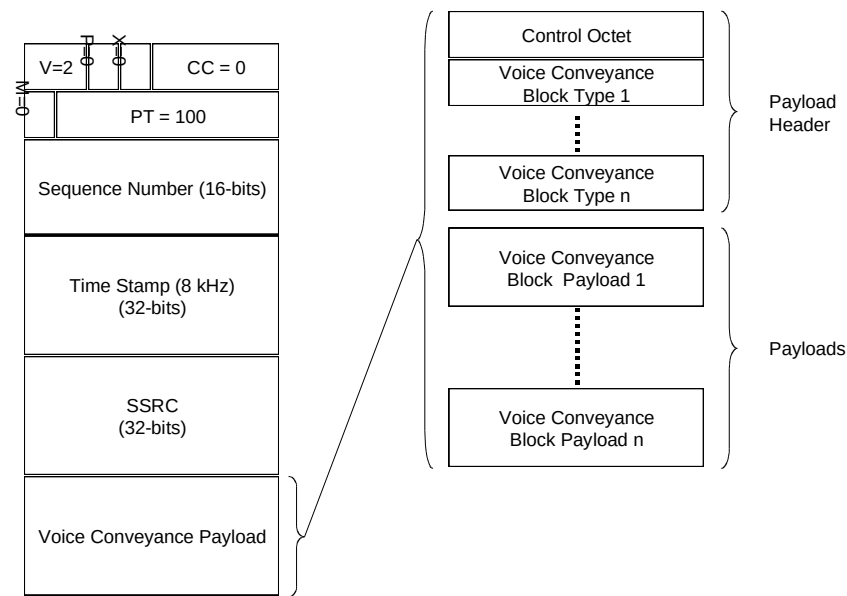


Figure 39 – Overview of DFSI Voice Conveyance Packet

8.4.3.1 Voice Conveyance Payload Header

The P25 RTP payload allows compact (often called compressed) and verbose (often called un-compressed) forms of the Voice Conveyance Payload header. All devices using this revision of the fixed station interface shall support the compact form (indicated by the C-bit = 1, see below). Support of the verbose form is optional. Any device not supporting the option shall discard any packets received with the indicator bit set to verbose form.

The Voice Conveyance Payload header shall consist of a control octet which indicates the number of Voice Conveyance blocks as illustrated in Figure 40 and Figure 41. The control octet will be followed by one or more block headers as shown in and described in 8.4.3.1.2 and 8.4.3.1.3 for the compact and verbose forms respectively.

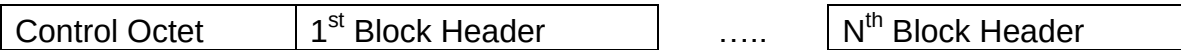
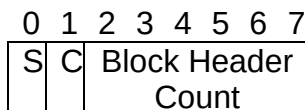


Figure 40 – DFSI Voice Conveyance Payload Header**8.4.3.1.1 Payload Header Control Octet.**

The payload header control octet shall contain 3 fields as illustrated in Figure 41.

**Figure 41 – Control Octet**

The bits of the control octet are defined as follows:

S – Signal bit. Has no special meaning for the DFSI. Any value may be sent. Receivers should ignore this flag.

C – Compact. Indicates the compact payload header.

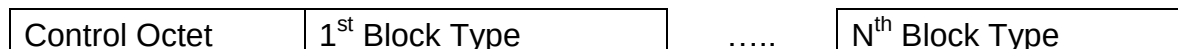
C =0 indicates the following payload headers use the verbose payload format (see Section 8.4.3.1.3).

C =1 indicates the following payload headers use the compact format (see section 8.4.3.1.2)..

Block Header Count – The number of Block Headers following the control octet. This is also the number of payloads in the packet.

8.4.3.1.2 Payload Header Compact Form.

The payload header has the control octet followed by one or more Block Headers. For the compact form, this Block Header is an octet that indicates the types of blocks in the remainder of the payload. This is shown in Figure 42. The Block Type is defined in Section 8.4.3.1.4.

**Figure 42 – DFSI Voice Conveyance Payload Header – Compact form****8.4.3.1.3 Payload Header Verbose Form**

The payload header has the control octet followed by one or more Block Headers. For the verbose form, this Block Header contains 4 octets providing the information shown in Figure 43. This format applies to FSI Protocol Version 2 and above.





Figure 43 – DFSI Voice Conveyance Block Header – Verbose form

In addition to the Block Type octet defined in Section 8.4.3.1.4, the block header contains the following fields.

Timestamp Offset - Each block header contains a 14 bit timestamp offset. This shall be in the same units as the RTP header timestamp. The sum of the RTP timestamp and the offset is the timestamp of the block.

Length - Each block header contains a 10 bit length indicator. This shall be the number of bytes contained in the corresponding payload. If a device receives a block type that it does not know how to process, the length allows it to skip that payload and continue processing the remaining blocks in the packet.

8.4.3.1.4 Voice Block Types

Each Block Header contains a Block Type octet which specifies the payload content. The block headers are followed by the block payloads. Figure 44 illustrates the structure of the block headers and block payloads.

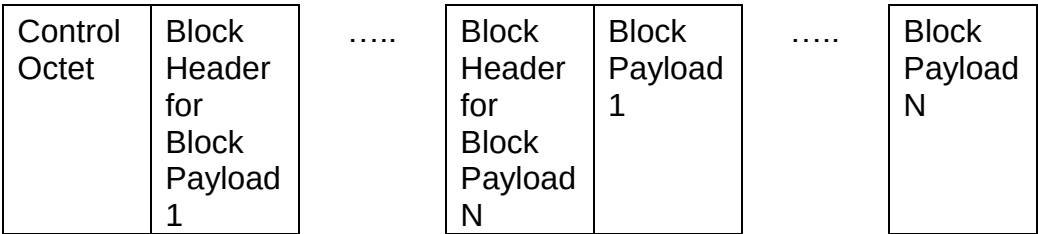


Figure 44 – Structure of Payload Header and Information Blocks

The block type octet is split into two fields, illustrated in Figure 45.

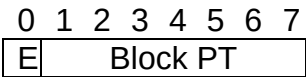


Figure 45 – Format of Block Type Field

E Indicates whether Block PT is a RTP payload type as defined by RFC 3550 ([7]) or a TIA-102 specific payload.

E = 0 Shall mean that block PT indicates a RTP payload defined by reference [7] is encapsulated in this Voice Conveyance Block Payload. The RTP payload types supported in this revision of the fixed station interface shall be: 0, μ law PCM audio. This encoding

scheme is described in RFC 3551 ([18]). The guidelines described in reference [18] are followed except that the payload is encapsulated in a Voice Conveyance Block Payload.

E = 1 Shall mean that the block PT indicates a block type specific to the voice conveyance payload. The block types supported by this version of the DFSI are defined in 8.4.4.

Block PT indicates the type of the corresponding payload. For E=0, Block PT shall be 0, indicating μ law PCM audio. For E=1, Block PT may take on the following values as shown in Table 13:

Table 13 - Block Payload Type FSI Protocol Versions

Payload Type	FSI Protocol Version
0 – CAI Voice	1 and 2
2 – Decoded Link Control Word	2
3 – Decoded Encryption Sync	2
4 – Decoded Low Speed Data	2
5 – Decoded Header Word	2
6 – Voice Header Part 1	1 and 2
7 – Voice Header Part 2	1 and 2
8 – Analog Start of Stream	2
9 – Start of Stream with NID	1 and 2
10 – End of Stream	1 and 2
12 – Receiver Report	1 and 2
13 – Voter Control	1
14 – Stream Acknowledge	1 and 2
20 – Transmitter Report	2
63-127 – Manufacturer Specific	1 and 2

Note: Version 2 means Version 2 and above.

The details of the P25 specific RTP Payloads are defined in 8.4.4

Any device that receives a block it cannot process, whether because it is an optional block that it does not support, it is a block that is not supposed to be on this interface, or it is a block type that is not defined for this version, shall discard the block and, if possible, continue processing the packet.

8.4.4 Voice Conveyance RTP Payload, Block PTs

In addition to the IETF defined Block PT type 0, μ law PCM audio (used with E=0), several more block types are needed by dispatch radio implementations. These are indicated with an E bit of 1. This provides for up to 128 additional Block Payload Types (Block PTs). Only a few Block PTs, described in the next sections, are required for the DFSI.

All of the following specific types are designated as the Block PT. Designation excludes the E bit.

The value of the block type will be equal to 128 + Block PT.

8.4.4.1 Block with PT= 0 – CAI Voice

Blocks with this PT shall contain the P25 Common Air Interface (CAI) Improved Multiband Excitation (IMBE) voice codec data along with additional data that is paired with a particular voice frame on the P25 CAI ([8]).

The IMBE data is divided into 8 decoded message vectors (U0-U7) as described in TIA-102.BABA ([11]). The message vectors U0-U6 have been decoded (error correction code removed). Vector U7 is unprotected on the CAI.

All CAI frames shall contain these 14 octets. Figure 46 defines the message format and content of the common fields for CAI blocks for all FSI Protocol versions.

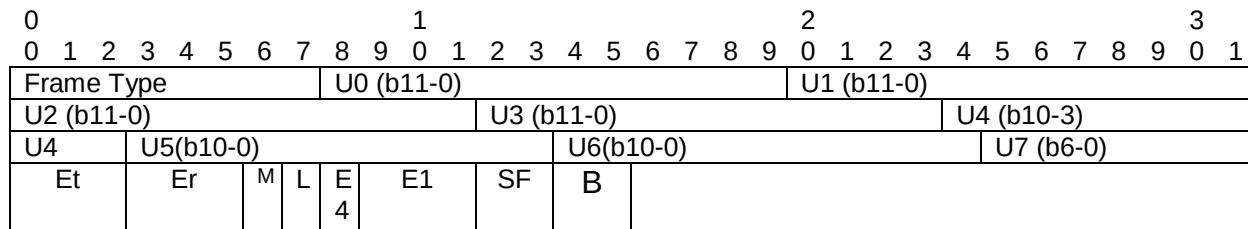


Figure 46 – Common Fields for CAI Blocks

A frame type field indicates which frame in a P25 super frame this block carries. This is necessary because different pieces of additional signaling are carried with each frame. The frame type field is called the PID/FT for Protocol Identifier and Frame Type.

The first (high order) 3 bits of the field shall constitute the PID. The next 5 bits signify a frame type associated with that protocol identifier. For this revision of the fixed station interface PID = 3 shall be supported. Frame type shall have a range of values. Hence the combined octet PID/FT shall have a range of values.

The composition of the blocks with each PID/FT shall be as follows in Table 14.

Table 14 – CAI Frame Types

Name	PID/FT	Description
IMBEFRAME1	\$62	IMBE Voice 1
IMBEFRAME2	\$63	IMBE Voice 2
IMBEFRAME3	\$64	IMBE Voice 3 + Link Control
IMBEFRAME4	\$65	IMBE Voice 4 + Link Control
IMBEFRAME5	\$66	IMBE Voice 5 + Link Control
IMBEFRAME6	\$67	IMBE Voice 6 + Link Control
IMBEFRAME7	\$68	IMBE Voice 7 + Link Control
IMBEFRAME8	\$69	IMBE Voice 8 + Link Control
IMBEFRAME9	\$6A	IMBE Voice 9 + Low Speed Data
IMBEFRAME10	\$6B	IMBE Voice 10
IMBEFRAME11	\$6C	IMBE Voice 11
IMBEFRAME12	\$6D	IMBE Voice 12 + Encryption Sync
IMBEFRAME13	\$6E	IMBE Voice 13 + Encryption Sync
IMBEFRAME14	\$6F	IMBE Voice 14 + Encryption Sync
IMBEFRAME15	\$70	IMBE Voice 15 + Encryption Sync
IMBEFRAME16	\$71	IMBE Voice 16 + Encryption Sync
IMBEFRAME17	\$72	IMBE Voice 17 + Encryption Sync
IMBEFRAME18	\$73	IMBE Voice 18 + Low Speed Data

The report count fields shall reflect the number of decoding errors detected when the over the air frame was recovered at a receiver and the error correcting code (Golay or Hamming) was decoded. These may be used for error mitigation when the voice is decoded. If the number of errors detected is greater than can be indicated in the report field, the field shall be set to the maximum value.

Et shall be the total number of errors detected in the frame. This shall be the parameter ε_T described by equation 95 in TIA-102.BABA ([11]) and limited to a maximum value of 7.

Er is a scaled version of ε_R described in equation 96 of reference [11]. Er shall be equal to $400 \times \varepsilon_R$ rounded down to the nearest integer and limited to a maximum of 7.

M is the mute frame flag, and L is the lost frame flag. The originating RFSS MAY set M and/or L values for any given frame based upon errors described in sections 7.7 and 7.8 of [11], or in accordance with local policy. When M is set to 1 and L is set to 1 the audio output can be muted. When M is set to 0 and L is set to 1 the previous audio output can be repeated.

E4 shall be the number of errors detected in vector U4, limited to a maximum of 1.

E1 shall be the number of errors detected in vector U1, limited to a maximum of 7.

SF indicates a 2-bit free running super frame counter that is incremented on the first frame of each super frame. For this issue of the fixed station interface, this behavior is to maintain consistency with the ISSI.

B indicates the busy status field as defined in TIA-102.BAAA-A ([8]).

In addition to the above 14 octets present on all IMBE frames, each of the following frame types specifies zero or more additional bytes that are appended to the end of the frame.

8.4.4.1.1 CAI Frames 1, 2, 10, and 11.

CAI Frames 1, 2, 10 and 11 have frame type fields \$62, \$63, \$6B, \$6C respectively. These frames shall contain only the basic 14 octets with no additional data. Figure 47 defines the message format and content for CAI frames 1, 2, 10 and 11 for all FSI Protocol versions..

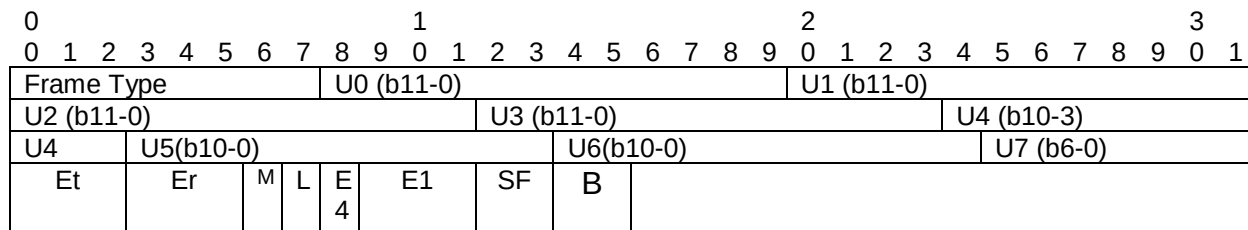


Figure 47 – Format for CAI Frames 1, 2, 10, and 11

8.4.4.1.2 CAI Frames 3 to 8 with Link Control Data

CAI frames 3 to 8 have frame type fields \$64 – \$69 respectively. They include an additional 4 bytes to carry Link Control data.

The Link Control field of a block shall carry four of the 24 RS encoded 6-bit link control code words as described in section 5.5 of TIA-102.BAAA-A ([8]). Refer to TIA-102.AABF-C ([10]) for detailed discussions of Link Control definitions.

The link control field shall contain code words:

0, 1, 2, and 3 for CAI frame 3;
 4, 5, 6, and 7 for CAI frame 4;
 8, 9, 10, and 11 for CAI frame 5;
 12, 13, 14, and 15 for CAI frame 6;
 16, 17, 18, and 19 for CAI frame 7; or
 20, 21, 22 and 23 for CAI frame 8.

The 7-bit STATUS field indicates the number of bit errors detected when the over the air Hamming code was decoded. The STATUS field is defined as follows:

$$\text{STATUS} = \text{ST3} * 3^3 + \text{ST2} * 3^2 + \text{ST1} * 3^1 + \text{ST0} * 3^0$$

Where

STn=error status of LCn

STn=0 means no errors

STn=1 means 1 error

STn=2 means erasure

Again, the “R” bits are reserved. Figure 48 defines the message format and content of the CAI Frame with a Link Control field for all FSI Protocol versions.

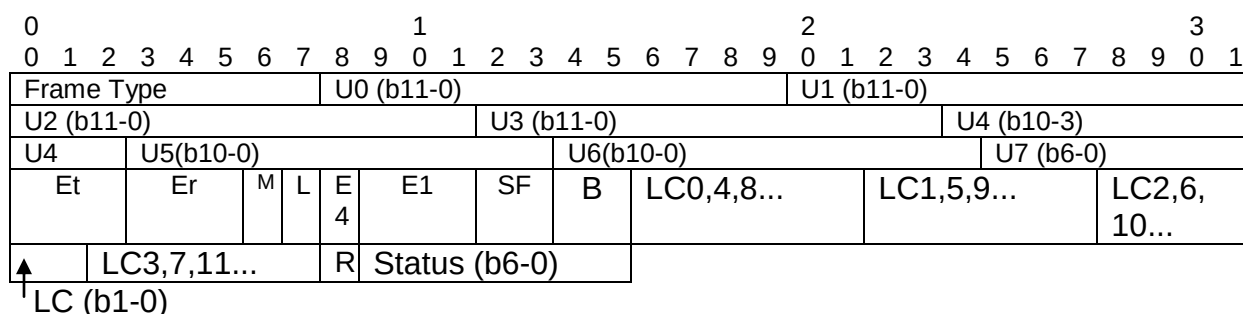


Figure 48 – CAI block with Link Control Word Field

8.4.4.1.3 CAI Frame 9 and 18 with Low Speed Data.

Frames 9 and 18 have Frame Type fields of \$6A and \$73 respectively. This field of a block shall contain two octets of low speed data, as described in section 5.6 of TIA-102.BAAA-A ([8]), either:

Octets 0 and 1 (CAI frame 9) or
Octets 2 and 3 (CAI frame 18).

The low speed data is accompanied by two 2-bit status words to indicate the number of errors decoded at the receiver. The binary values of this field are defined as follows:

00 = no errors

01 = 1 error

02 = 2 errors

03 = erasure – indicates that the decoder detected an uncorrectable combinations of errors in the code word.

Si – Status field S0 (CAI Frame 9), Status field S2 (CAI Frame 18)

Sj – Status field S1 (CAI Frame 9), Status field S3 (CAI Frame 18)

Figure 49 defines the message format and content of the CAI frame with a Low speed Data field for all FSI Protocol versions.

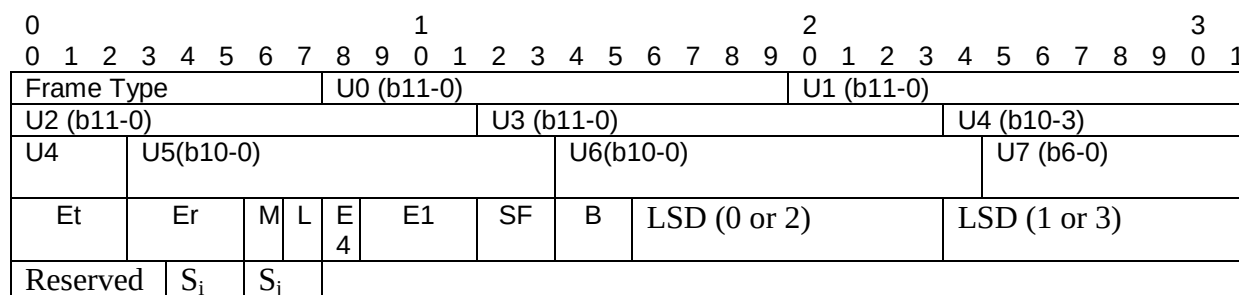


Figure 49 – CAI Block with a Low Speed Data Field

8.4.4.1.4 CAI Frames 12 to 17 with Encryption Sync

Frames 12 to 17 have Frame Type fields of \$6D to \$72 respectively and carry four of the 24 RS encoded 6-bit encryption sync code words described in section 5.4 of TIA-102.BAAA-A ([8]). The encryption sync field contains code words:

0,1,2 and 3 for Frame 12;
 4, 5, 6 and 7 for Frame 13;
 8, 9, 10 and 11 for Frame 14;
 12, 13, 14 and 15 for Frame 15;
 16, 17, 18 and 19 for Frame 16; or
 20, 21, 22 and 23 for Frame 17.

The 7-bit status field indicates the number of bit errors detected when the over the air Hamming code was decoded.

The STATUS field is defined as follows:

$$\text{STATUS} = \text{ST3} * 3^3 + \text{ST2} * 3^2 + \text{ST1} * 3^1 + \text{ST0} * 3^0$$

Where:

STn=error status of ESn

STn=0 means no errors

STn=1 means 1 error

STn=2 means erasure

Figure 50 defines the message format and content of the CAI frame with an Encryption Sync field for all FSI Protocol versions.

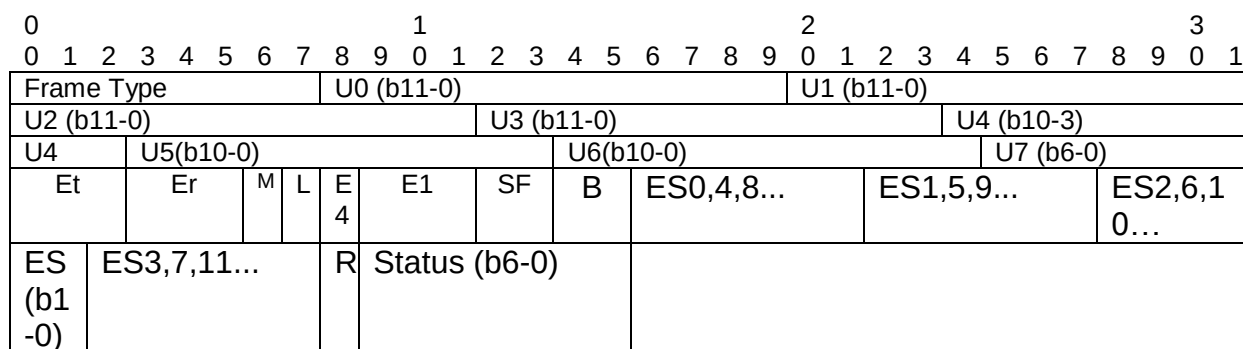
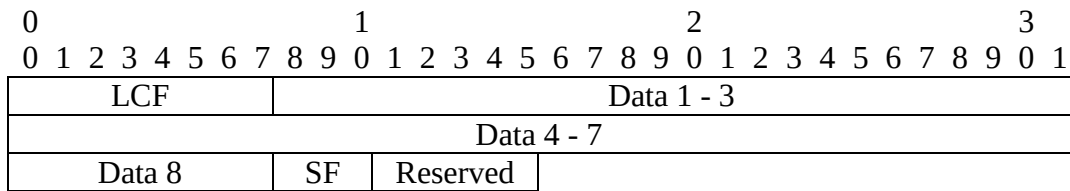


Figure 50 – CAI Blocks with Encryption Sync Field

8.4.4.2 Block with PT = 2 – Decoded Link Control Word

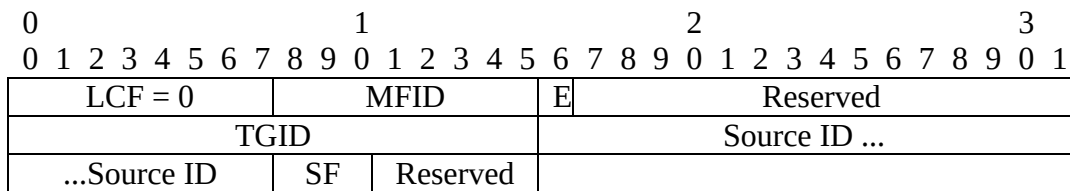
This is a complex subtype which may have any link control format (LCF) specified by TIA-102.BAAA-A [8]. The link control data consists of all 12 link control code words decoded and broken into constituent data fields. Refer to reference TIA-102.BAAA-A [8] for more details. Figure 51 defines the message format and content of the Decoded Link Control block for FSI Protocol Version 2 and above.

**Figure 51 – General Format of Link Control Word Block Payload**

LCF – Link Control Format, See TIA-102.BAAA-A [8] or TIA-102.AABF-C [10]
 Data – Contents of Link control. LCF dependent.

SF - indicates a 2-bit free running super frame counter that is incremented on the first frame of each super frame. For this issue of the fixed station interface, this behavior is to maintain consistency with the ISSI.

Figure 52 and Figure 53 illustrate examples of the group call and unit to unit Call Block Type 2 link control formats.

**Figure 52 – Example Format of Link Control Word Block Payload (LCF= 0)**

LCF – Link Control Format. 0 = Group Call

MFID – Manufacturer ID

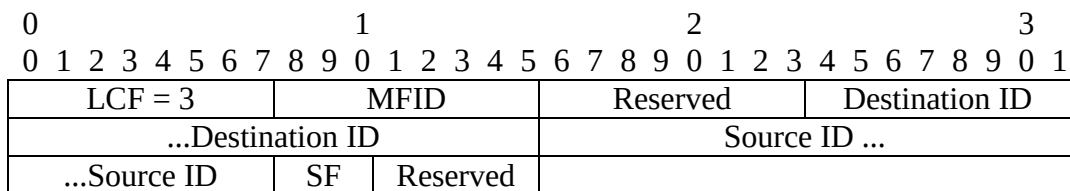
E – Emergency

TGID – Talk Group ID

Source ID – SUID of the talker

SF - indicates a 2-bit free running super frame counter that is incremented on the first frame of each super frame. For this issue of the fixed station interface, this behavior is to maintain consistency with the ISSI.

-OR-

**Figure 53 – Example Format of Link Control Word Block Payload (LCF= 3)**

LCF – Link Control Format. 3 = Unit Call

MFID – Manufacturer ID

Destination ID – SUID the call is directed to

Source ID – SUID of the talker

SF - indicates a 2-bit free running super frame counter that is incremented on the first frame of each super frame. For this issue of the fixed station interface, this behavior is to maintain consistency with the ISSI.

8.4.4.3 Block with PT = 3 – Decoded Encryption Sync Word

The Encryption Sync data consists of all 16 encryption sync code words decoded and broken into constituent data fields. Refer to reference TIA-102.BAAA-A [8] for more details. Figure 54 defines the message format and content of the Decoded Encryption Sync block for FSI Protocol Version 2 and above.

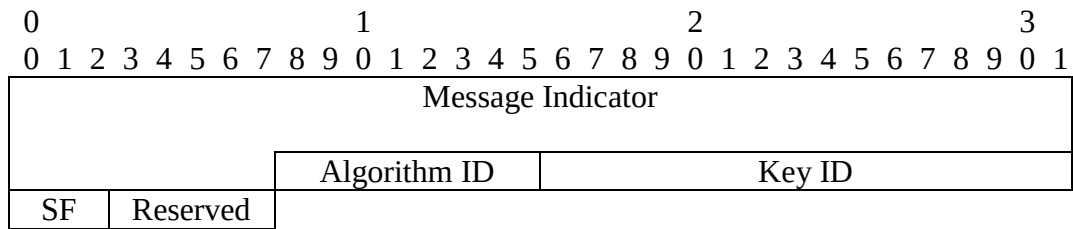


Figure 54 –Format of Encryption Sync Word Block Payload

Message Indicator – Encryption sync for the next super frame
Algorithm ID – Identifier of the algorithm used for encryption
Key ID – identifier of the key used for encryption
SF - indicates a 2-bit free running super frame counter that is incremented on the first frame of each super frame. For this issue of the fixed station interface, this behavior is to maintain consistency with the ISSI.

8.4.4.4 Block with PT = 4 – Decoded Low Speed Data

The low speed data consists of 2 decoded low speed data code words. The P25 logical link data unit (LDU) indicator (“L”) is set to 0 to indicate LDU1, 1 for LDU2. Refer to reference TIA-102.BAAA-A [8] for details on P25 low speed data and LDUs. Figure 55 defines the message format and content of the Decoded Low Speed Data block for FSI Protocol Version 2 and above.

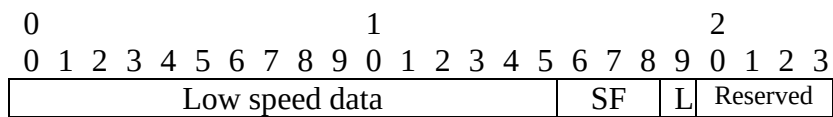


Figure 55 –Format of Low Speed Data Block Payload

Low speed data – user defined data carried with the voice stream
SF - indicates a 2-bit free running super frame counter that is incremented on the first frame of each super frame. For this issue of the fixed station interface, this behavior is to maintain consistency with the ISSI.
L – LDU 0 = LDU1, 1 = LDU2

8.4.4.5 Block with PT = 5 – Decoded Header Word

The Header word data consists of all 36 voice header data code words decoded and broken into constituent data fields. Refer to reference TIA-102.BAAA-A [8] for more details. Figure 56 defines the message format and content of the Decoded Header block for FSI Protocol Version 2 and above.

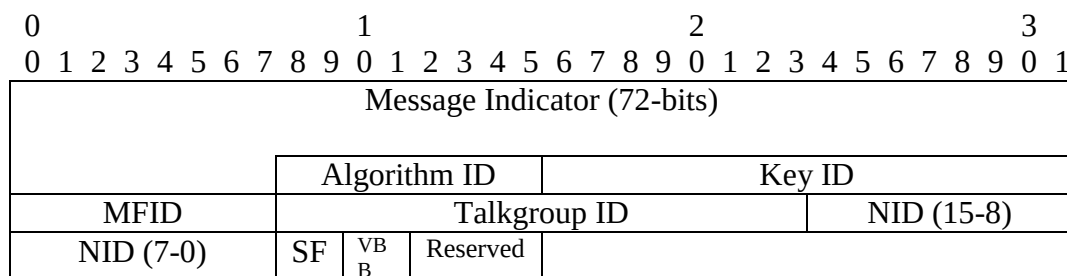


Figure 56 –Format of Decoded Header Block Payload

Message Indicator – Encryption sync for the next super frame

Algorithm ID – Identifier of the algorithm used for encryption

Key ID – Identifier of the key used for encryption

MFID – Manufacturer ID

Talkgroup ID – ID of the talking group

NID – NAC / DUID of the call

SF - Indicates a 2-bit free running super frame counter that is incremented on the first frame of each super frame. For this issue of the fixed station interface, this behavior is to maintain consistency with the ISSI.

VBB – Defined in TIA-102.BACA-A [5]. Not used in this interface.

8.4.4.6 Block with PT = 6 – Voice Header Part 1

Blocks of this type are sent at the start of a P25 voice stream. The contents shall be the information bits of the first 18 Header Data Unit Golay code words on the air interface.

If the fixed station subsystem receives a P25 voice stream from the air interface without a header data unit (such as due to an RF fade) it may send this voice stream on the fixed station interface without any voice header blocks. In this case the receiver of the DFSI voice stream may extract the header data from the link control and encryption sync words carried with the P25 voice stream.

The block also contains 3-bit status fields for each of the code words to indicate the number of errors detected in decoding the Golay code. The 3-bit status fields are split, with the two most significant bits sharing the same byte as the associated code word and the least significant bit lumped into one of three other bytes with the LSB's from other code words.

Fixed stations shall transmit within a Header code data unit on the air interface, the contents of a block of this type received on the DFSI.

For more details of the header fields see TIA-102.BAAA-A ([8]).

The frame type for a block with this PT shall be \$60.

Figure 57 defines the message format and content of the CAI Voice Header Part 1 block for all FSI Protocol versions.

0										1										2										3									
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1								
Frame Type = \$60										S0		G0				S1		G1				S2		G2															
S3		G3								S4		G4				S5		G5				S6		G6															
S7		G7								S8		G8				S9		G9				S10		G10															
S11		G11								S12		G12				S13		G13				S14		G14															
S15		G15								S16		G16				S17		G17				Reserved				S17		S16											
S15...S0																																							

Figure 57 – CAI Voice Header Part 1

8.4.4.7 Block with PT = 7 – Voice Header Part 2

Blocks of this type are sent following the Voice Header Part 1 block (PT = 6). The contents shall be the information bits of the last 18 Header Data Unit Golay code words on the air interface.

If the fixed station subsystem receives a P25 voice stream from the air interface without a header data unit (such as due to an RF fade) it may send this voice stream on the fixed station interface without any voice header blocks. In this case the receiver of the FSI voice stream may extract the header data from the link control and encryption sync words carried with the P25 voice stream. It also contains 3-bit status fields for each of the code words to indicate the number of errors detected in decoding the Golay code. The 3-bit status fields are split, with the two most significant bits sharing the same byte as the associated code word and the least significant bit lumped into one of three other bytes with the LSB's from other code words.

Fixed stations shall transmit within a Header code data unit on the air interface, the contents of a block of this type received on the fixed station interface.

For more details of the header fields see TIA-102.BAAA-A ([8]).

The frame type for blocks with this PT shall be \$61.

Figure 58 defines the message format and content of the CAI Voice Header Part 2 block for all FSI Protocol versions.

0										1										2										3									
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1								
Frame Type = \$61										S18	G18					S19	G19					S20	G20																
S21	G21					S22	G22					S23	G23					S24	G24																				
S25	G25					S26	G26					S27	G27					S28	G28																				
S29	G29					S30	G30					S31	G31					S32	G32																				
S33	G33					S34	G34					S35	G35					Reserved										S3	S3										
																																			5	4			
S33...S18																																							

Figure 58 – CAI Voice Header Part 2

8.4.4.8 Block with PT = 8 –Analog Start of Stream

Fixed stations and Hosts shall be capable of sending blocks of this type on the fixed station interface for Protocol Version 2 and above. This block type is used for analog transmissions with and without CTCSS/CDCSS in Protocol Version 2 and above.

Analog Streams use squelch codes to qualify the stream analogous to the NAC or talk group of the digital stream.

Figure 59 defines the message format and content of the Analog Start of Stream block for FSI Protocol Version 2 and above.

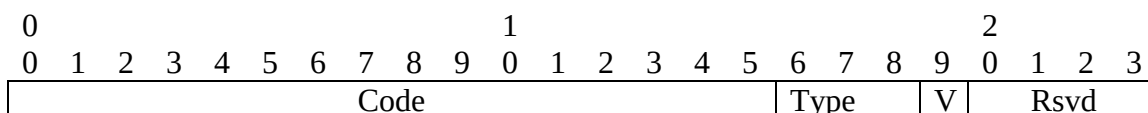


Figure 59 - Start of Stream Block

This block explicitly identifies the start of a stream.

Code – Squelch Code. For CTCSS, the value is the frequency in Decihertz.

This is simply ten times the frequency in Hertz, eliminating the decimal point. For CDCSS, the code is the actual CDCSS value. All values are represented in binary format.

Type – Type of stream that is starting.

1 indicates CTCSS,

2 indicates CDCSS,

3 indicates Carrier squelch

V – Valid. Applies to Analog codes. Bit is set if the code is valid and clear if the code is unknown. If a FSS detects a code, but cannot determine what that code is, but it is programmed to pass all audio to the host, it sets the valid bit to false to indicate to the FSH that the code for this stream could not be detected.

FS and Host using FSI Protocol Version 1 do not use this message. They use Block with PT = 9. The Block with PT = 9 does not define the contents of the fields in the Start of Stream block for analog streams. It depends on the accompanying audio blocks to indicate the stream type. When using protocol version 1, the squelch code, if provided, is configured at the station as the protocol does not define a means to carry the code over the FSI.

This block may also be used to convey the Code information throughout the stream. If used in that manner, the Code shall correspond to the data unit or portion thereof contained in the block that immediately follows the Start of Stream block.

Table 15 shows the known valid squelch codes. The mapping allows any CTCSS frequency or CDCSS code can be represented, even if not represented in the table.

Table 15 - Valid Squelch Codes

Type	Valid Codes								
1 (CTCSS)	Code (binary)	Code (decimal)	CTCSS Frequency	Code (binary)	Code (decimal)	CTCSS Frequency	Code (binary)	Code (decimal)	CTCSS Frequency
	0	0	None	10000001011	1035	103.5 Hz	11011001010	1738	173.8 Hz
	10100111110	670	67.0 Hz	10000110000	1072	107.2 Hz	11100000111	1799	179.9 Hz
	1010110101	693	69.3 Hz	10001010101	1109	110.9 Hz	11101000110	1862	186.2 Hz
	1011001111	719	71.9 Hz	10001111100	1148	114.8 Hz	11110001000	1928	192.8 Hz
	1011101000	744	74.4 Hz	10010100100	1188	118.8 Hz	11111110011	2035	203.5 Hz
	1100000010	770	77.0 Hz	10011001110	1230	123.0 Hz	100000010001	2065	206.5 Hz

	<u>1100011101</u>	<u>797</u>	79.7 Hz	<u>10011111001</u>	<u>1273</u>	127.3 Hz	<u>100000111011</u>	<u>2107</u>	210.7 Hz
	<u>1100111001</u>	<u>825</u>	82.5 Hz	<u>10100100110</u>	<u>1318</u>	131.8 Hz	<u>100010000101</u>	<u>2181</u>	218.1 Hz
	<u>1101010110</u>	<u>854</u>	85.4 Hz	<u>10101010101</u>	<u>1365</u>	136.5 Hz	<u>100011010001</u>	<u>2257</u>	225.7 Hz
	<u>1101110101</u>	<u>885</u>	88.5 Hz	<u>10110000101</u>	<u>1413</u>	141.3 Hz	<u>100011110011</u>	<u>2291</u>	229.1 Hz
	<u>1110010011</u>	<u>915</u>	91.5 Hz	<u>10110110110</u>	<u>1462</u>	146.2 Hz	<u>100100100000</u>	<u>2336</u>	233.6 Hz
	<u>1110110100</u>	<u>948</u>	94.8 Hz	<u>10111101010</u>	<u>1514</u>	151.4 Hz	<u>100101110010</u>	<u>2418</u>	241.8 Hz
	<u>1111001110</u>	<u>974</u>	97.4 Hz	<u>11000011111</u>	<u>1567</u>	156.7 Hz	<u>100111000111</u>	<u>2503</u>	250.3 Hz
	<u>1111101000</u>	<u>1000</u>	100.0 Hz	<u>11001010110</u>	<u>1622</u>	162.2 Hz	<u>100111101101</u>	<u>2541</u>	254.1 Hz
				<u>11010001111</u>	<u>1679</u>	167.9 Hz			
2 (CDCSS)	<u>Binary</u>	<u>Octal</u>	<u>Decimal</u>	<u>Binary</u>	<u>Octal</u>	<u>Decimal</u>	<u>Binary</u>	<u>Octal</u>	<u>Decimal</u>
	<u>10011</u>	<u>23</u>	19	<u>10010011</u>	<u>223</u>	147	<u>100100101</u>	<u>445</u>	293
	<u>10101</u>	<u>25</u>	21	<u>10010101</u>	<u>225</u>	149	<u>100100110</u>	<u>446</u>	294
	<u>10110</u>	<u>26</u>	22	<u>10010110</u>	<u>226</u>	150	<u>100101010</u>	<u>452</u>	298
	<u>11001</u>	<u>31</u>	25	<u>10100011</u>	<u>243</u>	163	<u>100101100</u>	<u>454</u>	300
	<u>11010</u>	<u>32</u>	26	<u>10100100</u>	<u>244</u>	164	<u>100101101</u>	<u>455</u>	301
	<u>11110</u>	<u>36</u>	30	<u>10100101</u>	<u>245</u>	165	<u>100110010</u>	<u>462</u>	306
	<u>100011</u>	<u>43</u>	35	<u>10100110</u>	<u>246</u>	166	<u>100110100</u>	<u>464</u>	308
	<u>100111</u>	<u>47</u>	39	<u>10101001</u>	<u>251</u>	169	<u>100110101</u>	<u>465</u>	309
	<u>101001</u>	<u>51</u>	41	<u>10101010</u>	<u>252</u>	170	<u>100110110</u>	<u>466</u>	310
	<u>101011</u>	<u>53</u>	43	<u>10101101</u>	<u>255</u>	173	<u>101000011</u>	<u>503</u>	323
	<u>101100</u>	<u>54</u>	44	<u>10110001</u>	<u>261</u>	177	<u>101000110</u>	<u>506</u>	326
	<u>110101</u>	<u>65</u>	53	<u>10110011</u>	<u>263</u>	179	<u>101001110</u>	<u>516</u>	334
	<u>111001</u>	<u>71</u>	57	<u>10110101</u>	<u>265</u>	181	<u>101010011</u>	<u>523</u>	339
	<u>111010</u>	<u>72</u>	58	<u>10110110</u>	<u>266</u>	182	<u>101010110</u>	<u>526</u>	342
	<u>111011</u>	<u>73</u>	59	<u>10111001</u>	<u>271</u>	185	<u>101011010</u>	<u>532</u>	346
	<u>111100</u>	<u>74</u>	60	<u>10111100</u>	<u>274</u>	188	<u>101100110</u>	<u>546</u>	358
	<u>1001100</u>	<u>114</u>	76	<u>11000110</u>	<u>306</u>	198	<u>101110101</u>	<u>565</u>	373
	<u>1001101</u>	<u>115</u>	77	<u>11001001</u>	<u>311</u>	201	<u>110000110</u>	<u>606</u>	390
	<u>1001110</u>	<u>116</u>	78	<u>11001101</u>	<u>315</u>	205	<u>110001010</u>	<u>612</u>	394
	<u>1010010</u>	<u>122</u>	82	<u>11010101</u>	<u>325</u>	213	<u>110010100</u>	<u>625</u>	404
	<u>1010101</u>	<u>125</u>	85	<u>11011001</u>	<u>331</u>	217	<u>110010111</u>	<u>627</u>	407
	<u>1011001</u>	<u>131</u>	89	<u>11011010</u>	<u>332</u>	218	<u>110011001</u>	<u>631</u>	409
	<u>1011010</u>	<u>132</u>	90	<u>11100011</u>	<u>343</u>	227	<u>110011010</u>	<u>632</u>	410
	<u>1011100</u>	<u>134</u>	92	<u>11100110</u>	<u>346</u>	230	<u>110100101</u>	<u>645</u>	421
	<u>1100011</u>	<u>143</u>	99	<u>11101001</u>	<u>351</u>	233	<u>110101100</u>	<u>654</u>	428
	<u>1100101</u>	<u>145</u>	101	<u>11101110</u>	<u>356</u>	238	<u>110110010</u>	<u>662</u>	434
	<u>1101010</u>	<u>152</u>	106	<u>11110100</u>	<u>364</u>	244	<u>110110100</u>	<u>664</u>	436
	<u>1101101</u>	<u>155</u>	109	<u>11110101</u>	<u>365</u>	245	<u>111000011</u>	<u>703</u>	451
	<u>1101110</u>	<u>156</u>	110	<u>11111001</u>	<u>371</u>	249	<u>111001010</u>	<u>712</u>	458
	<u>1110010</u>	<u>162</u>	114	<u>100001001</u>	<u>411</u>	265	<u>111010011</u>	<u>723</u>	467
	<u>1110101</u>	<u>165</u>	117	<u>100001010</u>	<u>412</u>	266	<u>111011001</u>	<u>731</u>	473
	<u>1111010</u>	<u>172</u>	122	<u>100001011</u>	<u>413</u>	267	<u>111011010</u>	<u>732</u>	474
	<u>1111100</u>	<u>174</u>	124	<u>100010011</u>	<u>423</u>	275	<u>111011100</u>	<u>734</u>	476
	<u>10000101</u>	<u>205</u>	133	<u>100011001</u>	<u>431</u>	281	<u>111100011</u>	<u>743</u>	483
	<u>10001010</u>	<u>212</u>	138	<u>100011010</u>	<u>432</u>	282	<u>111101100</u>	<u>754</u>	492

8.4.4.9 Block with PT = 9 – Start of Stream with NID

Fixed stations and Hosts shall be capable of sending blocks of this type on the fixed station interface. It shall contain the NID of the first data unit that the fixed station receives in a stream. This block type is used for both analog and digital transmissions in protocol Version 1, which does not support carrying CTCSS/CDCSS codes. It is used for digital transmissions in protocol Version 2 and above.

This block explicitly identifies the start of a stream.

NID – Comprised of the NAC and DUID. See TIA/-102.BAAA ([8]) for details.

When receiving an analog transmission the NID shall be zero (0).

Err Cnt – 4-bit count of the number of errors detected in the NID.

Figure 60 defines the message format and content of the Start of Stream block which is used by all FSI Protocol versions.

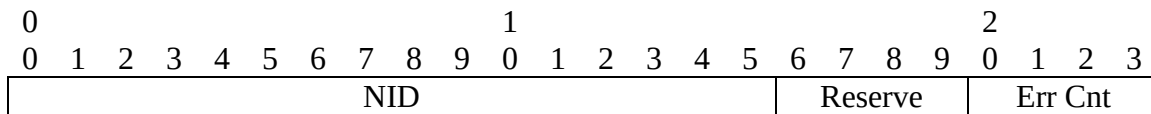


Figure 60 – Start of Stream Block

This block may also be used to convey the NID information throughout the stream. If used in that manner, the NID shall correspond to the data unit or portion thereof contained in the block that immediately follows the Start of Stream block.

8.4.4.10 Block with PT = 10 – End of Stream

Fixed stations shall send a block of this type when they detect the end of a voice stream on the air interface. This packet type explicitly identifies the end of a stream. No data apart from the RTP block header is defined. The FS & Host shall send a block of this type at the end of every voice stream for all protocol versions.

8.4.4.11 Block with PT = 11 – PTT Control Word

Blocks of this type are not supported for this version of the fixed station interface.

8.4.4.12 Block with PT = 12 – Receiver Report

This block is provided so that the fixed station subsystem can report the receiver status for each receiver. If multiple receivers are reported, multiple blocks may be included in one packet. Voter Reports are only sent on the inbound DFSI. Figure 61 defines the message format and content of the Receiver Report block for all FSI Protocol versions.

0										1					
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5
Receiver Number									D	Receiver Status					

Figure 61 – Receiver Report Block

Receiver number 0 is reserved for use in Block with PT = 13 – Voter Control. Receiver numbers 1 to 255 indicate the receiver that the report is for. Mapping of the receiver number to receiver names and physical location are determined by equipment configuration parameters.

D indicates if this receiver is Disabled.

D=0 means that the receiver is not disabled.

D=1 means that this receiver is inhibited because of a DISABLE command for this number (or all receivers) or a SELECT command for another receiver number.

Table 16 shows receiver status voter report code meanings.

Table 16 – Receiver Status Receiver Report Codes

Code	Mnemonic	Meaning
0	NO_SIGNAL	No usable signal is being received by this receiver.
1	SELECTED	This receiver is currently selected. The audio being sent originates from this receiver.
2	GOOD_P25	A P25 signal with the correct NAC being received by this receiver. This receiver is not currently selected by the voter.
3	GOOD_FM	An FM signal with the correct sub-audible signal (if applicable) is being received by this receiver. This receiver is not currently selected.
4	BAD_P25	A P25 signal is being received by this receiver but the NAC does not match the desired NAC.
5	BAD_FM	An analog FM signal is being received by this receiver but the desired sub-audible signal has not been detected.
6	NOT_EQUIPPED	No receiver is equipped for this number.
7	FAILED	The receiver with this number has failed.

8.4.4.13 Block with PT = 13 – Voter Control

This block is only used with FSI Protocol Version 1. Voter Control for FSI Protocol Version 2 and above shall be accomplished via the control plane message defined in section 8.3.1.12.

Voter control blocks are sent on the outbound fixed station interface. Figure 62 defines the Voter Control block message format and content for FSI Protocol Version 1.

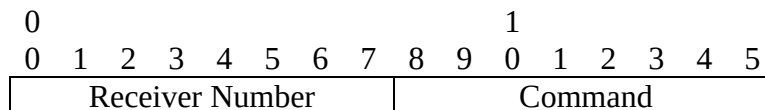


Figure 62 – Voter Control Block

Receiver number = 0 indicates that this command applies to all receivers. Receiver numbers 1 to 255 indicate the receiver that this command relates to. The receiver number used here shall be same as the receiver number used for the same physical receiver in Block with PT = 12 – Receiver Report.

Table 17 shows the meanings of the voter control command.

Table 17 – Voter Control Command

Code	Mnemonic	Meaning
0	REPORT	Send the status of this receiver now.
1	NORMAL	Forward speech from the receiver that has the best signal. This command cancels any SELECT or DISABLE command. Receiver number is not relevant to this command. Receiver number = 0 should be used with this command.
2	SELECT	Forward this receiver's speech regardless of how good the signal is. This command disables all other receivers. This command cannot be used with Receiver Number = 0.
3	DISABLE	Don't forward this receiver's speech regardless of how good it is.
4	ENABLE	Forward speech from this receiver if it has the best signal. This command cancels a DISABLE command for the specified Receiver. It has no effect on any SELECT command that may be in effect.

8.4.4.14 Block with PT = 14 Stream Acknowledge

This is sent on the inbound path by the FS for all Protocol versions indicating to the CFSH that the fixed station has started transmitting on the CAI. This is sent by the host on the outbound path for Protocol Version 2 and above to indicate that the NAC or Squelch Code has been received. No data apart from the RTP block header is defined.

8.4.4.15 Block with PT = 20 Transmitter Report

This block is provided so that the fixed station subsystem can report the status for each transmitter. If multiple transmitters are reported, multiple blocks may be included in one packet. Transmitter Reports are only sent on the inbound DFSI. Figure 63 defines the message format and content of the Transmitter Report block for FSI Protocol Version 2 and above.

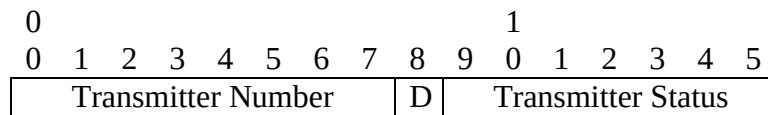


Figure 63 - Transmitter Report Block

Transmitter number 0 is reserved for use in Transmitter Control.

Transmitter numbers 1 to 255 indicate the transmitter that the report is for. Mapping of the transmitter number to transmitter names and physical location are determined by equipment configuration parameters.

D indicates if this transmitter is Disabled

D=0 means that the transmitter is not disabled.

D=1 means that this transmitter is inhibited because of a DISABLE command for this number (or all transmitters) or a SELECT command for another transmitter number.

Table 18 shows transmitter report status code meanings.

Table 18 - Transmitter Status Report Codes

Code	Mnemonic	Meaning
0	NO_SIGNAL	This transmitter is not currently transmitting.
1	SELECTED	This transmitter is currently selected. The audio being sent terminates on this transmitter.
6	NOT_EQUIPPED	No transmitter is equipped for this number.
7	FAILED	The transmitter with this number has failed.

8.4.4.16 Block PT = 63 to 127 – Manufacturer specific data

These packet types are provided so individual manufacturers can carry additional proprietary data within a stream. Equipment conforming to this document may include such blocks however they shall follow any standardized blocks which are sent in the same packet.

Equipment conforming to this document shall ignore blocks of these types whose MFID they do not understand. However equipment conforming to this document shall appropriately process any standardized blocks which precede a manufacturer specific block in the same packet.

The length field specifies the number of octets in this block following the length field. Beyond the mandatory P25 manufacturer’s identification (MFID) field and the length, the format of the internals of this packet are left open for any use. Manufacturers are encouraged to propose packet types, which may be of general use, to the TIA for inclusion in this profile. Figure 64 illustrates the format for manufacturer customizations.

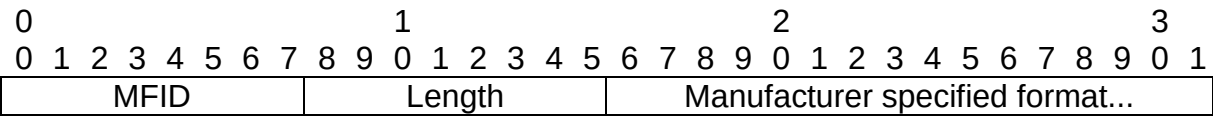


Figure 64 – Format for Manufacturer Customizations

8.4.5 Voice Conveyance Service Procedures

8.4.5.1 Connection of the Voice Conveyance Service.

8.4.5.1.1 Not Connected State

When the control service for a CFSH – Fixed Station Subsystem pair is in the not-connected state the voice conveyance service for that CFSH – Fixed Station Subsystem pair shall also be in the not connected state.

No voice conveyance packets are exchanged between a CFSH – Fixed Station Subsystem pair when they are in the not connected state.

The process of entering and leaving the connected state is described in 8.3.

8.4.5.1.2 Connected State

When the control service for a CFSH – Fixed Station Subsystem pair is in the connected state the voice conveyance service for that pair shall also be in the connected state.

The process of entering and leaving the connected state is described in 8.3.

When a CFSH – Fixed Station Subsystem pair are in the connected state they shall exchange voice conveyance packets according to the following procedures described in 8.4.5.2 through 8.4.5.8.

The setting of IP addresses, SSRC and UDP port number are described in 8.3.

8.4.5.2 Receiver Control

The procedures described in this subclause apply to all FSI Protocol versions.

The fixed station subsystem shall send an update of its state whenever its state changes. It shall send a report of its new state when it receives a FSC_RCV_CNTRL receiver control command. The fixed station subsystem shall also send periodic reports of its status. This allows for any reports which were missed when it changed state.

The Fixed Station Subsystem shall send a Block with PT = 12 – Receiver Report for each receiver in the subsystem at least as often as specified by the vote report interval configuration parameter.

The fixed station subsystem shall send a Block with PT = 12 – Receiver Report for the specified receiver (or all receivers if receiver number = 0) when it receives a Block with PT = 13 – Voter Control.

The fixed station subsystem shall send a Block with PT = 12 – Receiver Report when the indicated receiver's status changes.

The fixed station subsystem shall begin forwarding voice traffic from the receiver with the best signal when it receives a Block with PT = 13 – Voter Control with the command NORMAL. The method for determining the best signal is up to the manufacturer of the comparator / voter.

The fixed station subsystem shall forward all voice traffic from the specified receiver and stop forwarding voice traffic from all other receivers when it receives a Block with PT = 13 – Voter Control with command SELECT.

The fixed station subsystem shall stop forwarding voice packets from the specified receiver (or all receivers if the receiver number = 0) when it receives a Block with PT = 13 – Voter Control with command DISABLE.

If the fixed station subsystem receives a Block with PT = 13 – Voter Control specifying a receiver number that does not exist for that fixed station subsystem it shall send a Block with PT = 12 – Receiver Report specifying that receiver number with status NOT_EQUIPPED, unless the specified receiver number was 0.

8.4.5.3 Transmitter Status

It is not required that the Transmitter Status procedure be implemented. If implemented it shall comply with the following procedure. The procedures described in this subclause apply to FSI Protocol Version 2 and above.

The fixed station subsystem shall send an update of its transmitter state whenever its state changes. It shall send a report of its new state when it receives a FSC_TX_CNTRL transmitter control command. The fixed station subsystem shall also send periodic reports of its status. This allows for any reports which were missed when it changed state.

The Fixed Station Subsystem shall send a Block with PT = 20 Transmitter Report for each transmitter in the subsystem at least as often as specified by the transmitter report interval configuration parameter.

The fixed station subsystem shall send a Block with PT = 20 Transmitter Report when the indicated transmitter's status changes.

The Fixed Station Subsystem shall send a Block with PT = 20 Transmitter Report when a transmitter control message is received.

8.4.5.4 Starting Voice Transfer

8.4.5.4.1 Start of Stream

A Start of Stream packet shall be sent when a voice transmission starts on the FSI inbound to the host or outbound from the host. Protocol Version 1 uses PT=9 for analog and digital transmissions. Protocol Version 2 and above uses PT=9 for digital transmissions, and PT=8 for analog transmissions with or without CTCSS/CDCSS transmissions.

For FSI Protocol Version 2 and above, each packet sent following the start of stream shall include a Block with PT = 9 – Start of Stream with NID or PT = 8 – Start of Stream until a Block with PT = 14 Stream Acknowledge has been received. For FSI Protocol Version 2 and above, this procedure applies to both the CFSH and CFSS and is performed to ensure reliable carriage of the NID or Squelch code both from the station to the host and from the host to the station. For FSI Protocol Version 1 on the outbound path, each packet sent by the host following the start of stream shall include a Start of Stream PT=9 until a block with PT=14 has been received from the FS.

For FSI Protocol Version 2 and above, these packets may also include voice stream blocks even though a Block with PT = 14 Stream Acknowledge has not yet been received. The receipt of the Block with PT = 14 Stream Acknowledge should not delay the transmission of speech. The Block with PT = 14 Stream Acknowledge is provided so that the human operator of a console may be given an indication that the fixed station was successfully keyed.

Note that start of stream packets may occur at any time during a voice stream. The acknowledge mechanism is provided to allow optimization of link bandwidth once delivery of the initial packet is confirmed.

On the inbound path, a CFS may repeat the Start of Stream block periodically throughout the stream. CFSH equipment shall accept these repeated packets and may use them, but shall not depend on them being present in the stream. The recommended interval for repeating the Start of Stream block is to repeat the block with the packet corresponding to the start of each LDU1.

8.4.5.4.1.1NAC

If the CFSH intends the fixed station subsystem to use a locally configured NAC on the outbound CAI, the CFSH shall set the value of NAC to \$F7E.

If the CFSH intends the fixed station subsystem to use a specific NAC on the outbound air interface, the specific NAC shall be sent in the packet. This specific NAC shall not be \$F7E or \$F7F.

If the NAC in the start of stream message on the outbound DFSI is not \$F7E the fixed station shall use this NAC on the outbound CAI.

If the NAC in the start of stream message on the outbound DFSI is \$F7E the fixed station shall use a locally configured NAC on the outbound CAI.

On the inbound fixed station interface the fixed station subsystem shall insert the NAC received on the inbound CAI.

8.4.5.4.1.2DUID

The DUID shall correspond with the CAI data unit received first in the stream.

8.4.5.4.1.3Err Cnt

On the inbound fixed station interface the “Err Cnt” field of a Block with PT = 9 – Start of Stream, shall be the number of errors detected in the NID received on the CAI.

8.4.5.4.1.4Super Frame Counter

P25 voice streams include a super frame count. Senders shall reset their super frame counter to zero when they start a stream.

8.4.5.4.2 Acknowledging Start of Stream

For all FSI Protocol versions, when a fixed station subsystem receives an outbound packet containing a start of stream message it shall key the transmitter. When the transmitter is successfully keyed the fixed station subsystem shall send an inbound packet containing Block with PT = 14 Stream Acknowledge.

For all FSI Protocol versions, once the transmitter is successfully keyed, the fixed station subsystem shall send a Block with PT = 14 Stream Acknowledge in response to every start of stream that it receives.

For FSI Protocol Version 2 and above, when a CFSH receives an inbound packet from a CFS containing a start of stream message it SHALL send a Block with PT = 14 Stream Acknowledge in response.

For FSI Protocol Version 2 and above, the CFSH shall send a Block with PT = 14 Stream Acknowledge in response to every start of stream that it receives.

8.4.5.4.3 Acknowledging Start of Stream Messages

All start of stream messages are acknowledged to ensure that the NAC or squelch code is reliably carried across the network. Figure 65 depicts the start of stream message sequence for an inbound digital call. Figure 66 depicts the start of stream message sequence for an inbound analog call.

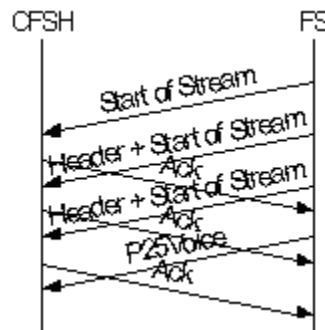


Figure 65- Start of stream message sequence - inbound digital

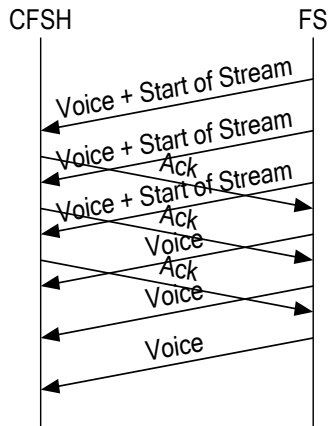


Figure 66 - Start of stream message sequence - Analog

8.4.5.4.4 Intercom Traffic

It is not required that intercom be implemented. If Intercom is implemented it shall comply with the following procedure. The following procedure applies to all FSI Protocol versions.

If an outbound voice stream starts without a Start of Stream (PT = 8 or PT = 9) the fixed station subsystem shall not key its transmitter but route the audio to its intercom facility if equipped.

Audio from the fixed stations intercom facility, if equipped, is routed to the inbound path of the DFSI. The inbound voice stream audio from the intercom is differentiated from the audio originating on the CAI since it starts without a Start of Stream (PT = 8 or PT = 9). It is expected that the users talking from the fixed station will identify themselves verbally. Verbal identification may be supplemented by the use of specific talk groups or P25 source address if the option to transfer intercom traffic as a P25 voice stream is being used.

Equipment implementing the DFSI shall support the transfer of intercom audio in analog transparent (G.711) mode.

Intercom traffic may be transferred as a P25 voice stream. This option would require that the fixed station include a vocoder.

8.4.5.5 The Header Packets

When a P25 voice stream is received from the CAI the header shall be sent on the DFSI as two fragments. This reduces delay since the first part of the header may be sent across the network before the entire header has been received on the CAI.

DFSI equipment sending an analog transparent stream shall not send any header packets. DFSI equipment that missed the header on the in-bound CAI should not send any header packets.

8.4.5.5.1 Sending Voice Headers on the Fixed Station Interface.

DFSI equipment shall send the first part of the P25 header in a packet containing a single Block with PT = 6 – Voice Header Part 1. As soon as the remainder of the header is received on the CAI, equipment shall send the second part of the header in a packet containing a single Block with PT = 7 – Voice Header Part 2.

Figure 67 illustrates the start of stream message sequence for outbound voice, and Figure 68 illustrates the start of stream message sequence with the first ACK lost.

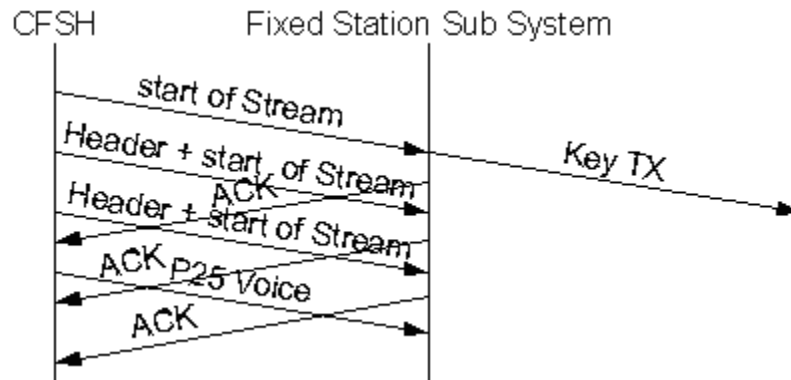


Figure 67 – Start of Stream Message Sequence – Outbound Voice for all FSI Protocol versions

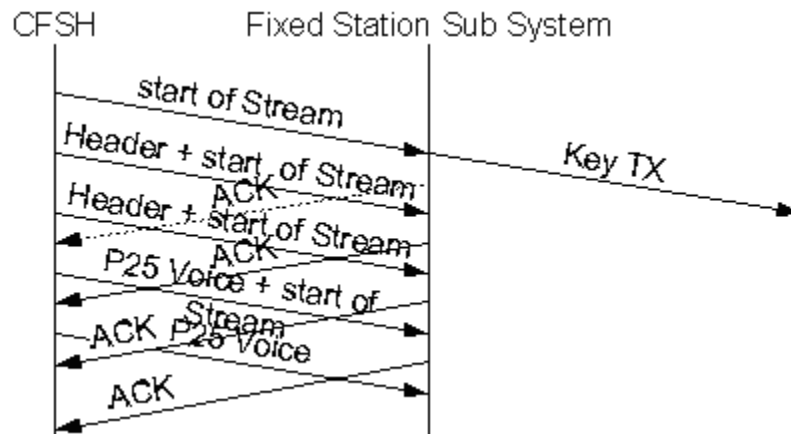


Figure 68 – Start of Stream with the First ACK Lost for all FSI Protocol versions

8.4.5.6 Transferring Voice

8.4.5.6.1 P25 IMBE Voice

Each frame of P25 voice to be sent on the fixed station shall be encapsulated as described in 8.4.4.1 Block with PT= 0 – CAI Voice. This shall be combined with any other blocks that need to be sent at the same time in an RTP packet. Each RTP packet may contain 1, 2, or 3 Block with PT= 0 – CAI Voice.

On the inbound path error correction is performed on the voice frames received from the CAI and the error counts are added to the error control fields of the block.

8.4.5.6.2 Super Frame Counter

Before sending a CAI frame type \$62 (IMBE voice 1) senders shall increment their super frame counter by 1. Senders shall insert the current value of their super frame counter in the SF field.

8.4.5.6.3 B Status Field

The Fixed station subsystem shall set the status symbols transmitted on the CAI to match the B field of the last status symbol received on the inbound CAI. The CSFH shall set the B field in the Block with PT= 0 – CAI Voice packets to match the status symbol desired on the outbound CAI.

8.4.5.7 Transport of Analog Transparent Voice

Analog transparent voice is a voice band signal typically sent to or from an analog FM RF transmitter or receiver. The fixed station is transparent to this base band signal and does not attempt to extract any meaning from it. In the receive direction the signal is demodulated, sampled and encoded. In the transmit direction it is decoded and modulated. Often this may be simply voice but it may include in band signaling between the analog subscriber units and the CSFH.

When in analog transparent mode DFSI equipment shall encode the analog audio signal using the μ law PCM scheme described in ITU-T Recommendation G.711 ([12]). Every 20 ms 160 PCM samples are assembled into an analog transparent voice packet consisting of a block of type 0 E=0 PCMU. The format of this block is described in RFC 3551 ([18]).

Analog voice may also be qualified by CTCSS or CDCSS squelch codes. If present and if the station is capable of decoding these codes, for FSI Protocol Version 2 and above they are carried with the voice in a PT=8 Analog Start of Stream block.

Figure 69 shows an example of the packet with analog transparent data block type for all FSI Protocol versions.

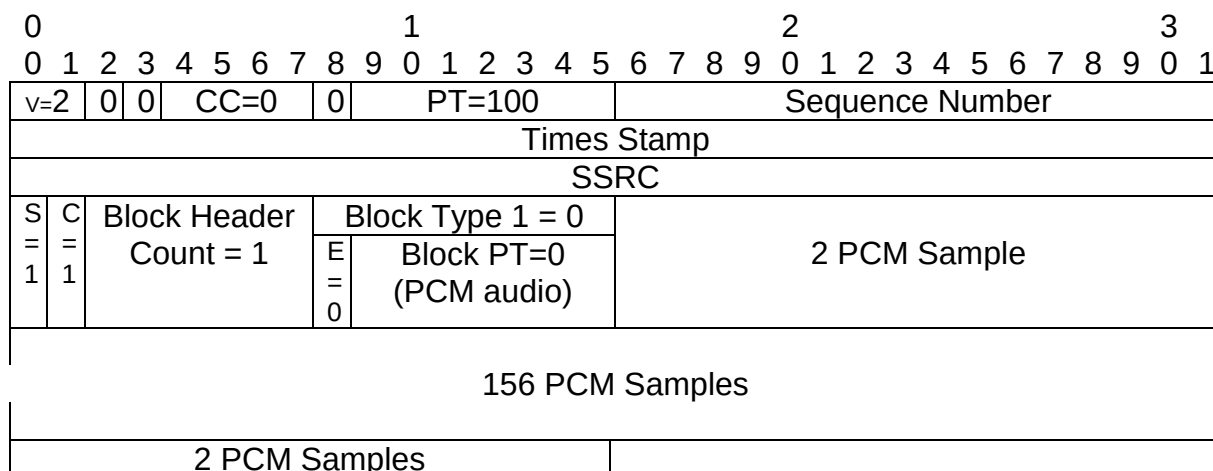


Figure 69 – Packet with Analog Transparent Data

8.4.5.8 Terminating a Voice Stream

This procedure applies to all FSI Protocol versions.

When a host or fixed station determines that a voice stream has ended it shall send four packets with a single Block with PT = 10 – End of Stream spaced 100 ms apart. If a new voice stream starts before all the end of stream packets have been sent, the remaining end of stream packets need not be sent and the new voice stream may be started immediately.

When a fixed station receives an end of stream packet it should de-key its transmitter. If a fixed station receives no voice packets for longer than a manufacturer specified timeout (typically 4 seconds) it should de-key its transmitter.

When a CFSH receives an end of stream packet, it shall terminate the stream in accordance with local policy.

Figure 70 illustrates the end-of-stream process for all FSI Protocol versions and Figure 71 illustrates the end-of-stream time-out for all FSI Protocol versions.

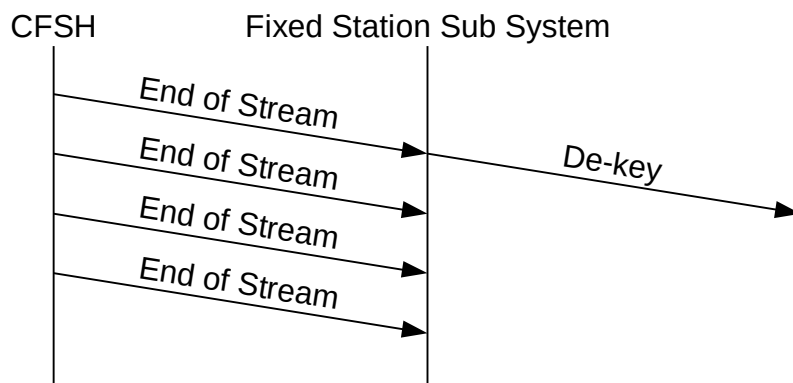


Figure 70 – End of Stream

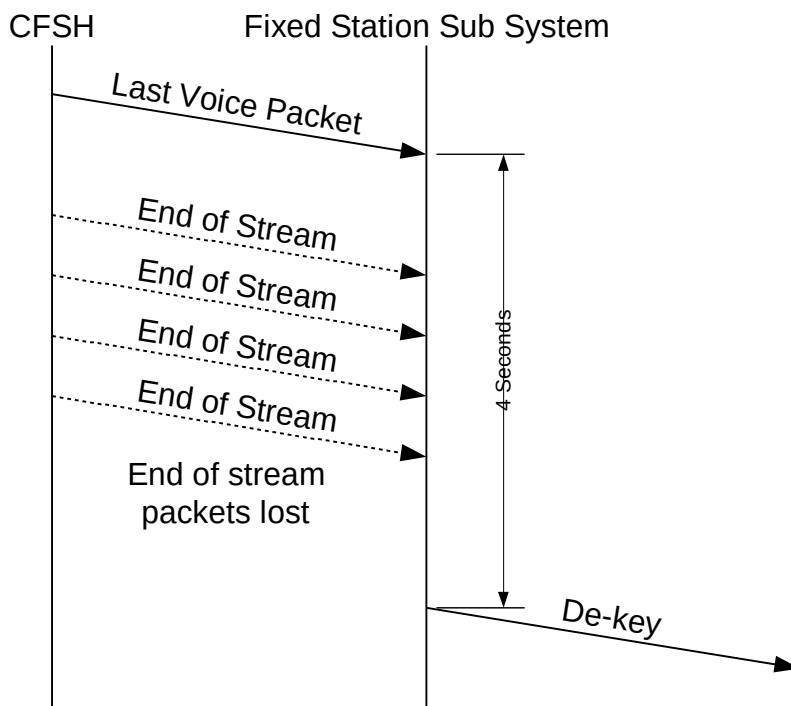


Figure 71 – End of Stream Time Out

9 Data Conveyance Procedures

9.1 Data Conveyance Connection Establishment Protocol Version 2

For Protocol Version 2, the CFSH shall establish the Data Conveyance connection only after it establishes the Protocol Version 2 Control Connection.

1. Upon detecting a Data Conveyance connectivity trigger while in the *CS Connected* state, the CFSH shall:
 - a. Send the FSC_CONNECT message indicating Data Conveyance connection to the CFSS that it is connected to via the Control Connection, applying the retransmission protocol of 8.3.2.3.
2. Upon receipt of the FSC_ACK response to the FSC_CONNECT,
 - a. If the status is CONTROL_ACK the CFSH shall
 - i. Activate the Data Conveyance connection based on local policy.
 - b. If the status is CONTROL_NAK_P_UNSUPP the CFSH shall:
 - i. Take actions as specified by local policy since protocol version is not relevant to the Data Conveyance connection.
 - c. If the status is CONTROL_NAK_V_UNSUPP, the CFSH shall retry the connect using the version associated with the response message version.
 - d. If the status is CONTROL_NAK_CONNFAIL the CFSH shall take actions as specified by local policy, this response indicates that a Data Conveyance connection is not supported by the CFSS.
3. Upon failure of the retransmission protocol on the FSC_CONNECT, the CFSH shall take actions as specified by local policy.
4. Upon receipt of FSC_CONNECT and if not in the *CS Connected* state and the connection type is Data Conveyance, the CFSS shall respond with a FSC_ACK with the Response Code of CONTROL_NAK_NO_CONTROL to indicate it cannot start the requested connection because it does not have a Control connection.
5. Upon receipt of FSC_CONNECT and it is in the *CS Connected* state and the connection type is Data Conveyance and the CFSS does not support a Data Conveyance connection, the CFSS shall respond with a FSC_ACK with the Response Code of CONTROL_NAK_CONNFAIL to indicate it cannot support the requested connection.
6. Upon receipt of FSC_CONNECT and it is in the *CS Connected* state and the connection type is Data Conveyance and the source IP address and Port of the FSC_CONNECT is not the same as the FSC_CONNECT that set up the Control Connection, the CFSS shall respond with a FSC_ACK with the Response Code of CONTROL_NAK_CONNECTED to indicate it is connected to a different CFSH.

7. Upon receipt of FSC_CONNECT and it is in the *CS Connected* state and the connection type is Data Conveyance and the CFSS supports a Data Conveyance connection and the source IP address and Port of the FSC_CONNECT is the same as the FSC_CONNECT that set up the Control Connection:
 - a. If the CFSS can support the Message Version it shall
 - i. Reply with FSC_ACK with the CONTROL_ACK Response Code
 - iii. Activate the Data Conveyance connection based on local policy.
 - b. If the CFSS cannot support the Message Version of the FSC_CONNECT request, it shall:
 - ii. Reply with FSC_ACK with the CONTROL_NAK_V_UNSUPP Response Code.

9.2 Data Conveyance Connection Detachment Procedures Protocol Version 2

1. Upon detecting a Data Conveyance disconnect trigger while it has an Data Conveyance connection with a CFSS, the CFSH shall:
 - a. Send the FSC_DISCONNECT message for the Data Conveyance connection to the CFSS, applying the retransmission protocol of 8.3.2.3.
2. Upon receiving the FSC_DISCONNECT message for the Data Conveyance connection while it does not have an active Data Conveyance connection, the CFSS shall respond with FSC_ACK with a Response Code of CONTROL_ACK.
3. Upon receiving a FSC_DISCONNECT message for an active Data Conveyance connection, the CFSS shall:
 - a. Reply with FSC_ACK with a status code of CONTROL_NAK_CONNECTED if the IP address and port of the sender of the FSC_DISCONNECT do not match those of the sender of the FSC_CONNECT that caused the activation of the Data Conveyance connection.
 - b. Otherwise,
 - i. Reply with FSC_ACK with a status code of CONTROL_ACK and,
 - ii. Deactivate the Data Conveyance connection per local policy.
4. Upon receiving a FSC_ACK with a status code of CONTROL_ACK in response to sending a FSC_DISCONNECT for a Data Conveyance connection to a CFSS , a CFSH shall:
 - a. Deactivate the Data Conveyance connection per local policy.
 - b. Discontinue the transmission of any and all packets on the Data Conveyance connection.

9.3 Data Conveyance Message

The primary purpose of the Data Conveyance Message is to convey CAI Data Packets as defined by reference [8] across the FSI. The format of a Data Conveyance message is shown in Figure 72, and is comprised of a Data Conveyance header and a Data Conveyance Payload.

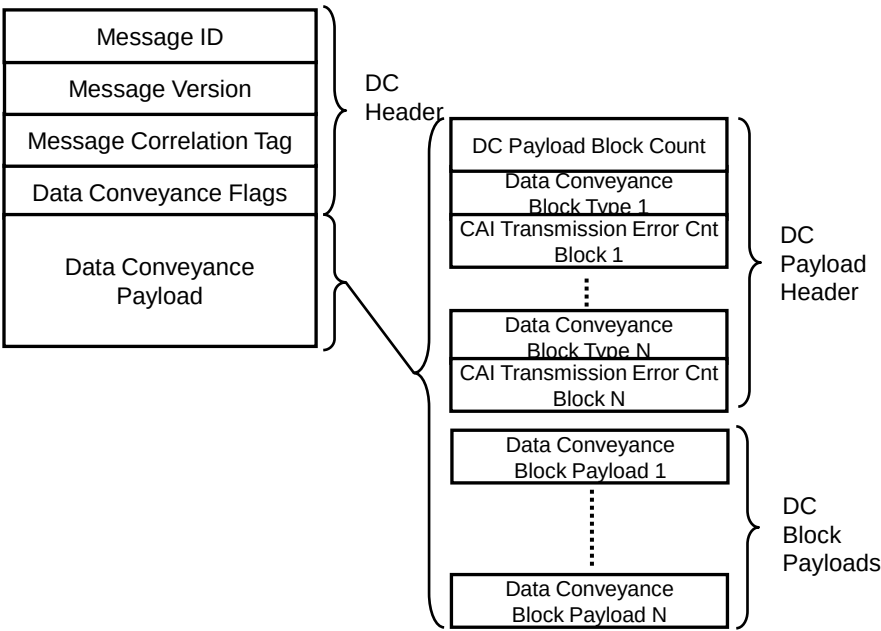


Figure 72 – FSI Data Conveyance Message

The Data Conveyance message is bi-directional, meaning that both the CFSH and the CFSS construct inbound and outbound messages according to this message structure.

For inbound CAI data blocks, it is the responsibility of the CFSS to demodulate each CAI block, decode the FEC, insert the CAI data blocks into a Data Conveyance message and forward the message to the CFSH Data Conveyance socket.

For outbound CAI data blocks, the CFSH delivers the Data Conveyance messages to the CFSS Data Conveyance socket. It is the responsibility of the CFSS to take each DC Block Payload, extract the CAI Blocks, add the FEC, and modulate each CAI Block over the air.

9.3.1 Data Conveyance Header

The Data Conveyance Header, reference Figure 73, includes a Message ID field, a Message Version field, a Message Correlation Tag field, and a set of Data

Conveyance Flags. The first three octets match the base header of all control messages, and to this end, the Message ID value space is shared between the Control and Data Conveyances. The Data Conveyance Flags octet is not present in the Control connection messages.

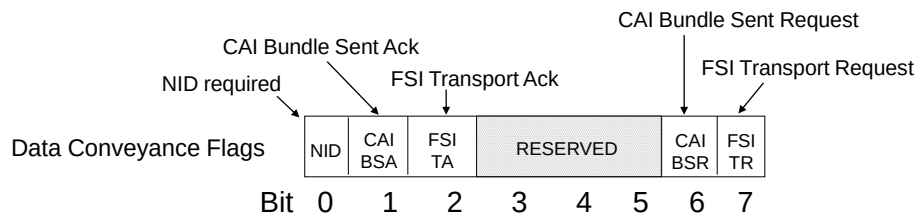
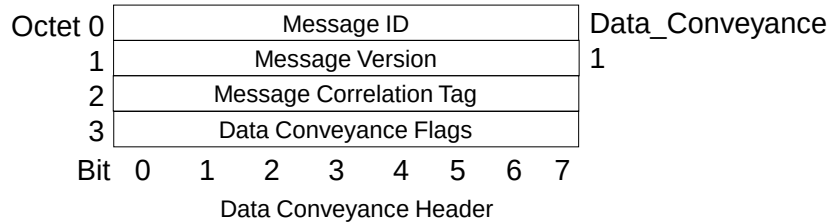


Figure 73 – Data Conveyance Message Header

The usage and format of the Message ID, Message Version and Message Correlation Tag fields are unchanged from the Control connection messages. The Message ID for a Data Conveyance message is \$40, and the Message Version field is \$01.

The Data Conveyance Flags field includes the capability for the CFSH to request a FSI Transport acknowledgment, or a CAI Bundle Sent acknowledgment. The CFSS is responsible for responding to the FSI Transport or CAI Bundle Sent acknowledgment requests.

The FSI Transport Request flag (FSI TR) provides an indication to the CFSS that the CFSH requests an FSI Transport Acknowledgment be returned when the current Data Conveyance message is received over the FSI. The CAI Bundle Sent Request flag (CAI BSR) is an indication to the CFSS that the CFSH requests an CAI Bundle Sent Acknowledgment be returned when the current Data Conveyance message is sent over the CAI.

When the CFSS receives a Data Conveyance message from the CFSH with the FSI TR flag asserted, the CFSS responds by sending a Data Conveyance header to the CFSH with the FSI TA bit asserted. The Message Correlation Tag field is set to the Message Correlation Tag field received in the associated Data Conveyance message that requested the FSI Transport Ack.

When the CFSS receives a Data Conveyance message from the CFSH with the CAI BSR flag asserted, the CFSS responds by sending a Data Conveyance header to the CFSH with the CAI BSA bit asserted after the CAI data blocks in the received Data Conveyance message have been sent over the air. The Message Correlation Tag field is set to the Message Correlation Tag field received in the associated Data Conveyance message that requested the CAI Bundle Sent Ack.

The Data Conveyance Flags include the capability for the CFSS to provide an indication to the CFSH that the CFSS requires a Network ID. This is accomplished by the CFSS sending a Data Conveyance header to the CFSH with the NID flag asserted. The value of the Message Correlation Tag will be that of the original outbound Data Conveyance message that failed to supply the NID. Upon receiving a DC message from the CFSS with the NID flag asserted, the CFSH may resend the Data Conveyance message to the CFSS with a Start of Stream DC Block Payload (NID) inserted.

The NID, CAI BSA and FSI TA flags are not asserted by the CFSH, and are zero in all outbound Data Conveyance messages.

The CAI BSR, and FSI TR flags are not asserted by the CFSS, and are zero in all inbound Data Conveyance messages.

The Reserved Data Conveyance Flag bits are for future use and are set to zero for both inbound and outbound Data Conveyance messages.

9.3.2 Data Conveyance Payload

The Data Conveyance Payload consists of a DC Payload Header and one or more DC Block Payloads. The following sub-sections describe the components of a Data Conveyance Payload.

9.3.2.1 Data Conveyance Payload Header

The Data Conveyance Payload Header consists of the Data Conveyance Payload Block Count and one or more Data Conveyance Block Types and CAI Transmission Error Counts.

9.3.2.1.1 Data Conveyance Payload Block Count

The DC Payload Block Count octet is shown in Figure 74.

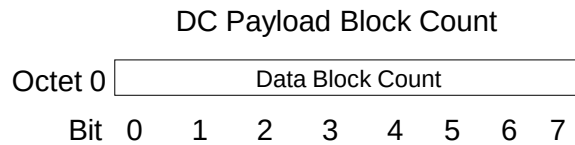
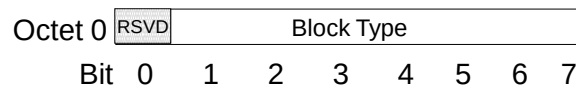


Figure 74 – DC Payload Block Count for FSI Data Conveyance Message

The DC Payload Block Count indicates the total number of Block Payloads in the Data Conveyance message. More information on the data block count for Data Conveyance messages can be found in section 9.4.1.2.

9.3.2.1.2 Data Conveyance Block Type

Figure 75 shows the format of the Data Conveyance Block Type Octet.



Data Conveyance Block Type Octet

Figure 75 – Data Conveyance Block Type Octet

The Data Conveyance Block Type Octet contains 2 fields; the “RSVD” field and the “Block Type” field. The “RSVD” field of the DC Block Type Octet is Reserved and not used in Data Conveyance messages. The “Block Type” field is set to one of the Data Conveyance Block Type values shown in Table 19.

A series of Data Conveyance Block Type octets provides the order and type of Data Conveyance Block Payloads that follow.

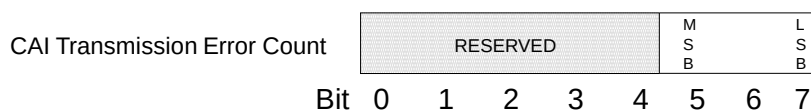
Table 19 - Data Conveyance Block Types

Data Conveyance Block Type	Value
Start of Stream with NID	09
End of Stream	10
Receiver Report	12
Transmitter Report	20
Confirmed Data Header Block	31
Confirmed Data Block	32
Confirmed Data Last Block	33
Confirmed Response	34
Unconfirmed Data Header Block	35
Unconfirmed Data Block	36
Unconfirmed Data Last Block	37

The DFSI Packet Data interface is present only when the Conventional FNE Data model [BAEA] is employed such that there is an FNE present. This means that the DFSI is only used with asymmetric addressing (not used with enhanced (also called symmetric) addressing). The CFSS does not repeat asymmetric addressed Packet Data messages on its own because they only contain the source Logical Link ID (unit ID). The CFSS would be repeating the message back to the source SU. The CFSH can process the asymmetrically addressed message, determine the destination Logical Link ID (unit ID), and use the CFSS to transmit the received Packet Data message to the destination SU.

9.3.2.1.3 CAI Transmission Error Count

Figure 76 shows the CAI Transmission Error Count octet.

**Figure 76 - CAI Transmission Error Count**

The number of FEC errors detected by the CFSS for the particular block in the received CAI transmission is provided to the CFSH in inbound Data Conveyance messages using the CAI Transmission Error Count parameter. This parameter is

limited to a maximum of 7. The CAI Transmission Error Count parameter is set to %000 for outbound Data Conveyance blocks and inbound Data Conveyance blocks not decoded on the CAI (i.e., Start of Stream).

9.3.2.2 Data Conveyance Block Payloads

The Data Conveyance Block Payloads shall match the order and type specified in the Data Conveyance Payload Header. A Data Conveyance Block Payload is defined for each of the Data Conveyance Block Types listed in Table 19.

The format of the Data Conveyance Start of Stream Block Payload is unchanged from the Voice Conveyance Start of Stream Type (PT = 9). However, a new DUID (Data Unit Identifier) value of %1100 is used to indicate Packet Data as defined in [8] Common Air Interface.

The Data Conveyance End of Stream Block uses the same block type as the Voice Conveyance End of Stream Type (PT = 10). This block has no payload.

The Receiver Report Block Payload follows the format and usage of the Voice Conveyance Receiver Report block (PT=12).

The Transmitter Report Block Payload follows the format and usage of the Voice Conveyance Transmitter Report block (PT=20).

A Confirmed Data Header Block Payload follows the format of the Confirmed Data Packet Header Format as described in [8] Common Air Interface.

A Confirmed Data Block Payload follows the format of the Confirmed Data Packet Data Block Format as described in [8] Common Air Interface.

A Confirmed Data Last Block Payload follows the format of the Confirmed Data Packet Last Data Block Format as described in [8] Common Air Interface.

A Confirmed Response Block Payload follows the format of the Response Packet Format as described in [8] Common Air Interface.

An Unconfirmed Data Header Block Payload follows the format of the Unconfirmed Data Packet Header Format as described in [8] Common Air Interface.

An Unconfirmed Data Block Payload follows the format of the Unconfirmed Data Block Format as described in [8] Common Air Interface.

An Unconfirmed Data Last Block Payload follows the format of the Unconfirmed Data Last Block Format as described in [8] Common Air Interface.

9.4 Data Conveyance Procedures

When a CFSH – CFSS pair are in the connected state for Data Conveyance, they exchange Data Conveyance messages according to the following procedures.

9.4.1 Transfer of Data Conveyance messages

For inbound Data traffic, the CFSS is responsible for demodulating and decoding inbound CAI data blocks, constructing Data Conveyance messages as described in section 9.3, and sending them to the CFSH. An example is shown in Figure 77.

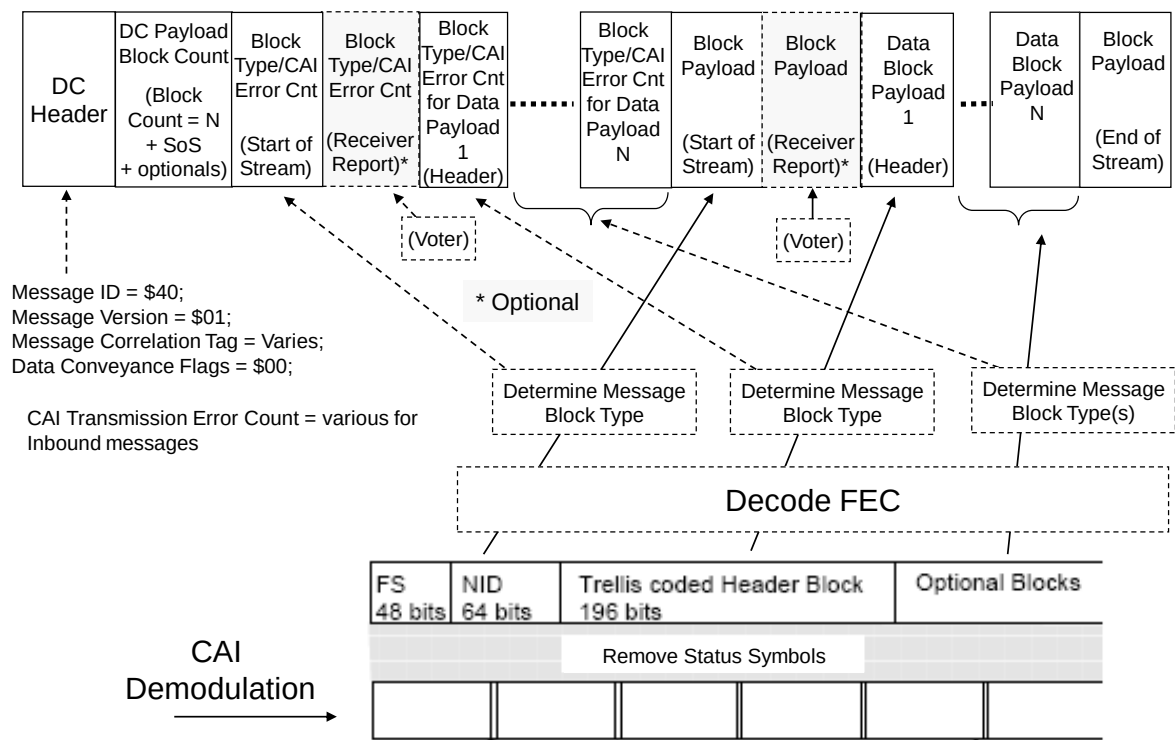


Figure 77 – Inbound Data Conveyance Message Assembly

For outbound Data traffic, The CFSH is responsible for constructing Data Conveyance messages as defined in section 9.3, and sending them to the CFSS.

The CFSS receives the outbound Data Conveyance messages and is responsible for encoding, inserting status bits, and modulating the CAI Blocks over the air. An example of this is shown in Figure 78.

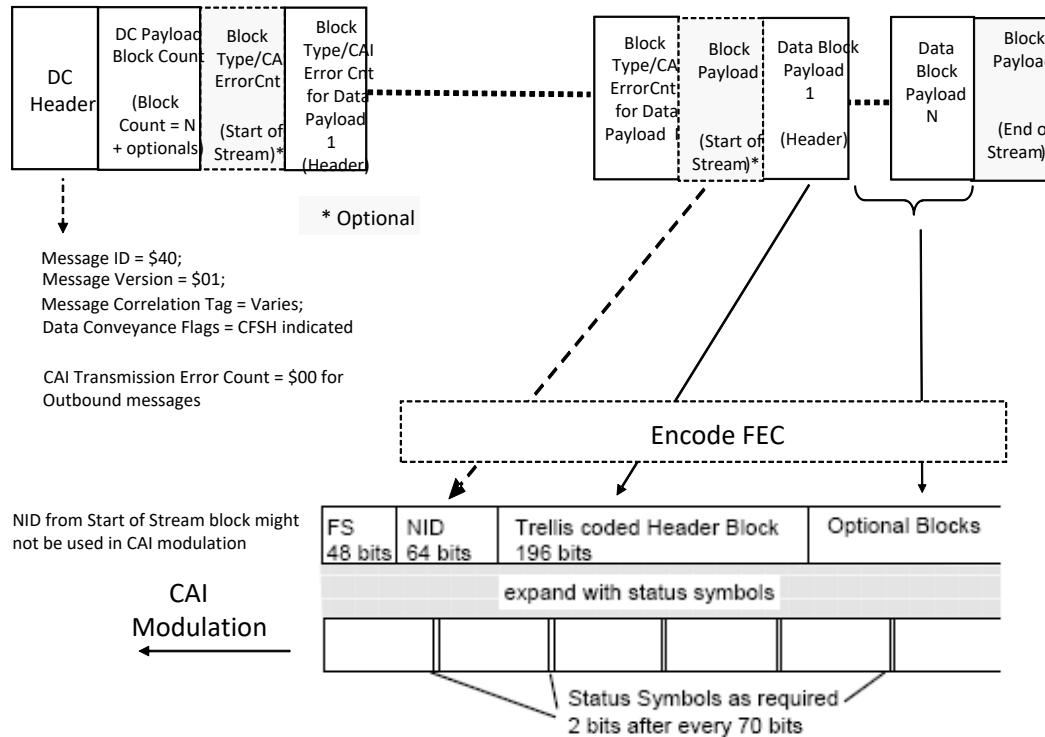


Figure 78 – Outbound Data Conveyance Message Modulation

A single DC message (or a related series of DC messages) shall contain no more than one Data Header Block and no more than one Data Last Block. The number of Data Blocks between a Data Header Block and Data Last Block may vary.

The CFSH is responsible for processing inbound DC messages that contain a Confirmed Response Block Payload.

The CFSH is also responsible for providing any outbound retransmissions and generating acknowledgments for inbound Data Conveyance messages.

The number of FEC errors detected by the CFSS in each block of the inbound CAI transmission is provided to the CFSH in an inbound Data Conveyance message using the CAI Transmission Error Count Field found in the Data Conveyance Payload Header.

9.4.1.1 Start of Stream for Data

The Data Conveyance message supports the transfer of the Network ID between the CFSH and the CFSS in the form of a Data Conveyance Start of Stream Block Payload.

At the beginning of an inbound Data transfer, a single Data Conveyance Start of Stream Block Payload shall be inserted by the CFSS in order to provide the Network ID (NID) to the CFSH.

On the outbound path, a Data Conveyance Start of Stream Block Payload may be inserted by the CFSH.

When a Start of Stream Block Payload is present, it shall:

1. Appear in the same Data Conveyance message as a Confirmed Data Header Block, Unconfirmed Data Header Block, or Confirmed Response Block Payload, and
2. Appear prior to the Confirmed Data Header Block, Unconfirmed Data Header Block, or Confirmed Response Block Payload in the Data Conveyance message payload.

The CFSS shall be configured to self-manage a local NID only, to utilize the NID provided by the CFSH if available otherwise self-manage, or use NID from CFSH only.

If the CFSS is configured to self-manage a local NID only:

1. The CFSS shall be provisioned with a default NID to use as part of self-managed NID algorithm.

2. When the CFSS modulates the CAI Blocks of a Data Conveyance message on the CAI, the CFSS uses the self-managed NID.

If the CFSS is configured to utilize the NID provided by the CFSH if available, otherwise self-manage:

1. The CFSS shall be provisioned with a default NID to use as part of self-managed NID algorithm.
2. When the CFSS modulates the CAI Blocks of a Data Conveyance message on the CAI, the CFSS shall use the NID provided in the Start of Stream Block Payload for the CAI data fragment, if available.
3. When the CFSS modulates the CAI Blocks of a Data Conveyance message on the CAI and a NID is not provided in the Start of Stream Block Payload for the CAI data fragment, the CFSS uses the self-managed NID.

If the CFSS is configured to utilize the NID provided by the CFSH only:

1. The CFSS shall only modulate CAI Blocks on the CAI if it has received a Start of Stream Block Payload for the CAI data fragment in progress.
2. When the CFSS modulates the CAI Blocks of a Data Conveyance message on the CAI, the CFSS shall use the NID provided in the Start of Stream Block Payload for the CAI data fragment.

If the CFSS is configured to utilize the NID provided by the CFSH only and the NID is not present in the outbound Data Conveyance message, the CFSS may respond by sending a Data Conveyance header to the CFSH with the NID flag asserted. The value of the Message Correlation Tag will be that of the original outbound Data Conveyance message that failed to supply the NID.

9.4.1.2 Number of Block Payloads

The Data Block Count field in a Data Conveyance message payload represents the total number of Data Conveyance Block Payloads in the Data Conveyance message (i.e. Sum of CAI data blocks, Start of Stream block, Receiver Report and Transmitter Report blocks).

A packet data message can be constructed of up 128 CAI data blocks (Start of Stream block, Receiver Report, and Transmitter Report blocks do not count towards this limit). If the number of CAI data blocks in the packet data message exceeds the Max_CAI_Data_Blocks parameter for either inbound or outbound DC messages,

then multiple DC messages are sent for that packet data message. Each DC message sent in a series shall contain the negotiated inbound or outbound Max_CAI_Data_Blocks number of CAI data blocks, except the final DC message of the sequence or an individual DC message which may have less than the Max_CAI_Data_Blocks number of CAI data Block Payloads.

There is no requirement to pad-fill the final DC message of a sequence or individual DC message if the number of CAI Data Blocks in the payload is less than the negotiated Inbound or Outbound Max_CAI_Data_Blocks value.

The maximum number of CAI data blocks in an inbound DC message is determined by taking the lower of the CFSH and CFSS specified inbound max values (CFSH Inbound Max_CAI_Data_Blocks parameter from the FSC_CONNECT message, and CFSS Inbound Max_CAI_Data_Blocks parameter from the FSC_ACK message).

The maximum number of CAI data blocks in an outbound DC message is similarly determined by taking the lower of the CFSH and CFSS specified outbound max values (CFSH Outbound Max_CAI_Data_Blocks parameter from the FSC_CONNECT message, and CFSS Outbound Max_CAI_Data_Blocks parameter from the FSC_ACK message).

For example, if the FSC_CONNECT message specifies a CFSH Inbound Max_CAI_Data_Blocks parameter value of 100, and the corresponding FSC_ACK from the CFSS specifies a CFSS Inbound Max_CAI_Data_Blocks value of 16, the resulting maximum number of CAI data blocks in any single inbound DC message would be 16.

Similarly, if the FSC_CONNECT message specifies a CFSH Outbound Max_CAI_Data_Blocks parameter value of 4, and the corresponding FSC_ACK from the CFSS specifies a CFSS Outbound Max_CAI_Data_Blocks value of 128, the resulting maximum number of CAI data blocks in any single outbound DC message would be 4.

9.4.1.3 End of Stream for Data

The Data Conveyance message supports the End of Stream block. The End of Stream block has no payload.

At the end of an inbound Data transfer, a single Data Conveyance End of Stream Block shall be inserted by the CFSS in order to inform the CFSH of the end of the received data stream.

On the outbound path, the CFSH may insert a Data Conveyance end of Stream Block.

When an End of Stream Block is present, it shall:

1. Appear in the same Data Conveyance message as a Confirmed Data Header Block (when there are no blocks to follow), Unconfirmed Data Header Block (when there are no blocks to follow), Confirmed Response Block Payload, Confirmed Data Last Block, or Unconfirmed Data Last Block, and,
2. Appear after the Confirmed Data Header Block (if there are no blocks to follow), Unconfirmed Data Header Block (if there are no blocks to follow), Confirmed Response Block Payload, Confirmed Data Last Block Payload, or Unconfirmed Data Last Block Payload in the DC message payload. In other words, it is the last block in the Data Conveyance message payload.

10 Informative Annex

10.1 Reserved Words

As of the date of this document, the following payload Types are reserved for use on other TIA Interfaces.

Table 20 - Block Payload Type Reservations

Payload Type	Defined in
1 – Transmission Control Packet Type	TIA-102.BACA-B
11 –PTT Control Word	TIA-102.BACA-B
15 – Console PTT Control Word	TIA-102.BACA-B
16 - Conv RF PTT Control Word	TIA-102.BACE
17 - Conv Console PTT Control Word	TIA-102.BACE
18 - Conv Console Control Word	TIA-102.BACE
19 - Conv Link Control Word	TIA-102.BACE
21 - Station Settings for Console Transmission	TIA-102.BACE
22 - Station Control Command	TIA-102.BACE
23 - Station Control Report Request	TIA-102.BACE
24 - SBC Command	TIA-102.BACE
25 - Voter Command	TIA-102.BACE
26 - Voter Update	TIA-102.BACE
27 - Connection Request	TIA-102.BACE
28 - Connection Heartbeat	TIA-102.BACE
29 - Control ACKnowledgement	TIA-102.BACE
30 - Disconnect Request	TIA-102.BACE
31-62 – Reserved for future expansion	

- End -

Copyright Telecommunications Industry Association
Provided by IHS under license with TIA
No reproduction or networking permitted without license from IHS Markit

THE TELECOMMUNICATIONS INDUSTRY ASSOCIATION

TIA represents the global information and communications technology (ICT) industry through standards development, advocacy, tradeshow, business opportunities, market intelligence and world-wide environmental regulatory analysis. Since 1924, TIA has been enhancing the business environment for broadband, wireless, information technology, cable, satellite, and unified communications.

TIA members' products and services empower communications in every industry and market, including healthcare, education, security, public safety, transportation, government, the utilities. TIA is accredited by the American National Standards Institute (ANSI).



**TELECOMMUNICATIONS
INDUSTRY ASSOCIATION**

**1320 N. Courthouse Road, Suite 200
Arlington VA, 22201, USA**

**+1.703.907.7700 MAIN
+1.703.907.7727 FAX**